

Hemodynamic disorders and Cardiovascular diseases

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Outline

Hemodynamic disorders

- Edema and effusion
- Hyperemia and congestion
- Hemorrhage
- Thrombosis
- Disseminated intravascular coagulation (DIC)
- Embolism
- Infarction
- Shock***

Heart diseases

- Congenital heart disease***
- Congestive heart failure***
- Ischemic heart disease
- Hypertensive heart disease
- Valvular heart disease
- Myocarditis
- Tumors of the heart

Vascular diseases

- Vascular structure
- Vascular injury and response
- Blood pressure regulation
- Hypertension
- Atherosclerosis
- Aneurysm and dissection
- Vasculitis
- Disorder of vein
- Vascular tumor

HEMODYNAMIC DISORDERS

Edema and effusion

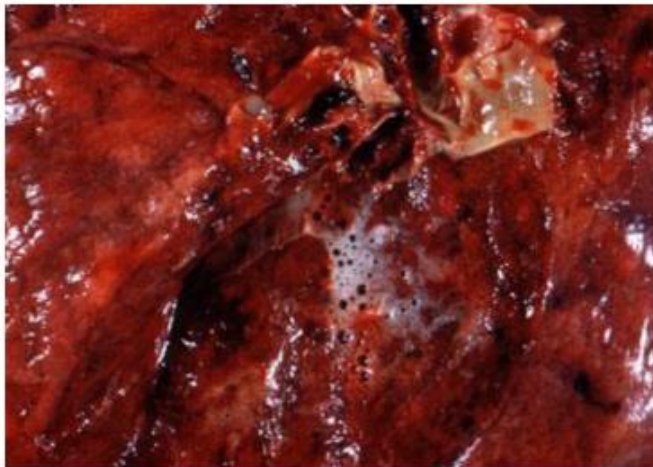
- **Edema** = accumulation of fluid in **interstitial spaces**

e.g. Subcutaneous edema
Periorbital edema
Brain edema

- **Effusion** = accumulation of fluid in **body cavities**

e.g. Hydrothorax (pleural effusion)
Ascites (peritoneal effusion)

Transudate (non-inflammatory) = protein-poor
Exudate (inflammatory) = protein-rich



Pulmonary edema



Hydrothorax



Chylothorax

Edema and effusion

- Factors influencing fluid movement across capillary walls

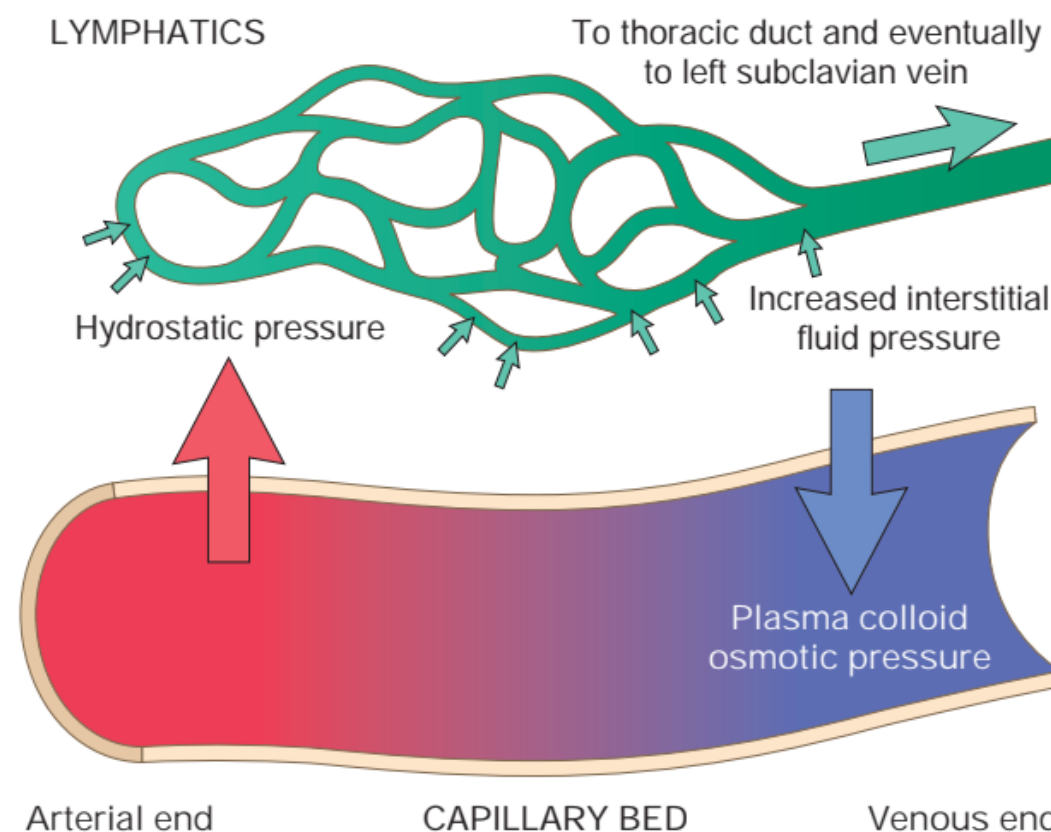
- Hydrostatic pressure
- Colloid osmotic pressure
- Lymphatic drainage

- **Causes of edema**

1. Increased hydrostatic pressure
2. Reduced plasma osmotic pressure
3. Lymphatic obstruction
4. Increased vascular permeability

5. **Sodium and water retention**

Increased hydrostatic pressure (due to intravascular fluid volume expansion) and diminished vascular colloid osmotic pressure (due to dilution).



Factors influencing fluid movement across capillary walls. Normally, hydrostatic and osmotic forces are nearly balanced so that there is little net movement of fluid out of vessels. Many different pathologic disorders are associated with increases in capillary hydrostatic pressure or decreases in plasma osmotic pressure that lead to the extravasation of fluid into tissues. Lymphatic vessels remove much of the excess fluid, but if the capacity for lymphatic drainage is exceeded, tissue edema results.

Edema and effusion

Increased Hydrostatic Pressure

Impaired Venous Return

- Congestive heart failure
- Constrictive pericarditis
- Ascites (liver cirrhosis)
- Venous obstruction or compression
 - Thrombosis
 - External pressure (e.g., mass)
- Lower extremity inactivity with prolonged dependency

Arteriolar Dilation

- Heat
- Neurohumoral dysregulation

Reduced Plasma Osmotic Pressure (Hypoproteinemia)

- Protein-losing glomerulopathies (nephrotic syndrome)
- Liver cirrhosis
- Malnutrition
- Protein-losing gastroenteropathy

Lymphatic Obstruction

- Inflammatory
- Neoplastic
- Postsurgical
- Postirradiation

Sodium Retention

- Excessive salt intake with renal insufficiency
- Increased tubular reabsorption of sodium
 - Renal hypoperfusion
 - Increased renin-angiotensin-aldosterone secretion

Inflammation

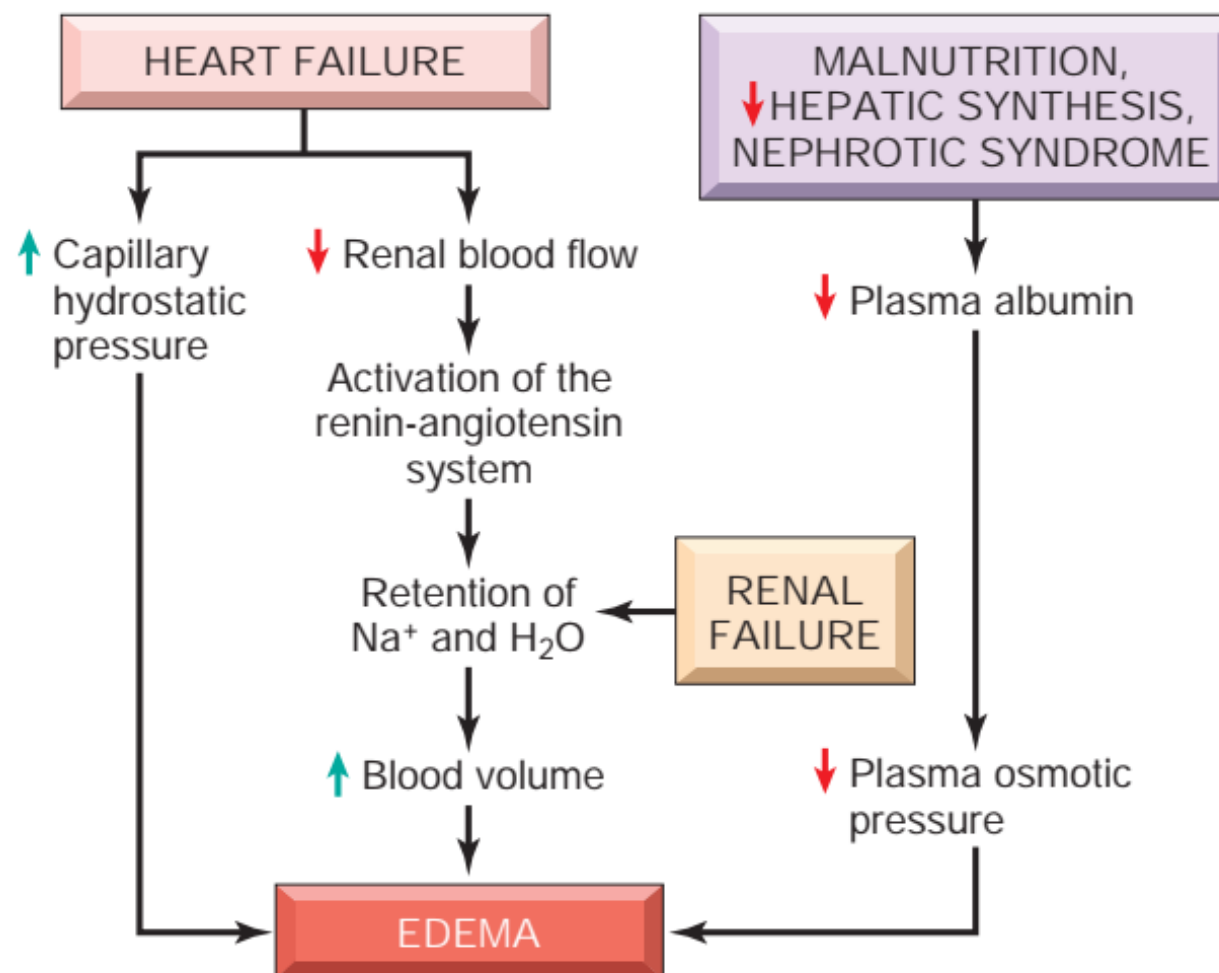
- Acute inflammation
- Chronic inflammation
- Angiogenesis

Modified from Leaf A, Cotran RS: *Renal Pathophysiology*, ed 3, New York, 1985, Oxford University Press, p 146.

Under normal circumstances **albumin** accounts for almost half of the total plasma protein; it follows that conditions leading to **inadequate synthesis** or **increased loss of albumin** from the circulation are common causes of reduced plasma oncotic pressure.

Edema and effusion

Mechanisms of systemic edema in heart failure, renal failure, malnutrition, hepatic failure, and nephrotic syndrome.



Hyperemia and congestion

Hyperemia and congestion both stem from **increased blood volumes within tissues**, but have different underlying mechanisms and consequences.

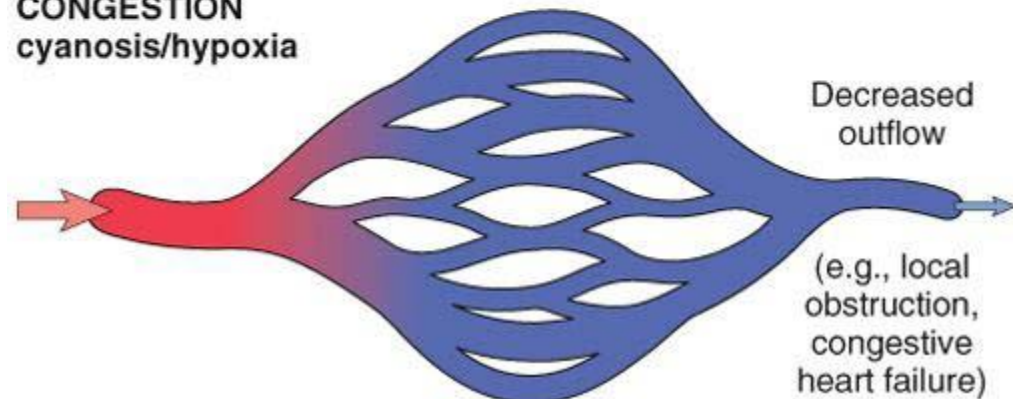
HYPEREMIA erythema



Hyperemia

- **active** process
- arteriolar dilation (e.g., at sites of inflammation or in skeletal muscle during exercise)
- increased blood flow
- tissues turn **red (erythema)**: **oxygenated** blood ↑

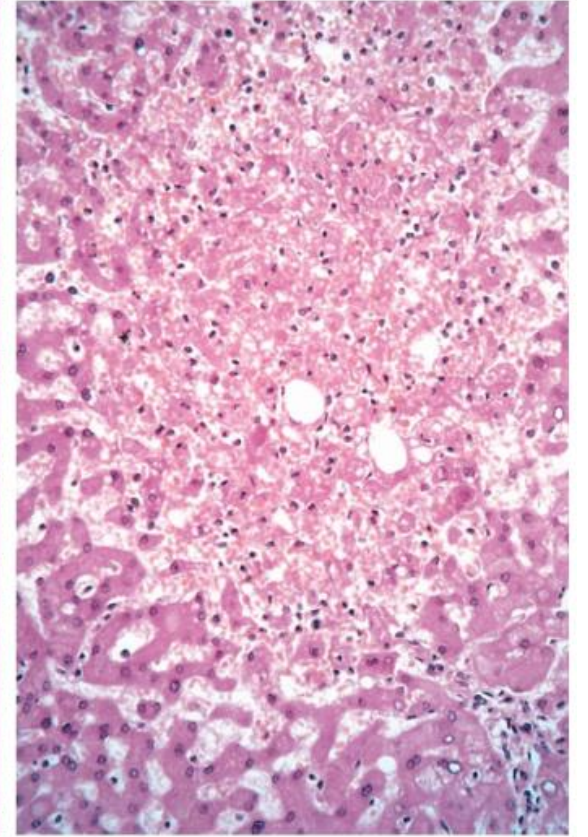
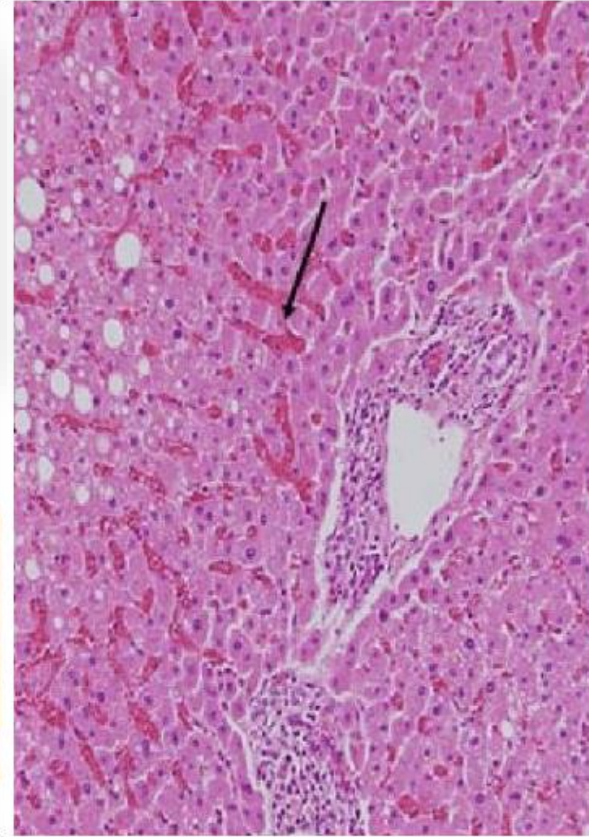
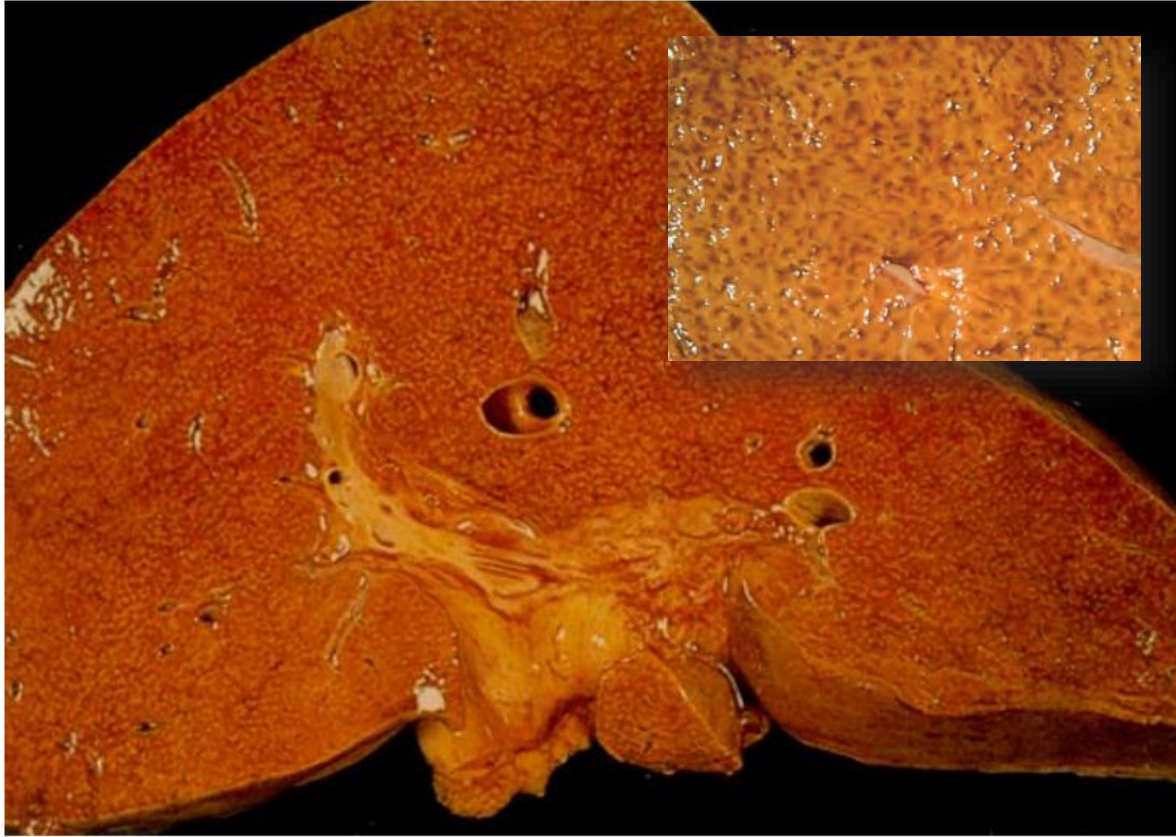
CONGESTION cyanosis/hypoxia



Congestion

- **passive** process
- reduced venous outflow (e.g. cardiac failure or isolated venous obstruction)
- **blue-red** color (**cyanosis**): **deoxygenated** blood ↑

Hyperemia and congestion



Chronic congestion of liver, with hemorrhagic necrosis
(nutmeg liver)

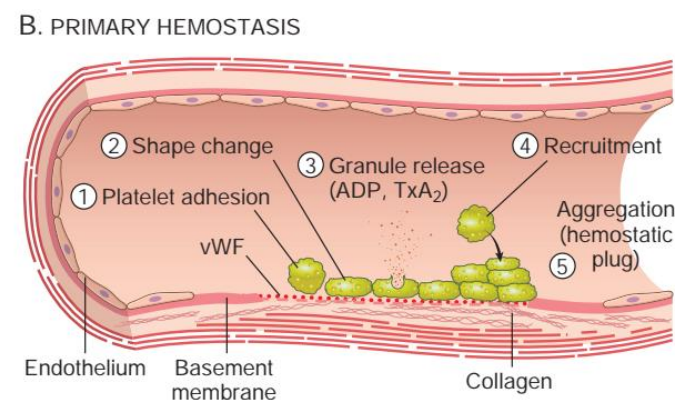
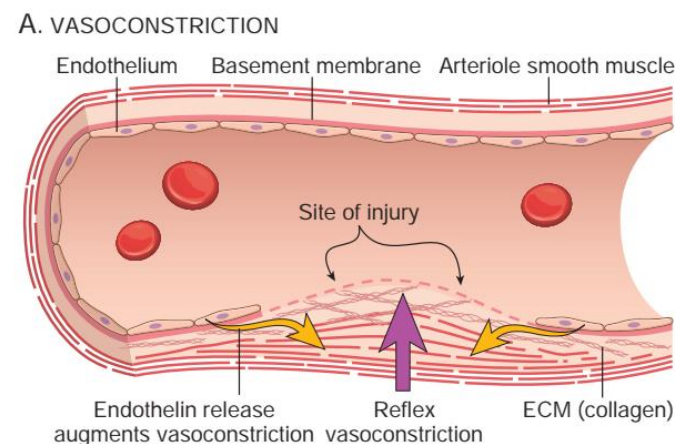
Normal hemostasis

- Hemostasis depends on three general components:

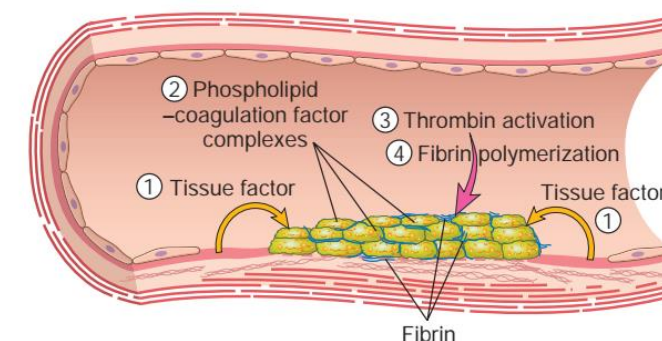
- Vascular wall
- Platelets
- Coagulation pathways

- **Events in hemostasis**

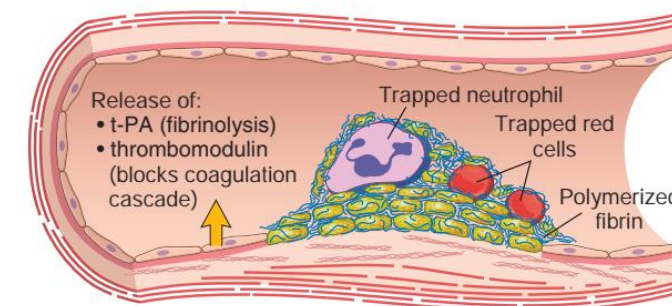
- A. Vasoconstriction: reduces blood flow to the injured area
- B. Primary hemostasis: the formation of the platelet plug
- C. Secondary hemostasis: deposition of fibrin
- D. Antithrombotic counter-regulation: Clot stabilization and resorption



C. SECONDARY HEMOSTASIS

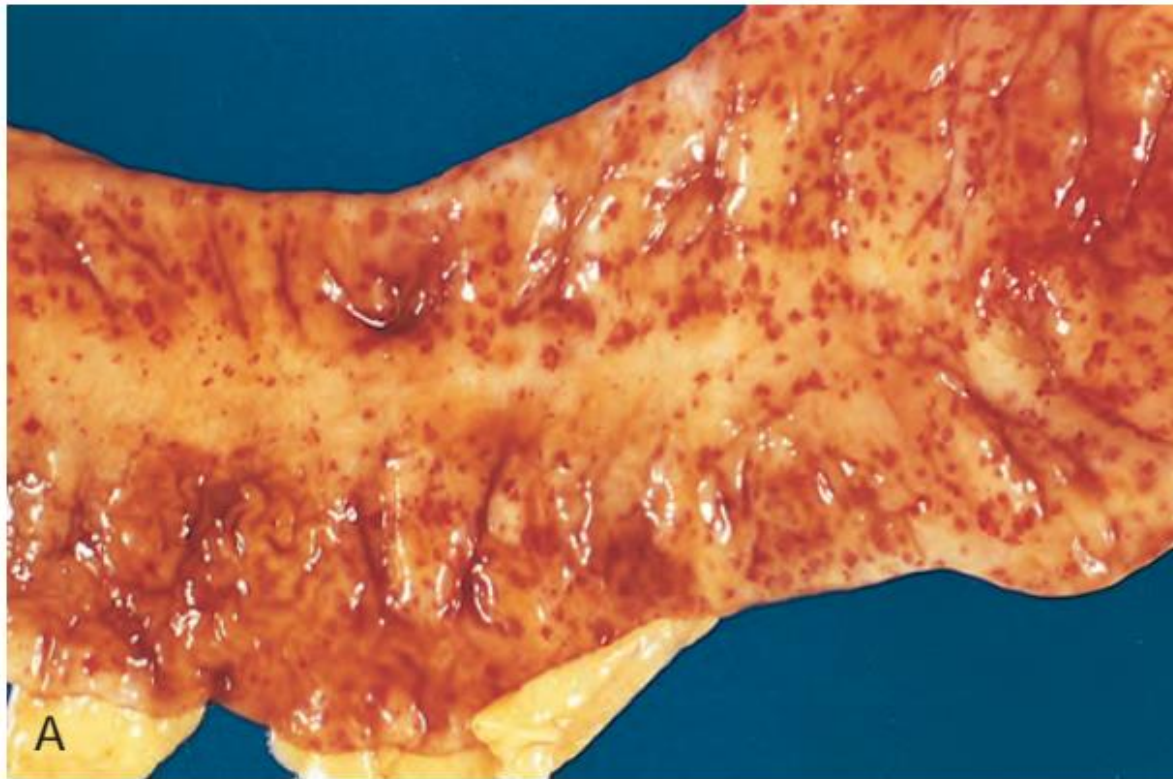


D. THROMBUS AND ANTITHROMBOTIC EVENTS

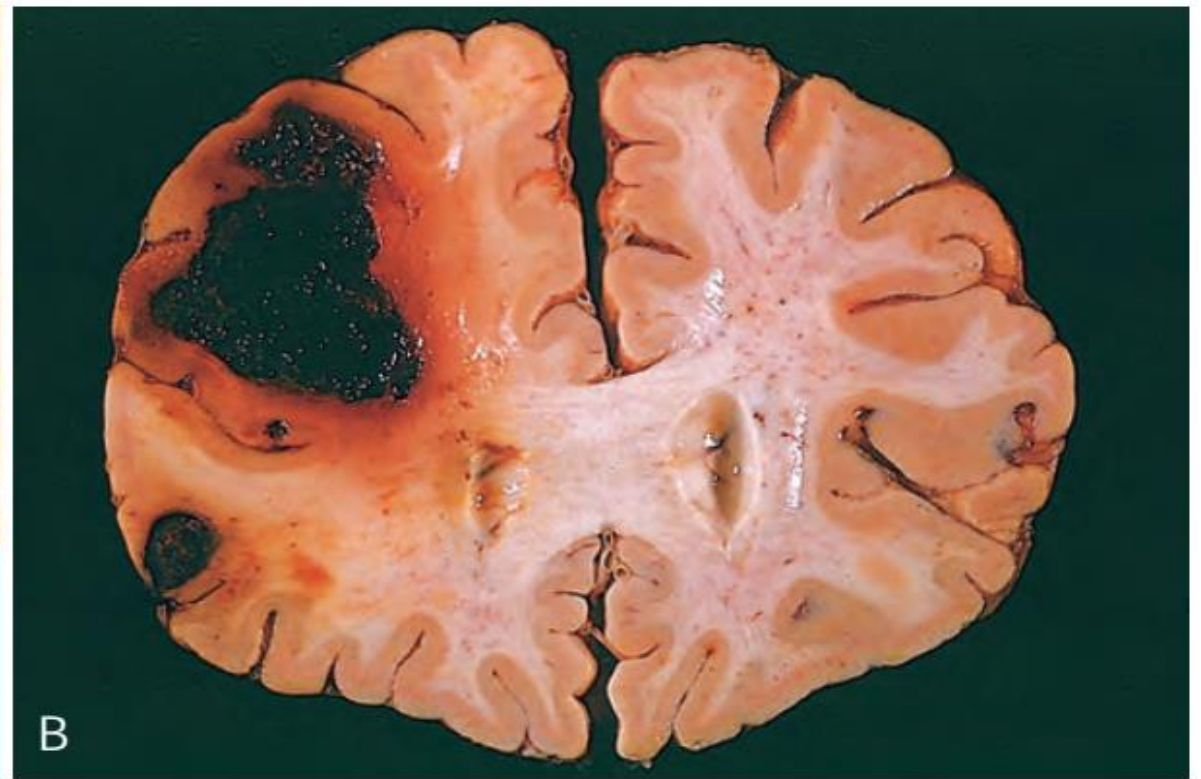


Hemorrhage (bleeding)

Extravasation of blood due to **defects** in **vessel walls**, **platelets**, or **coagulation factors**



Colonic mucosal hemorrhage

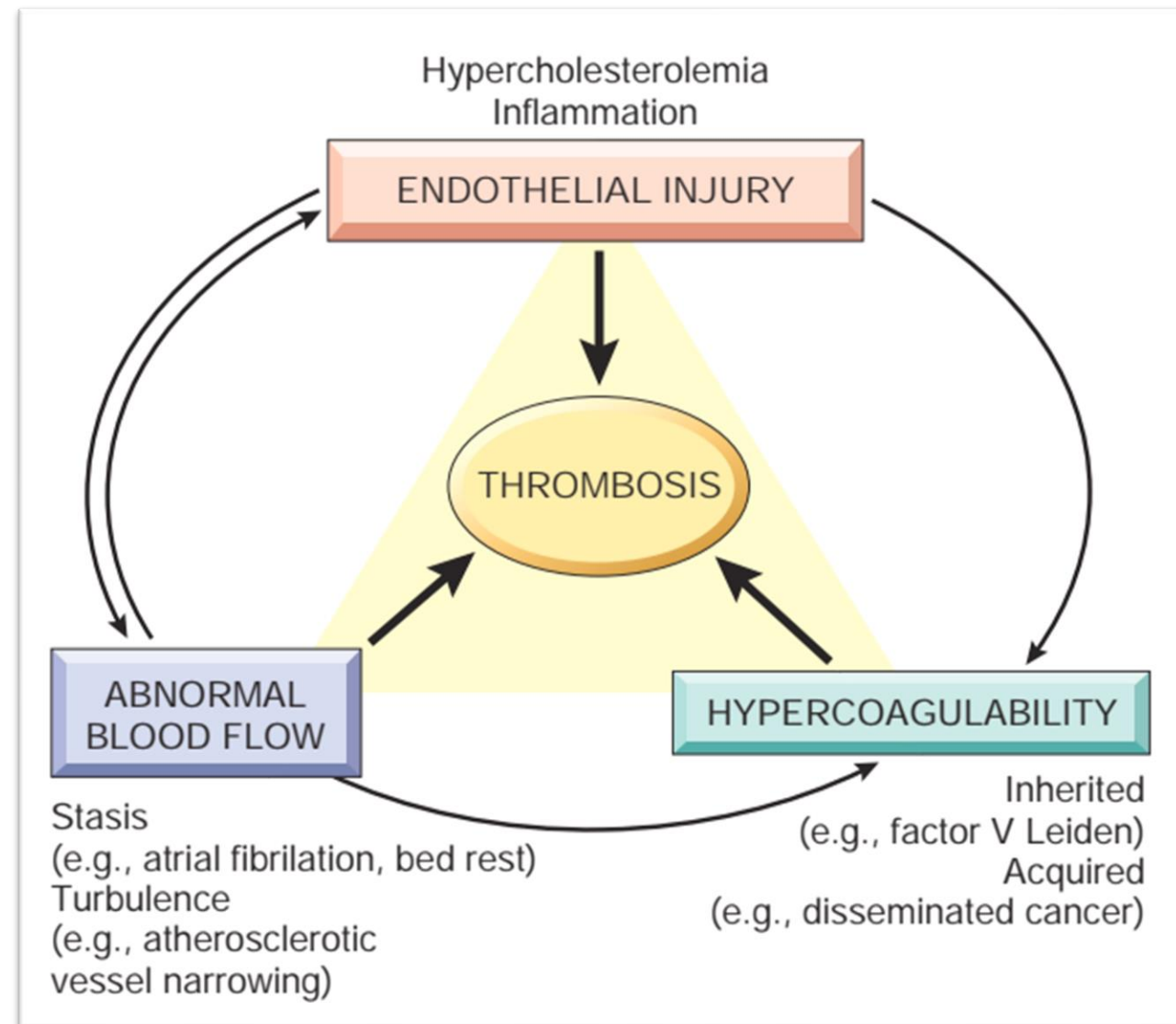
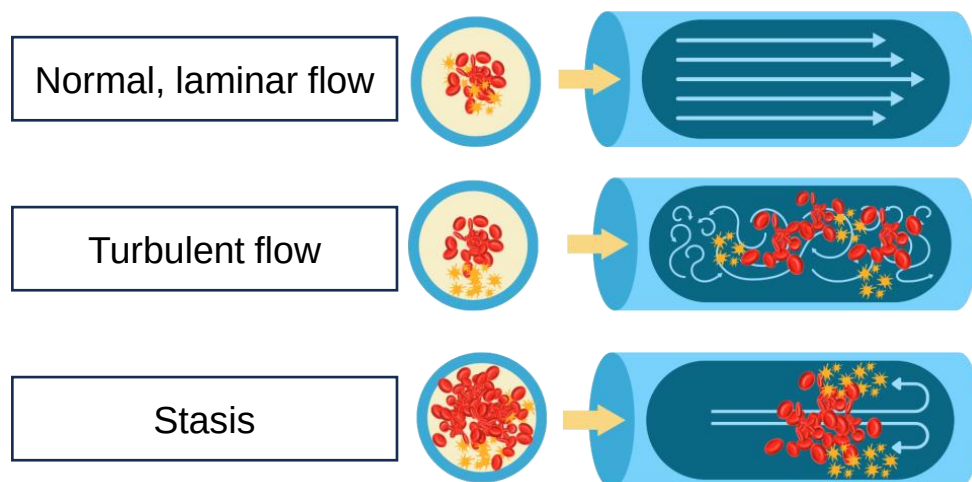


Intracerebral hemorrhage

Thrombosis (Clotting)

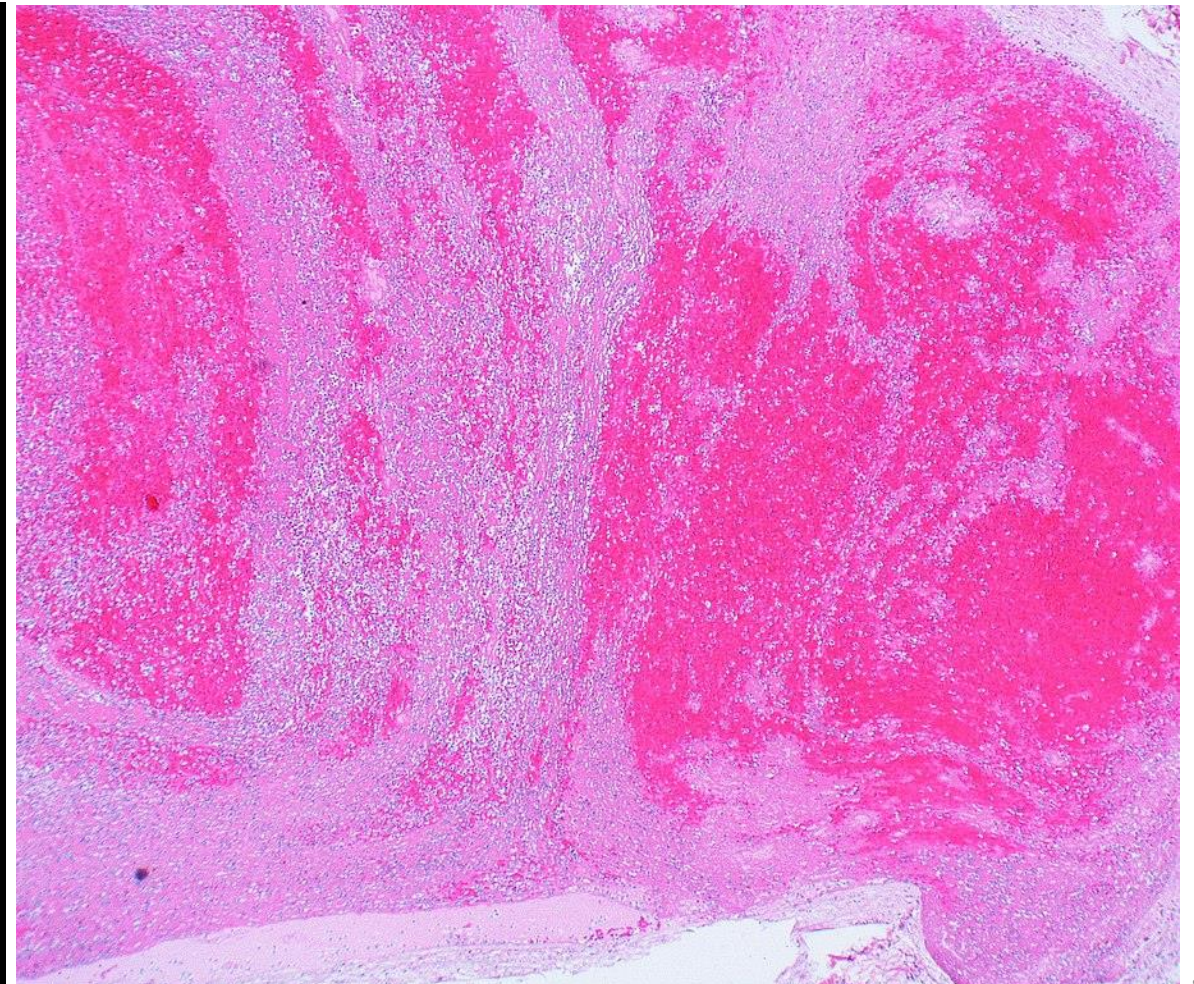
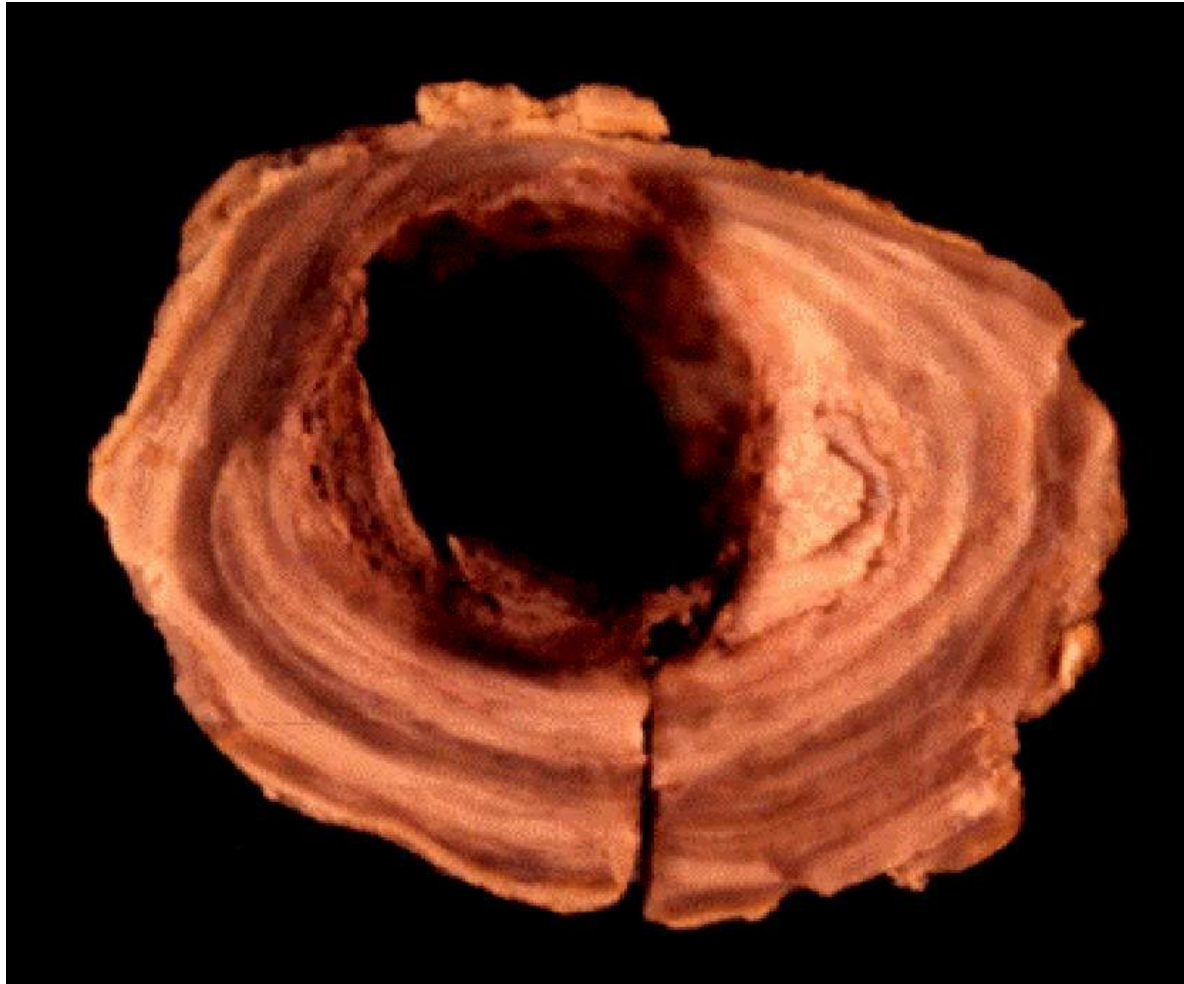
- Formation of thrombus(clot) by dysregulation of normal hemostasis
- The primary abnormalities that lead to thrombosis are

- 1) endothelial injury
- 2) stasis or turbulent blood flow
- 3) hypercoagulability of the blood



Thrombosis

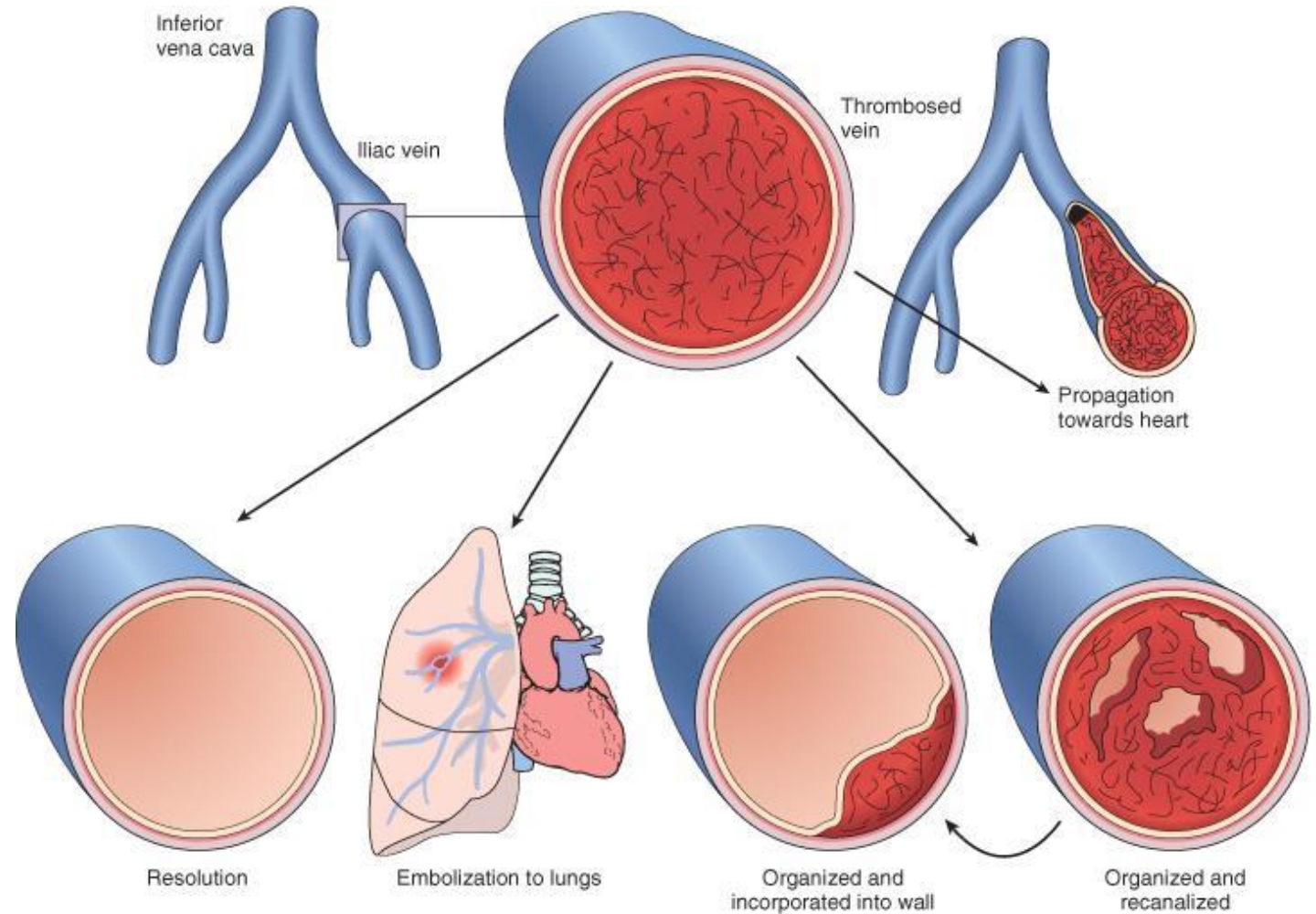
- **Lines of Zahn** → **Pale** (platelet and fibrin) and **Dark** (red blood cell)



Thrombosis

Fate of the Thrombus

- **Propagation:** accumulate additional platelets and fibrin
- **Embolization:** dislodge and travel to other sites in the vasculature
- **Dissolution:** rapid shrinkage and total disappearance
- **Organization and recanalization:** ingrowth of endothelial cells, smooth muscle cells, and fibroblasts → reestablish the continuity of the original lumen



Disseminated intravascular coagulation (DIC)

- DIC is **widespread thrombosis** within the microcirculation → **organ ischemia**
- It is not a specific disease but rather a complication of a large number of conditions associated with systemic activation of thrombin.
- Systemic activation of the coagulation leads to consumption of coagulation factors and platelets
→ **Widespread hemorrhage**

Condition	Examples
Severe infectious diseases	Gram-positive or -negative organisms, malaria, hemorrhagic fevers
Malignancy	Solid tumors (eg, adenocarcinomas) Acute promyelocytic leukemia or monocytic leukemia
Trauma	Multitrauma Brain injury Burns
Obstetrical complications	Abruptio placentae Amniotic fluid embolism
Vascular malformations	Kasabach-Merrit syndrome Giant hemangiomas Other vascular malformations Large aortic aneurysms
Severe immunologic reactions	Transfusion reaction
Heat stroke	
Post-cardiopulmonary resuscitation	

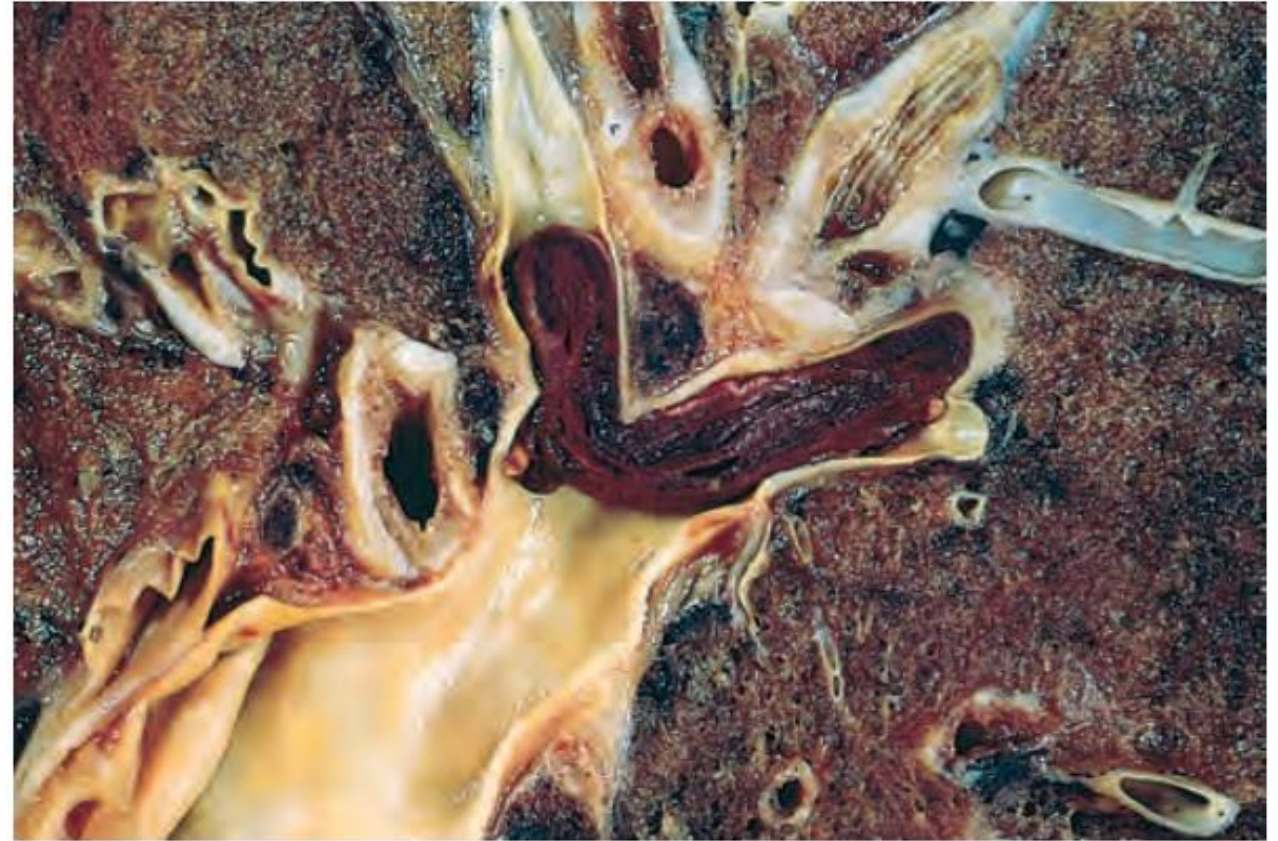
Embolism

- A detached **intravascular solid, liquid, or gaseous mass** that is carried by the blood from its point of origin to a distant site → tissue **dysfunction or infarction**
- **Examples:**
 - Thromboembolism
 - Amniotic fluid embolism
 - Air embolism
 - Fat embolism
 - Tumor embolism
 - Foreign body embolism

Pulmonary embolism (PE)

- The most common form of thromboembolic disease >> Originate from DVT
- Incidence: 100-200 cases per 100,000 people in the United States
- Serious disorder: causes about 100,000 deaths per year in the United States
- Most pulmonary emboli (60% to 80%) are clinically silent because they are small
- Sudden death, acute right heart failure or cardiovascular collapse

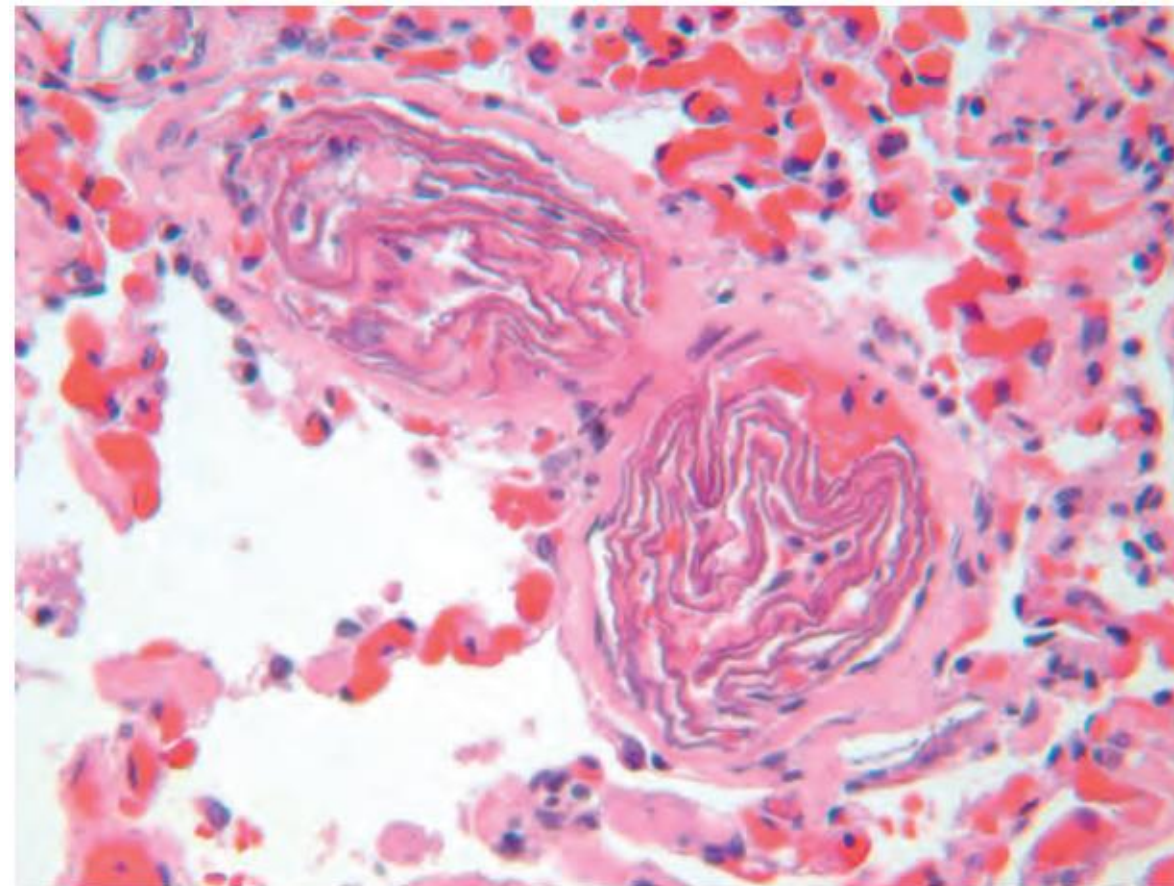
Rarely, a venous embolus passes through an interatrial or interventricular defect and gains access to the systemic arterial circulation (paradoxical embolism)



straddle the pulmonary artery bifurcation
(saddle embolus)

Amniotic fluid embolism

- The 5th most common cause of maternal mortality worldwide
- Complication of labor and the immediate postpartum period
- Incidence: approximately 2 to 6 in 100,000 deliveries
- Mortality rate: up to 80%
- The onset is characterized by **sudden** severe dyspnea, cyanosis, and shock, followed by neurologic impairment ranging from headache to seizures and coma, and by DIC

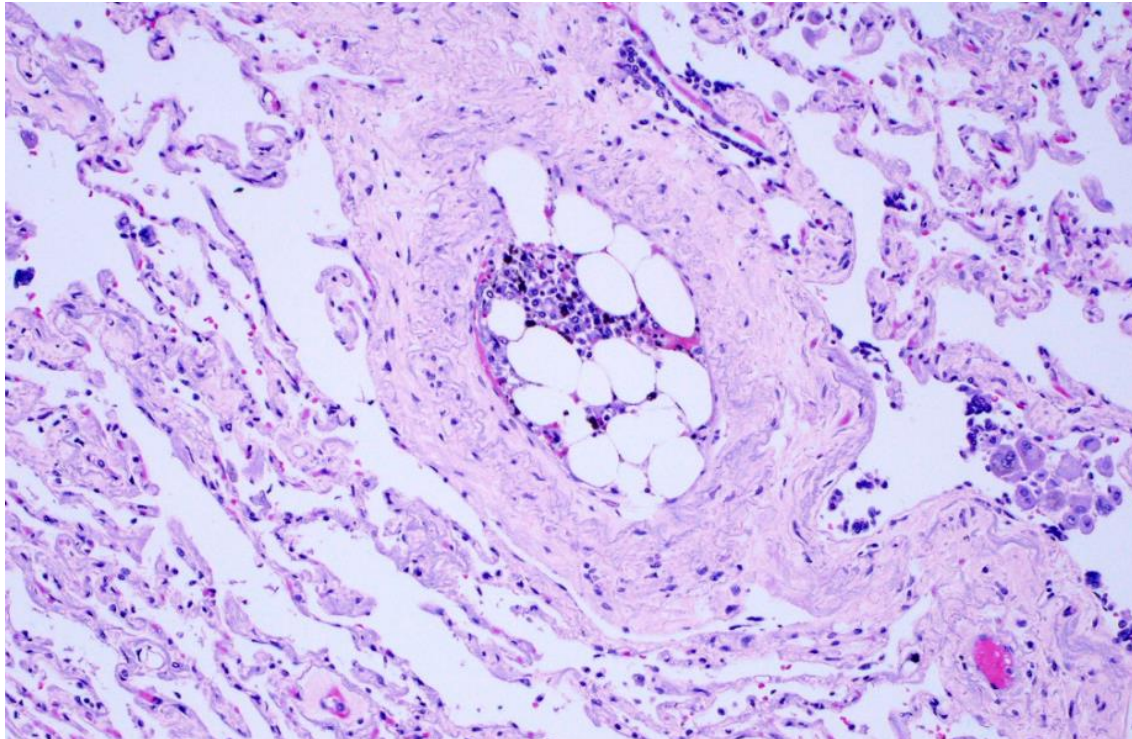


Presence of squamous cells shed from fetal skin, lanugo hair, fat from vernix caseosa, and mucin derived from the fetal respiratory or gastrointestinal tract **in the maternal pulmonary microvasculature**

Embolism

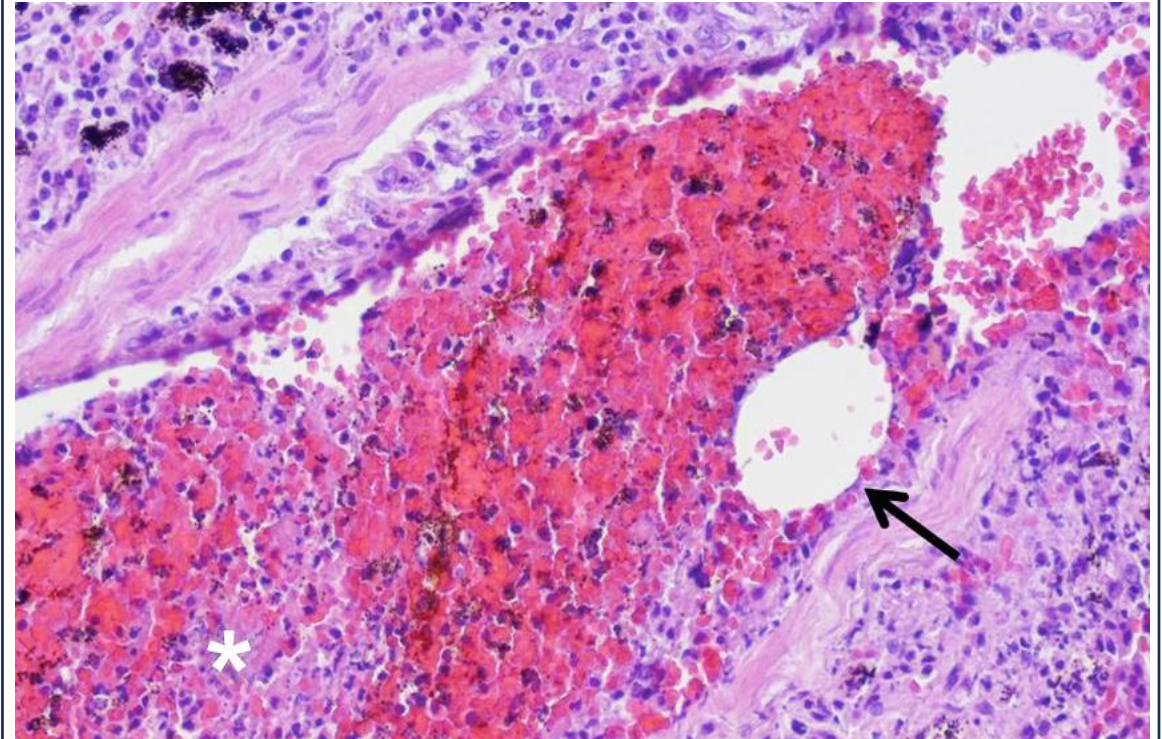
Fat embolism

- Fat globules or hematopoietic bone marrow
- After fractures of long bones or soft-tissue trauma and burns

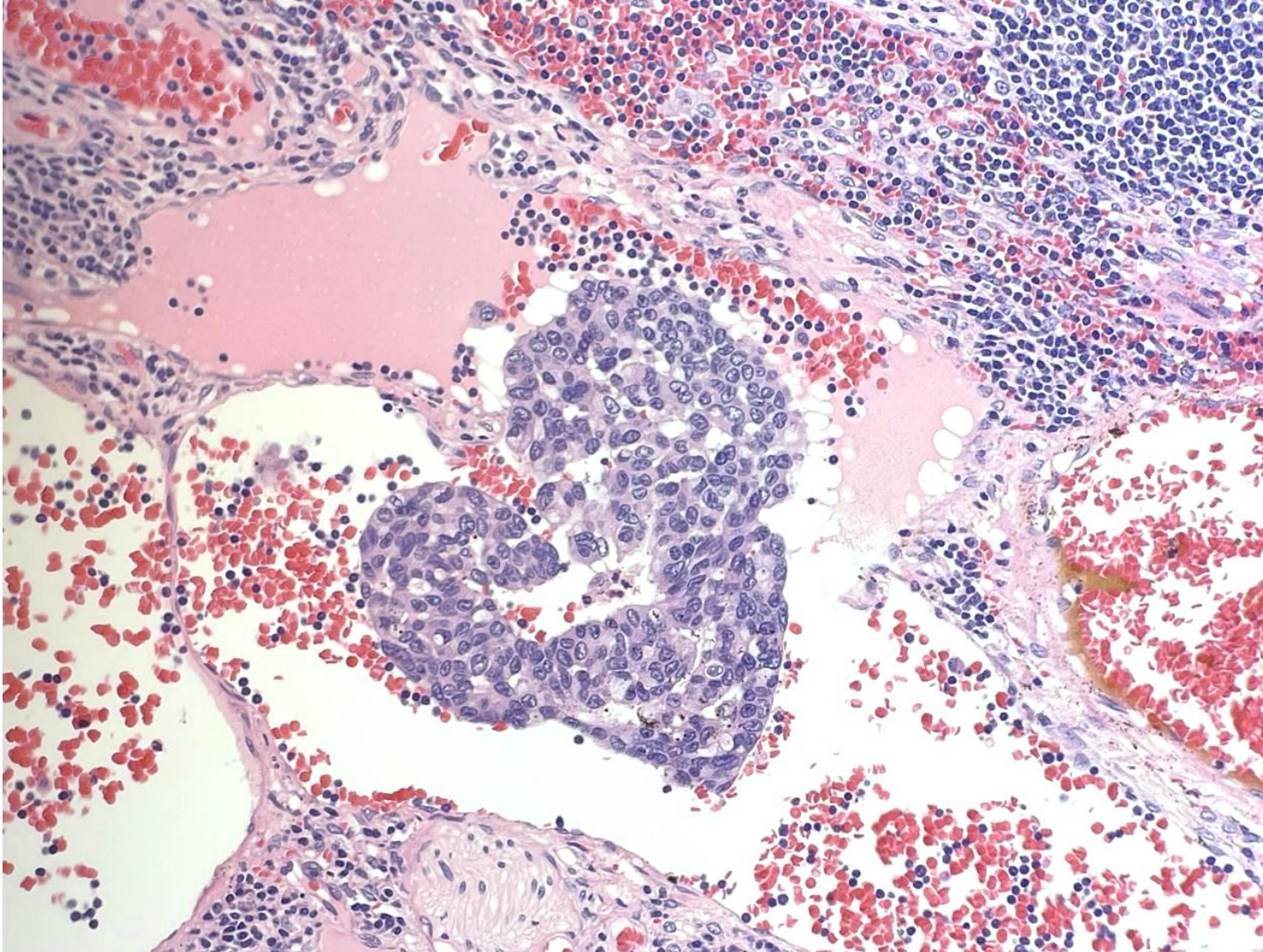


Air embolism

- Communication between the vasculature and outside air



QUIZ



Embolism ที่เห็นนี้
มีชื่อว่า???

.....

Infarction

An area of **ischemic necrosis** most commonly caused by **arterial occlusion** (typically due to thrombosis or embolization); **venous outflow obstruction** is a less frequent cause.

Red infarct

- 1) Venous occlusions
- 2) Loose or spongy organs (e.g., lung)
- 3) Organs with dual circulations (e.g., lung and small intestine)
- 4) Reestablished flow

White infarct

- 1) Arterial occlusions
- 2) Solid organs
- 3) Organs with end-arterial circulation (e.g., heart, spleen, and kidney)



(A) Hemorrhagic, roughly wedge-shaped pulmonary red infarct.
(B) Sharply demarcated white infarct in the spleen.

Infarction

Factors That Influence Development of an Infarct

1) Anatomy of the vascular supply

- Dual blood supply organ (lung, liver, hand, forearm) → relatively resistant to infarction

2) Rate of occlusion

- Slowly developing occlusions → provide time for development of collateral pathways of perfusion → are less likely to cause infarction

3) Tissue vulnerability to hypoxia

- Neurons → 3 to 4 minutes
- Myocardial cells → 20 to 30 minutes

Shock

A state of **circulatory failure** that **impairs tissue perfusion** and leads to cellular hypoxia.

	Hypovolemic	Distributive	Cardiogenic	Obstructive
History	Bleeding Severe dehydration	Sepsis Anaphylaxis* Spinal cord injury*	Acute MI Arrhythmia	Massive PE Pneumothorax Tamponade
Pathophysiology	Volume loss	Vasodilation	Pumping failure	Flow obstruction
Cardiac output	Decreased	Increased	Decreased	Decreased
SVR	Increased	Decreased	Increased	Increased
Extremities	Cold	Warm	Cold	Cold

*Less commonly, shock can occur in the setting of

- Spinal cord injury (neurogenic shock)
- IgE-mediated hypersensitivity reaction (anaphylactic shock)

Shock: stage of shock *(Hypovolemic and cardiogenic)*

Initial nonprogressive stage

- Compensatory mechanisms:
 - Catecholamines
 - Antidiuretic hormone
 - Renin-angiotensin-aldosterone axis
 - Sympathetic stimulation
- Symptoms:
 - Tachycardia
 - Anuria
 - Cold skin and pallor

Progressive stage

Tissue hypoperfusion
and hypoxia



Anaerobic metabolism



Lactic acidosis



Peripheral vasodilatation



Decreased venous return



Decreased cardiac output

Irreversible stage

- Widespread cell injury
- Multiple organ failure
- Expire

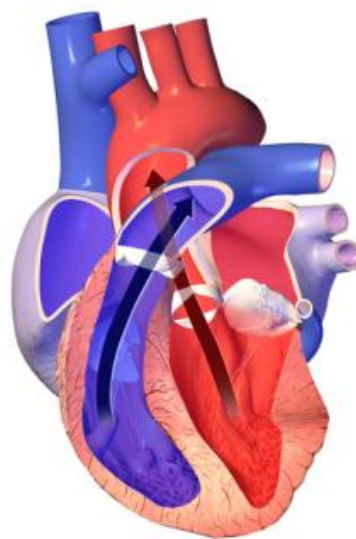
HEART DISEASES

Congestive heart failure (CHF)

- CHF is defined as the condition in which a **heart cannot pump blood to adequately meet the metabolic demands of peripheral tissues**
- Common end stage of many forms of chronic heart disease
- Acute hemodynamic stresses can all precipitate sudden CHF

Systolic dysfunction

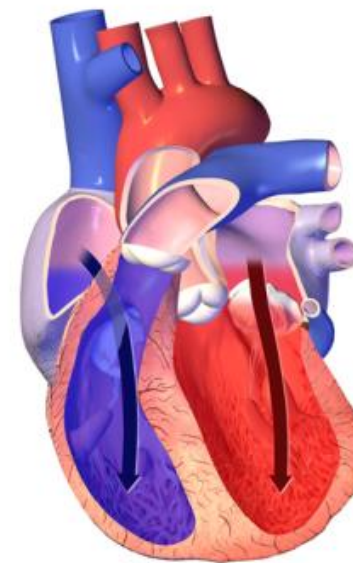
- Inadequate myocardial contraction
- Decreased ejection fraction (EF)



Systole
(pumping)

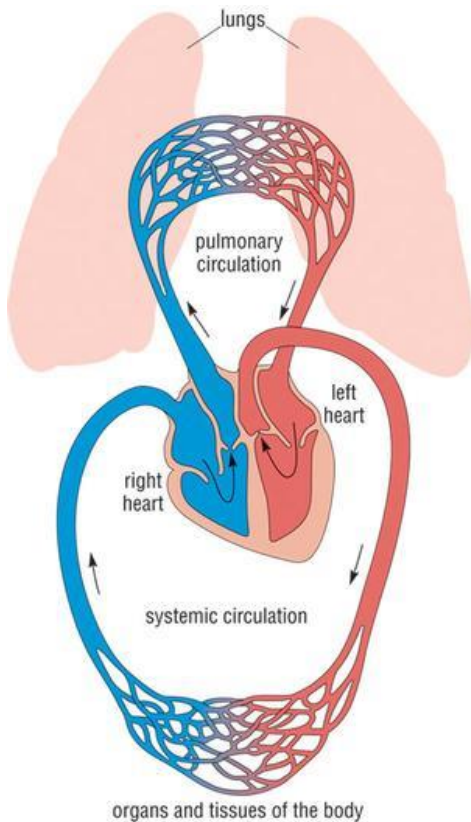
Diastolic dysfunction

- Inability of the heart to adequately relax and fill

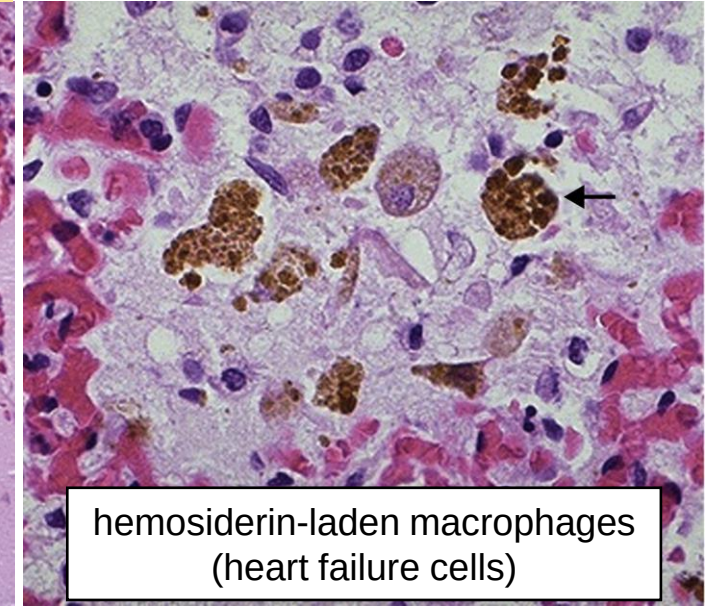
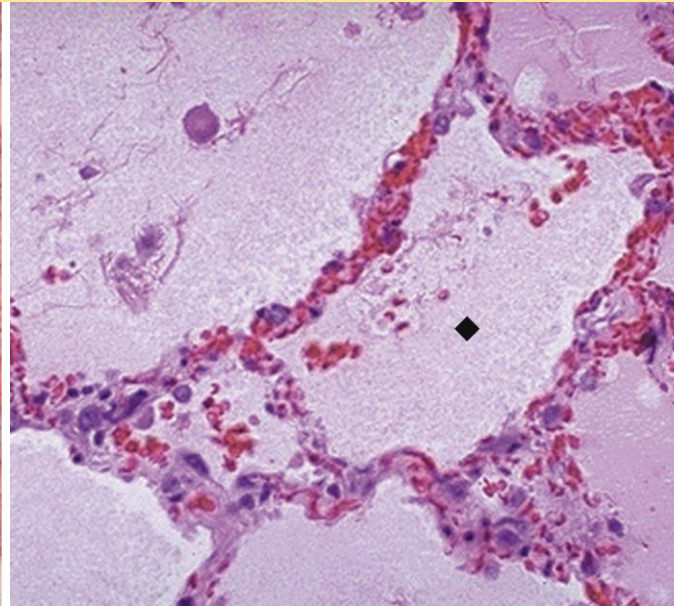


Diastole
(filling)

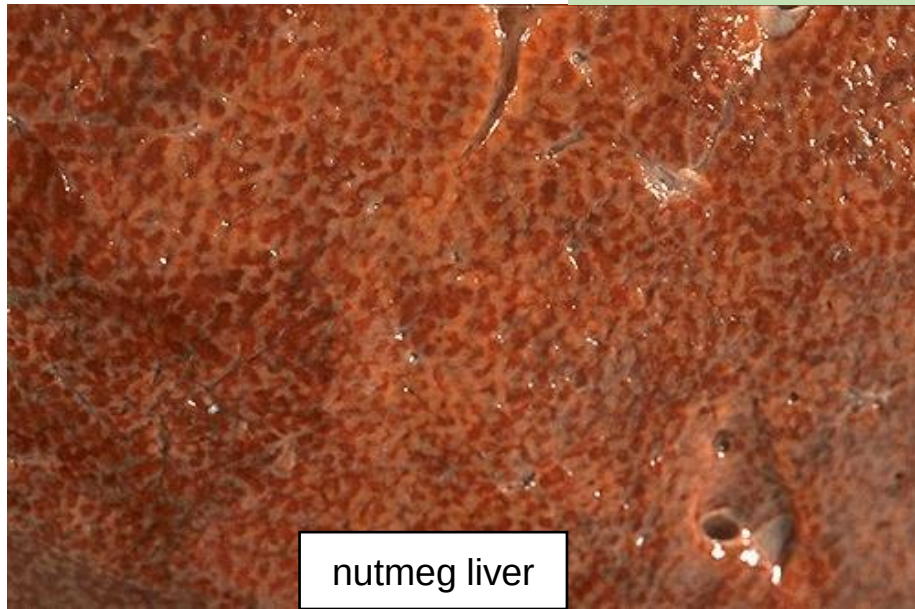
Congestive heart failure (CHF)

	Right-sided CHF		Left-sided CHF
Cause	- Secondary to left-sided CHF - Primary pulmonary hypertension, or recurrent pulmonary thromboembolism (Cor pulmonale)		- Ischemic heart disease - Hypertension - Aortic and mitral valvular diseases - Primary myocardial diseases
Pathology	- Congestion of liver, spleen, GI tract, kidneys, brain. - Effusion of pleural cavity, pericardial cavity, peritoneal cavity - Edema of subcutaneous tissue		<u>Lung</u> - Pulmonary edema and congestion <u>Kidney</u> - Decreased renal perfusion - Activation of RAAS system - Salt and water retention >> edema - Pre-renal azotemia <u>Brain</u> - Hypoxic encephalopathy
Clinical features	peripheral and dependent edema (Pretibial edema)		dyspnea, orthopnea, paroxysmal nocturnal dyspnea

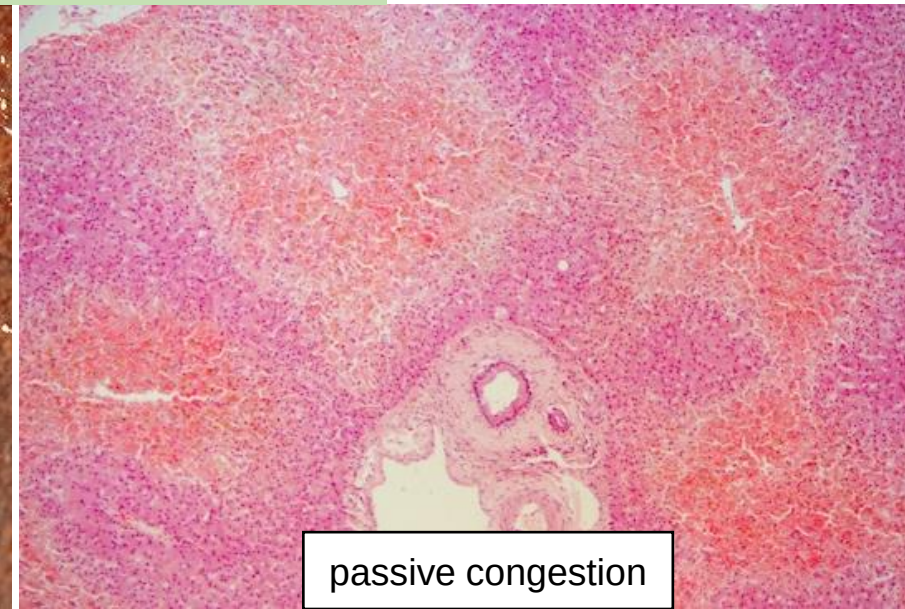
Pulmonary edema



Hepatic congestion



nutmeg liver

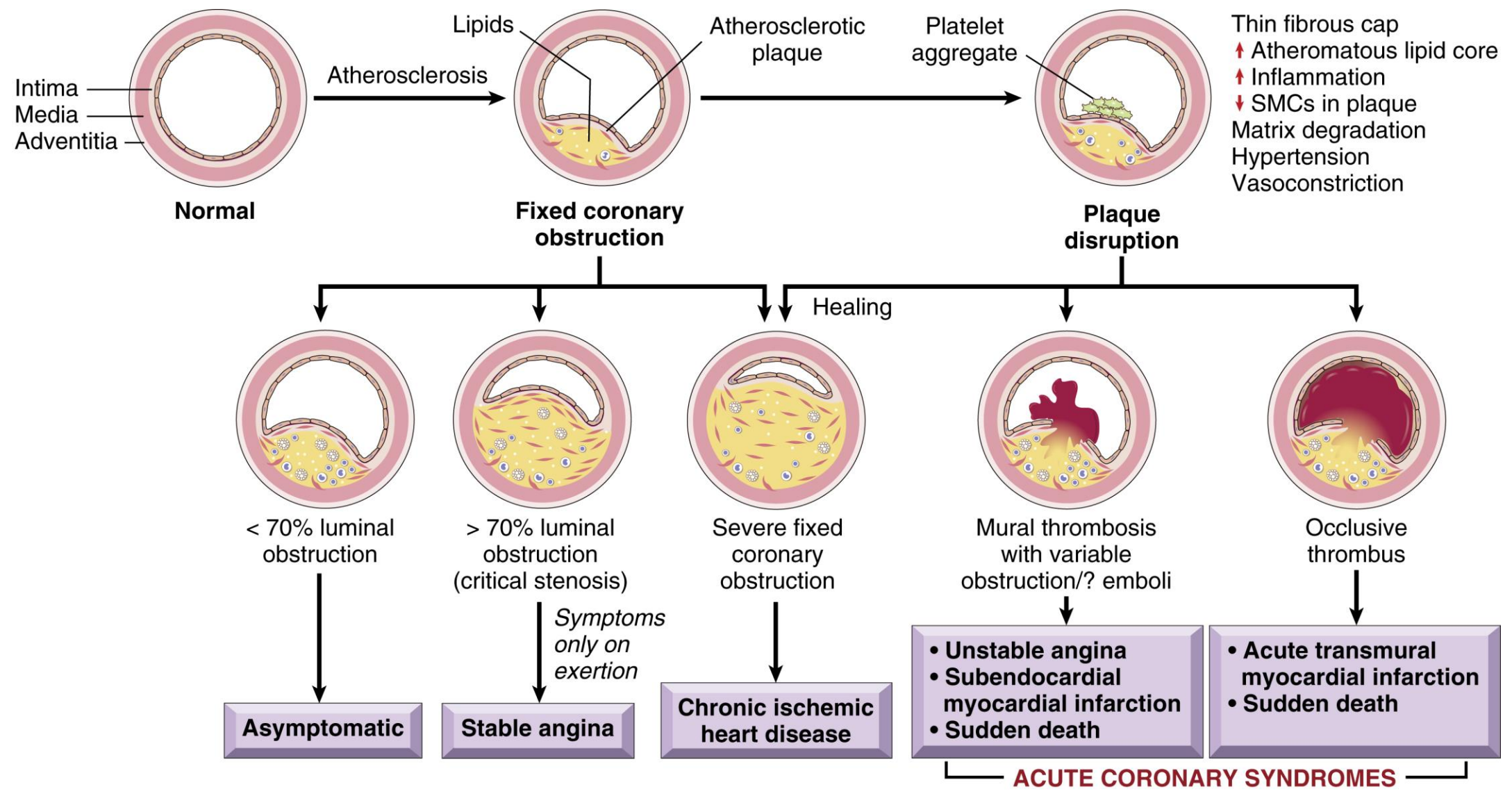


passive congestion

Ischemic heart disease (IHD)

- A group of related entities resulting from **myocardial ischemia**
 - More than 90% of IHD, results from **atherosclerosis** in coronary arteries (coronary artery disease, CAD)
 - Clinical manifestations of IHD:
 - Angina pectoris (intermittent chest pain)
 - Myocardial infarction (MI)
 - Chronic IHD with heart failure
 - Sudden cardiac death
- an **imbalance** between myocardial **supply** (perfusion) and cardiac **demand** for oxygenated blood
- Stable angina – associated with physical activity, emotional excitement, or psychological stress
 - Prinzmetal/variant angina – occurs at rest, caused by coronary artery spasm
 - Unstable angina - progressively lower levels of physical activity or even occurring at rest

Sequential progression of coronary artery lesions



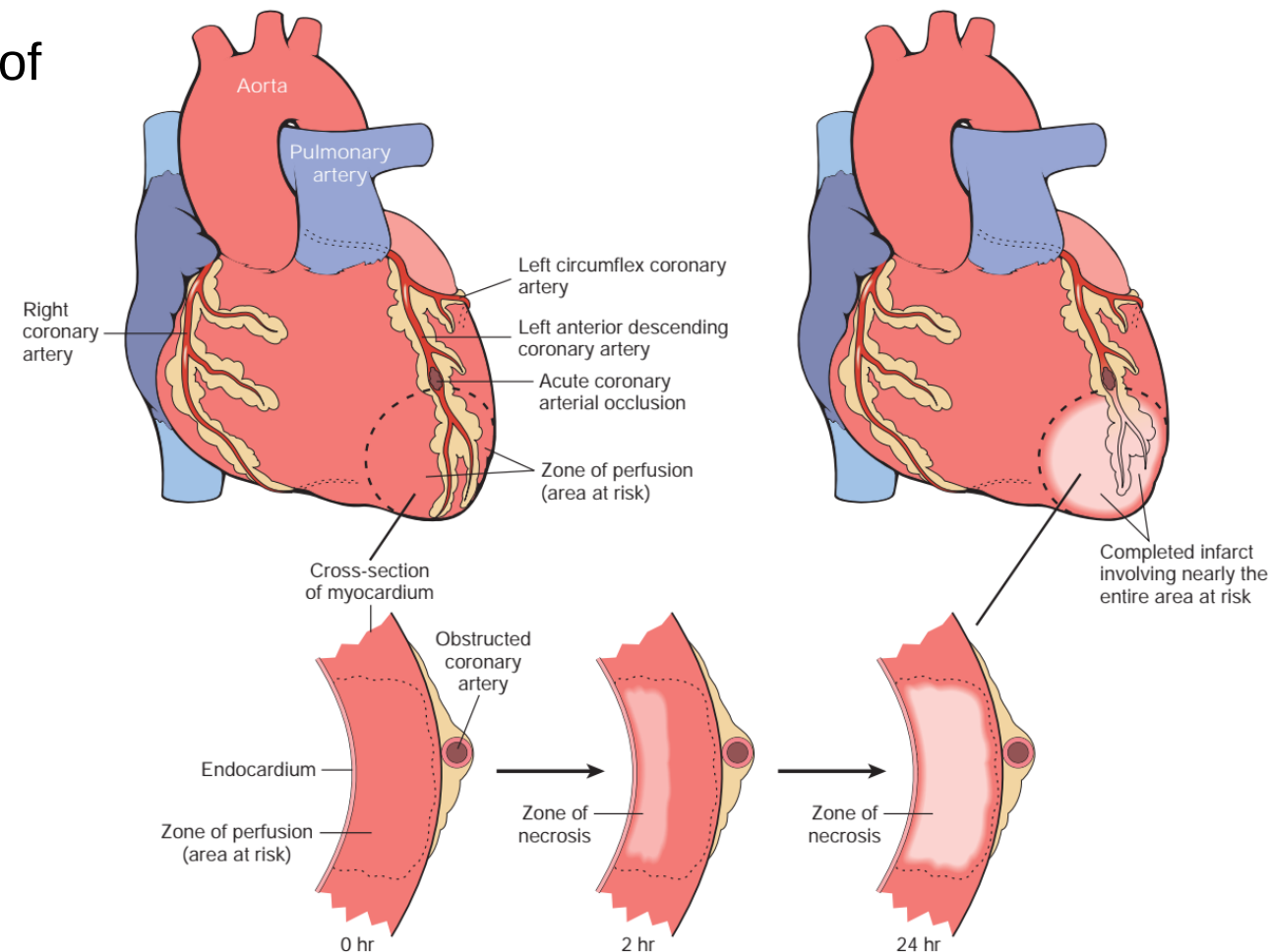
Myocardial infarction (MI)

- **Necrosis of myocytes** resulting from ischemia
- Caused by **acute thrombotic obstruction** of a coronary artery due to rupture of an atherosclerotic plaque

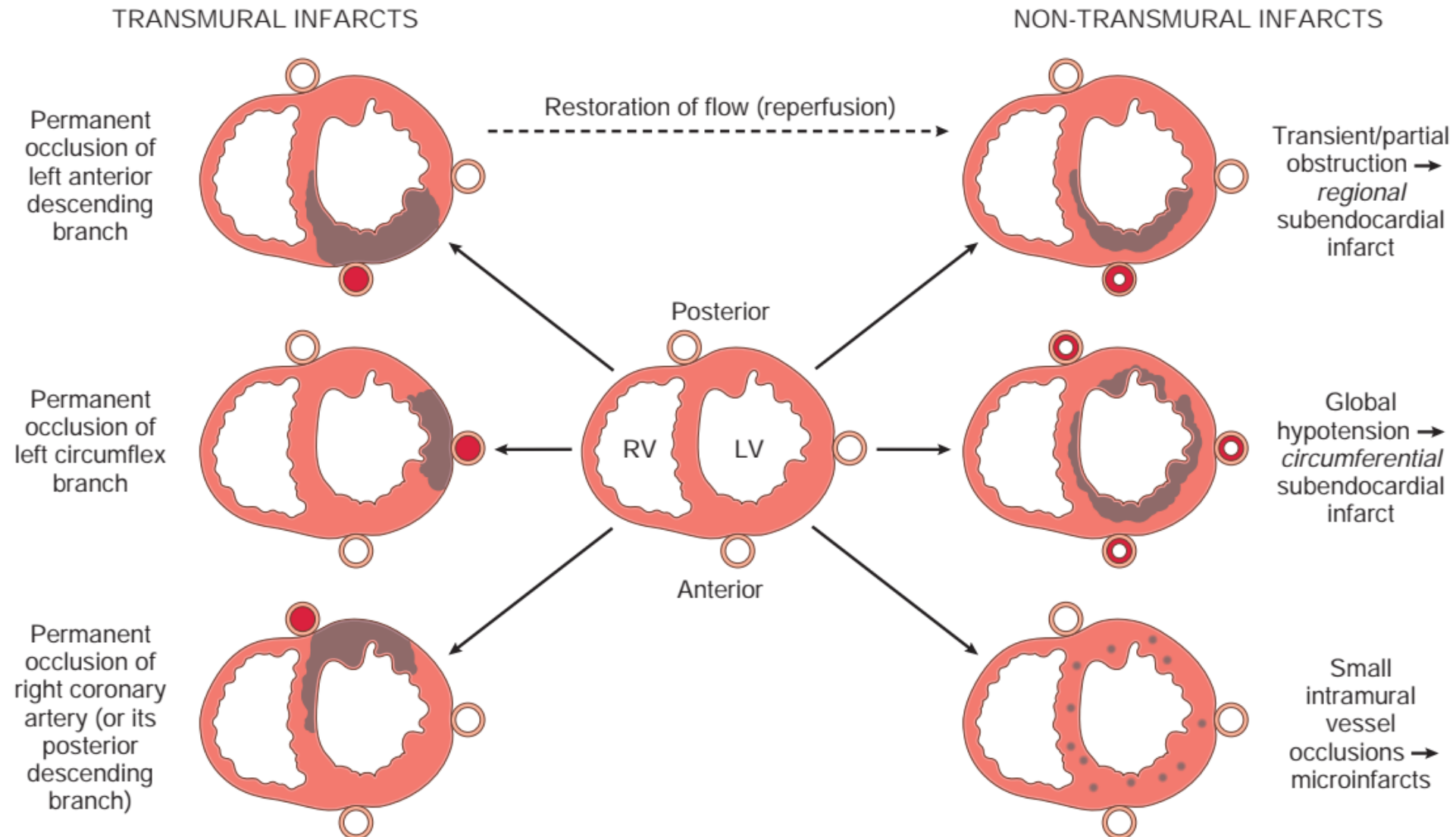
Table 12.4 Approximate Time of Onset of Key Events in Ischemic Cardiac Myocytes

Feature	Time
Onset of ATP depletion	Seconds
Loss of contractility	<2 minutes
ATP reduced to 50% of normal	10 minutes
ATP reduced to 10% of normal	40 minutes
★ Irreversible cell injury	20–40 minutes
Microvascular injury	>1 hour

ATP, Adenosine triphosphate.



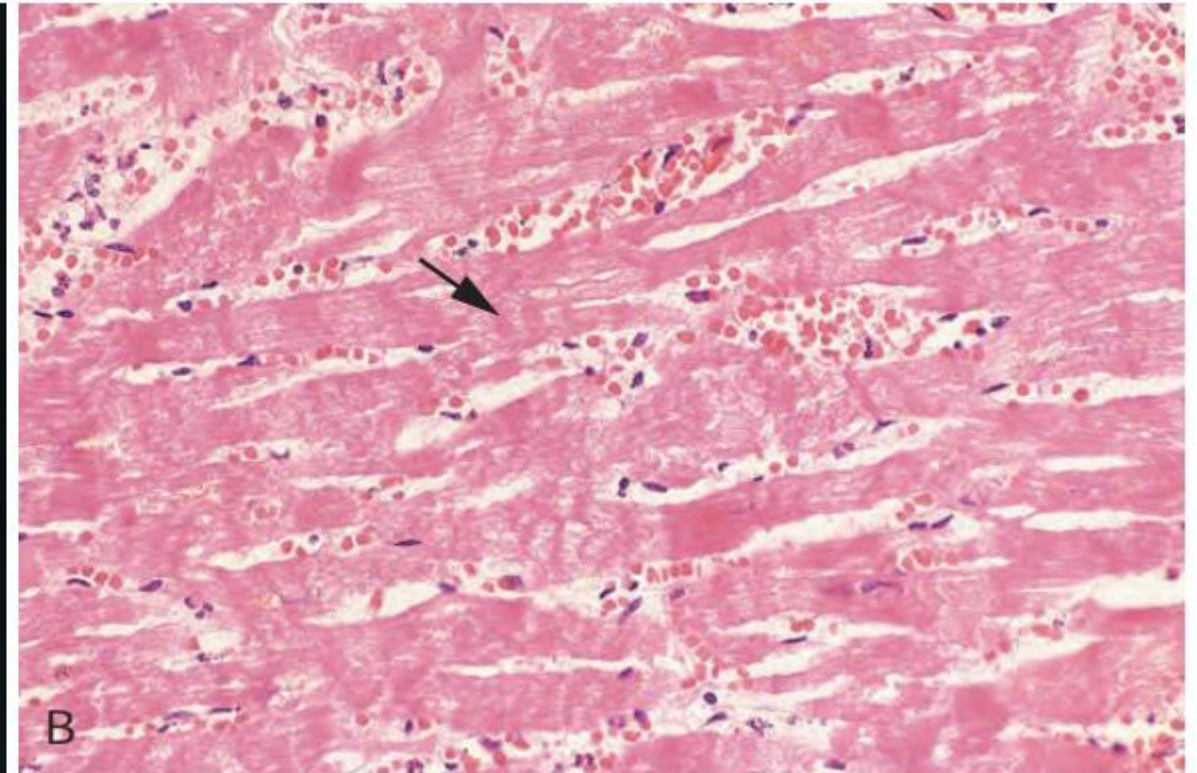
Patterns of Infarction



Myocardial infarction (MI)



Large, densely hemorrhagic, anterior wall acute myocardial infarction



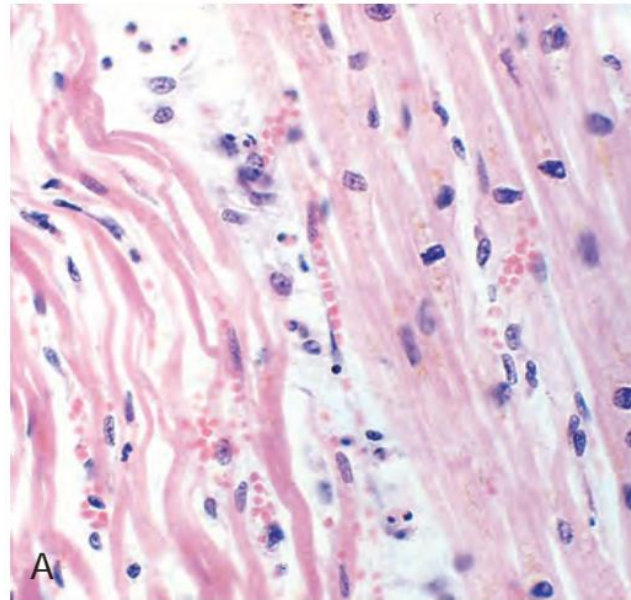
Myocardial necrosis with hemorrhage and contraction bands, visible as dark bands spanning some myofibers (arrow)

Evolution of Morphologic Changes in Myocardial Infarction

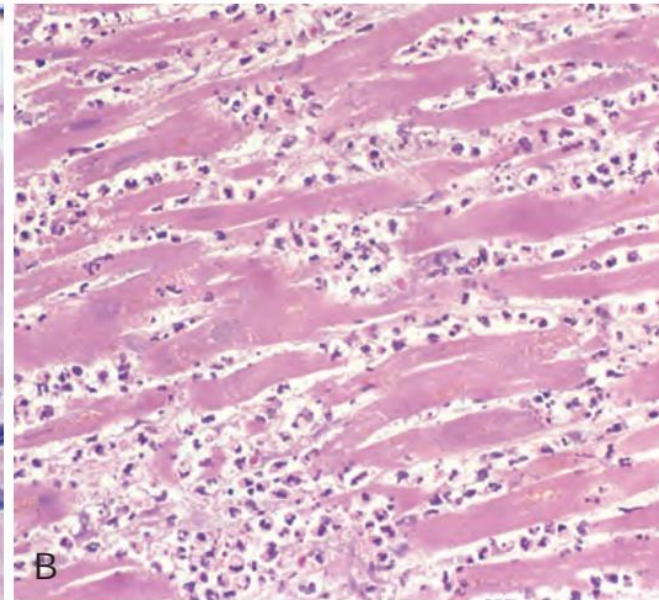
Time	Gross Features	Light Microscope	Electron Microscope
Reversible Injury			
0–½ hour	None	None	Relaxation of myofibrils; glycogen loss; mitochondrial swelling
Irreversible Injury			
★ ½–4 hours	None	Usually none; variable waviness of fibers at border	Sarcolemmal disruption; mitochondrial amorphous densities
4–12 hours	Dark mottling (occasional)	Early coagulative necrosis; edema; hemorrhage	
12–24 hours	Dark mottling	Ongoing coagulative necrosis; pyknosis of nuclei; myocyte hypereosinophilia; marginal contraction band necrosis; early neutrophilic infiltrate	
★ 1–3 days	Mottling with yellow-tan infarct center	Coagulative necrosis, with loss of nuclei and striations; brisk interstitial infiltrate of neutrophils	
3–7 days	Hyperemic border; central yellow-tan softening	Beginning disintegration of dead myofibers, with dying neutrophils; early phagocytosis of dead cells by macrophages at infarct border	
7–10 days	Maximally yellow-tan and soft, with depressed red-tan margins	Well-developed phagocytosis of dead cells; granulation tissue at margins	
★ 10–14 days	Red-gray depressed infarct borders	Well-established granulation tissue with new blood vessels and collagen deposition	
2–8 weeks	Gray-white scar, progressive from border toward core of infarct	Increased collagen deposition, with decreased cellularity	
>2 months	Scarring complete	Dense collagenous scar	

Evolution of Morphologic Changes in Myocardial Infarction

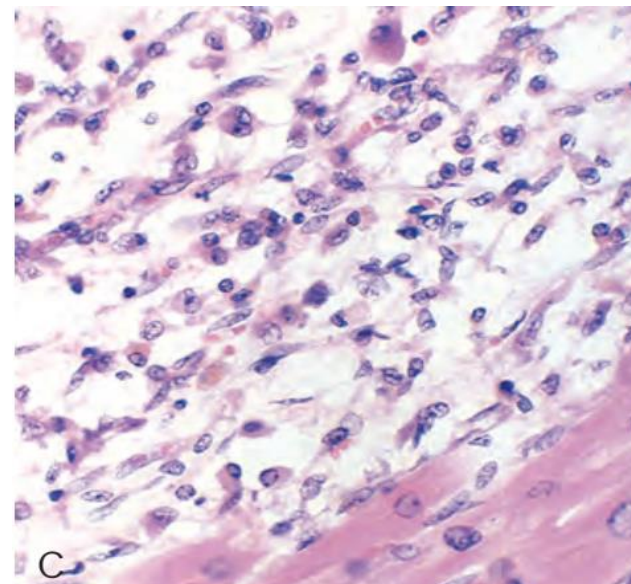
12 hrs. - 1 day
coagulative necrosis
and wavy fibers



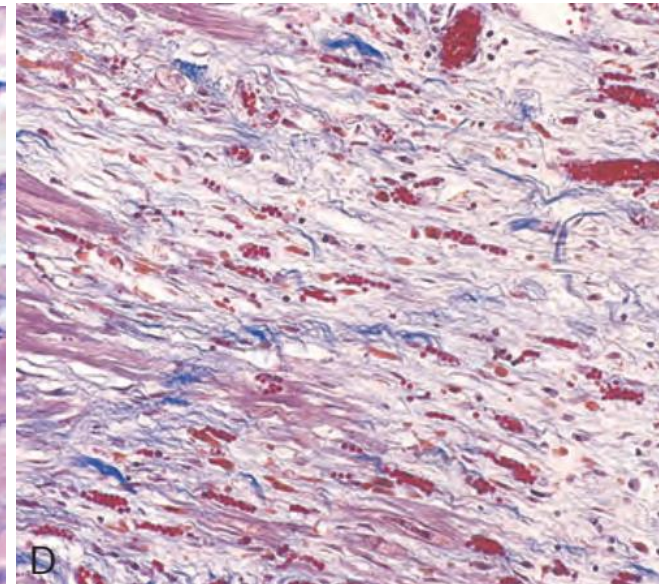
3 - 4 days
neutrophils infiltrate



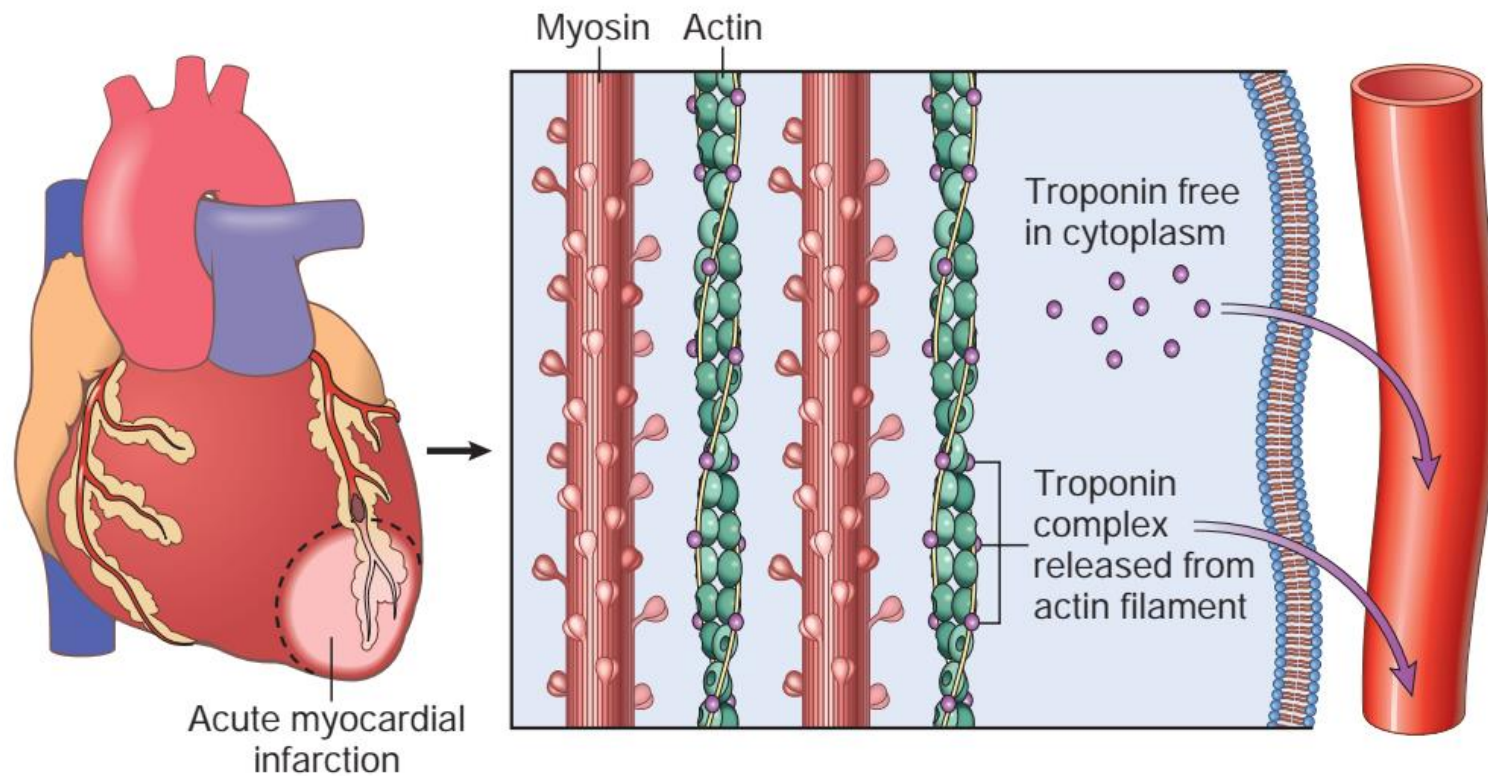
7 - 10 days
removal of necrotic
myocytes by
phagocytosis



2 weeks
increased collagen
deposition



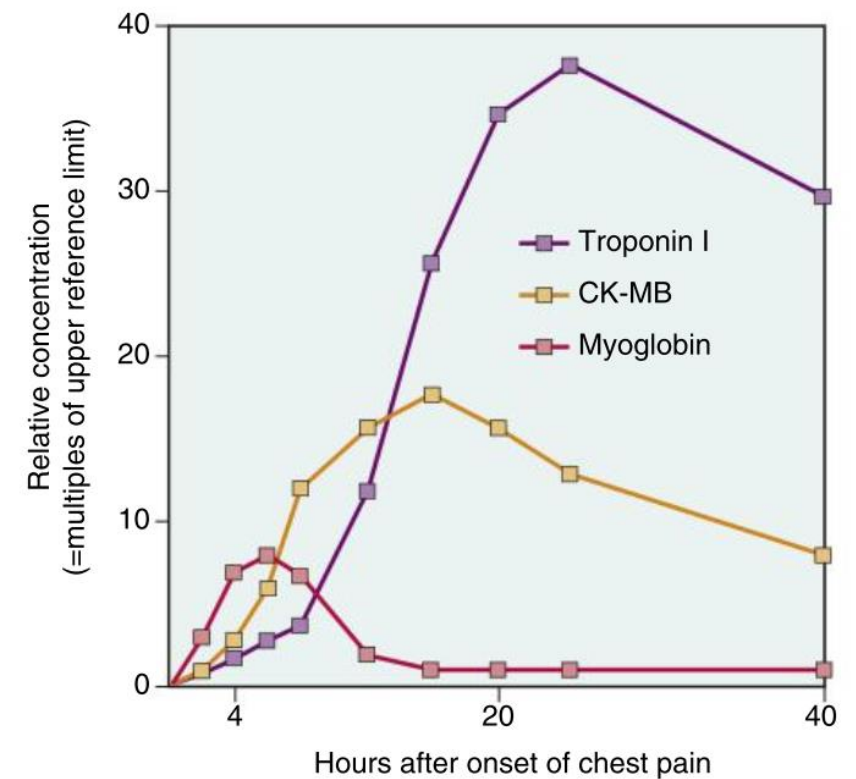
Cardiac biomarker for diagnosis of myocardial infarction



1. Onset of myocardial infarction

2. Plasma membrane of necrotic myocytes becomes leaky

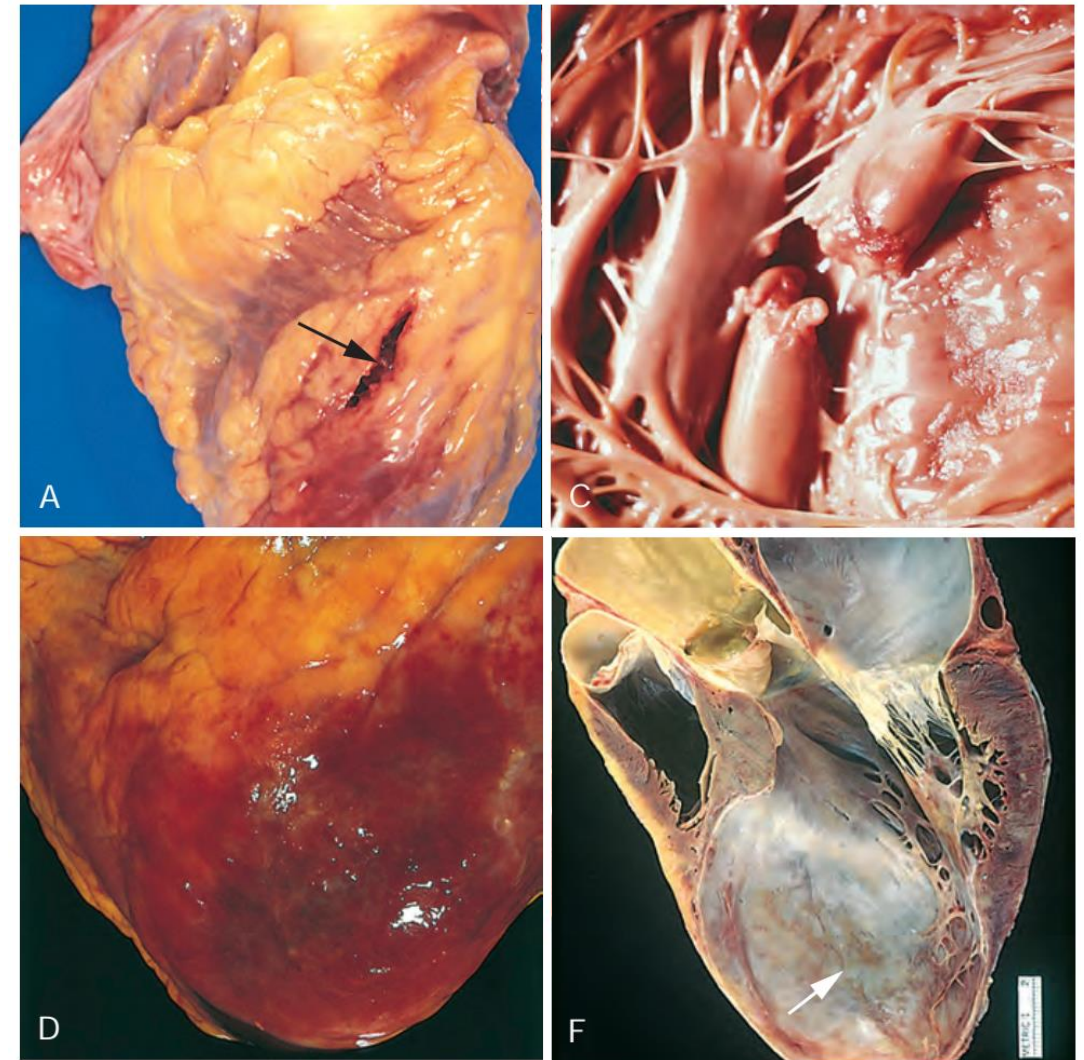
3. Troponin leaks out of cell into circulation



4. Biomarker for diagnosis of myocardial infarction

Consequences and Complications of Myocardial Infarction

- Arrhythmias - cause sudden death, ventricular fibrillation (serious)
- Contractile dysfunction - hypotension, pulmonary congestion, and pulmonary edema
- Myocardial rupture - 1% to 5%, but is frequently fatal
- Papillary muscle dysfunction
- Pericarditis
- Chamber dilation
- Mural thrombus
- Progressive heart failure

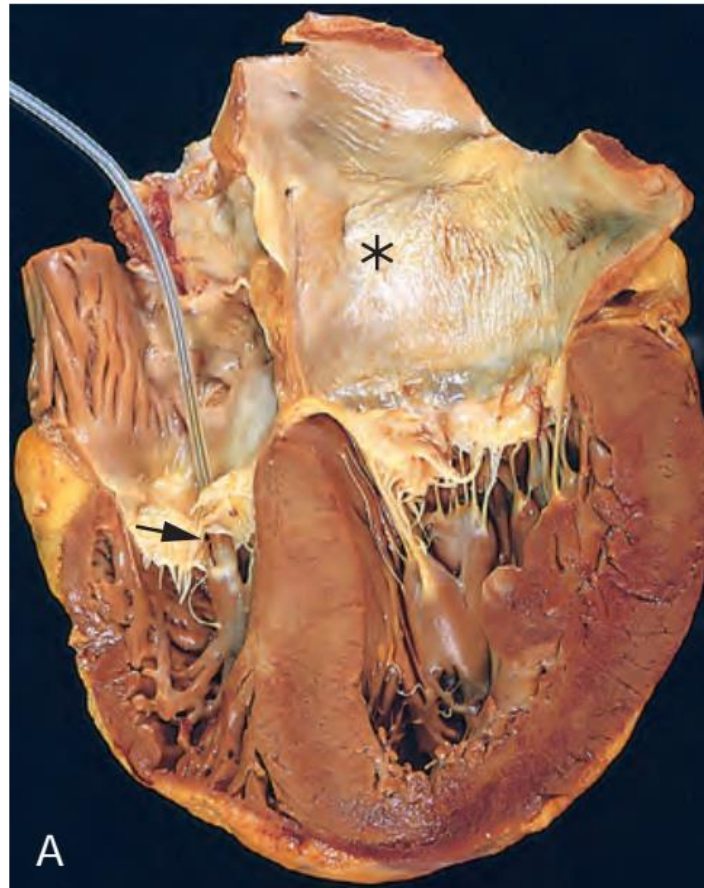


(A) Myocardial rupture (C) Rupture of papillary muscle
(D) Fibrinous pericarditis (F) Large apical left ventricular aneurysm

Hypertensive heart disease (HHD)

- A consequence of the increased demands placed on the heart by **hypertension**, causing pressure overload and ventricular hypertrophy

Systemic
(Left-Sided)
HHD



Pulmonary
(Right-Sided)
HDD



Valvular heart disease

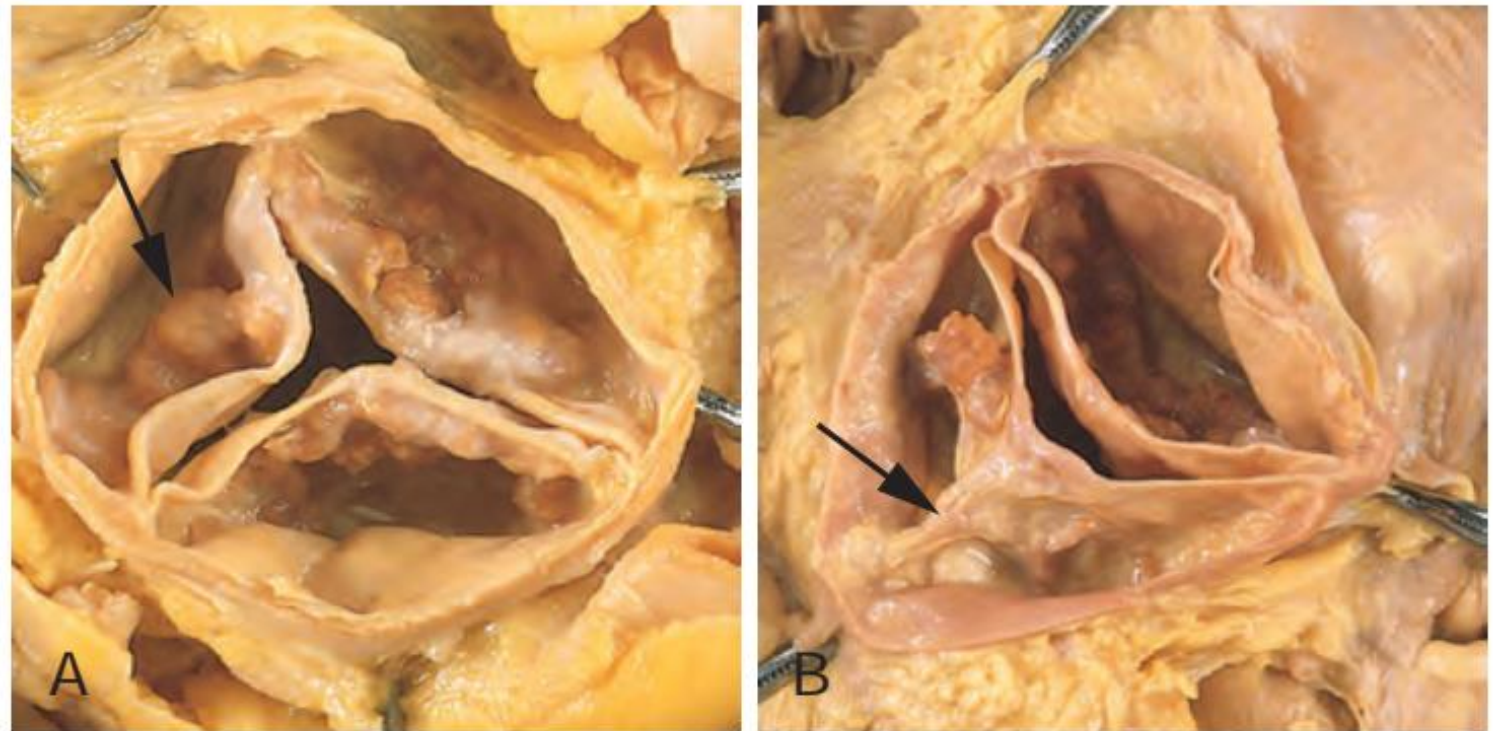
- **Stenosis** - failure of a valve to **open** completely
- **Insufficiency (regurgitation/incompetence)** - failure of a valve to **close** completely
- Abnormal flow produces abnormal heart sounds called **murmurs**, severe lesions can even be externally palpated as **thrills**

1. Calcific aortic stenosis
2. Mitral valve prolapse (Myxomatous degeneration)
3. Rheumatic heart disease
4. Infective endocarditis

Mitral Valve Disease	Aortic Valve Disease
Mitral Stenosis	Aortic Stenosis
★ Postinflammatory scarring (rheumatic heart disease)	★ Postinflammatory scarring (rheumatic heart disease) ★ Senile calcific aortic stenosis Calcification of congenitally deformed valve
Mitral Regurgitation	Aortic Regurgitation
Abnormalities of leaflets and commissures Postinflammatory scarring ★ Infective endocarditis ★ Mitral valve prolapse “Fen-phen”–induced valvular fibrosis	Intrinsic valvular disease ★ Postinflammatory scarring (rheumatic heart disease) ★ Infective endocarditis Aortic disease Degenerative aortic dilation Syphilitic aortitis Ankylosing spondylitis Rheumatoid arthritis Marfan syndrome
Abnormalities of tensor apparatus Rupture of papillary muscle Papillary muscle dysfunction (fibrosis) Rupture of chordae tendineae Abnormalities of left ventricular cavity and/or annulus Left ventricular enlargement (myocarditis, dilated cardiomyopathy) Calcification of mitral ring	

Calcific aortic stenosis

- The most common cause of aortic stenosis
- **Incidence increased with age**, and most patients present at age of 70 years or greater
- Risk factors: bicuspid aortic valve, male sex, high LDL cholesterol, hypertension and smoking
- Clinical features: asymptomatic, left ventricular hypertrophy, ischemic heart disease, congestive heart failure

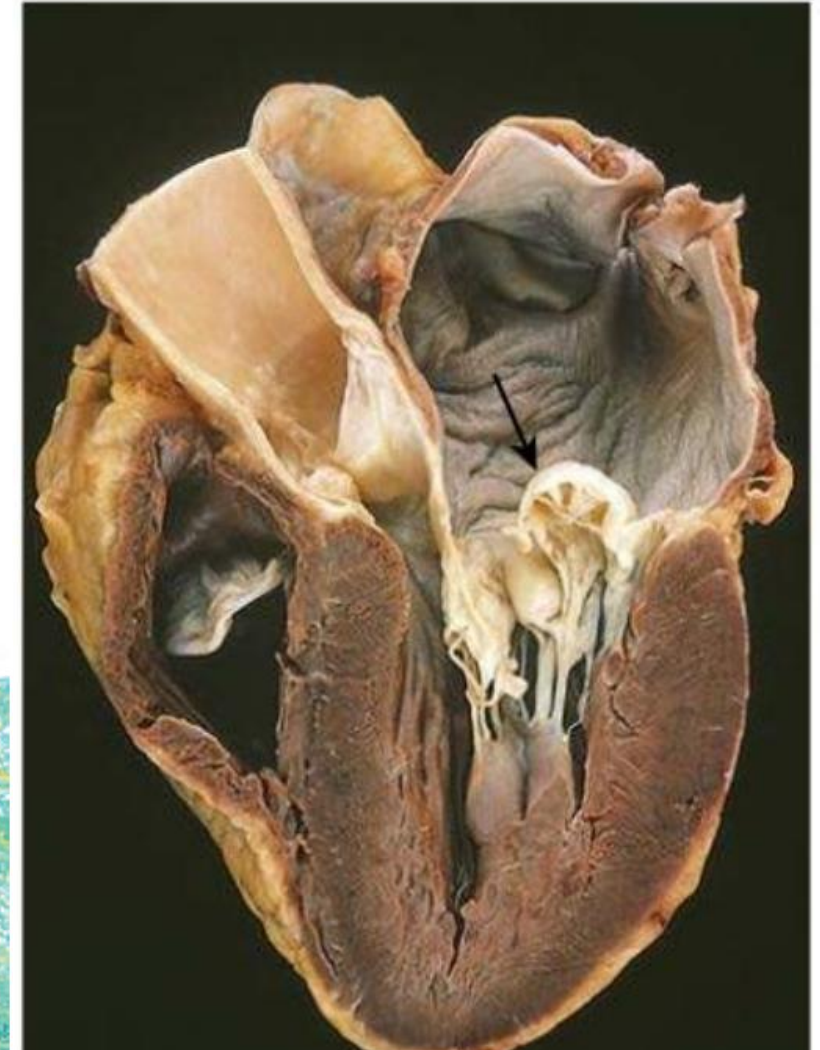
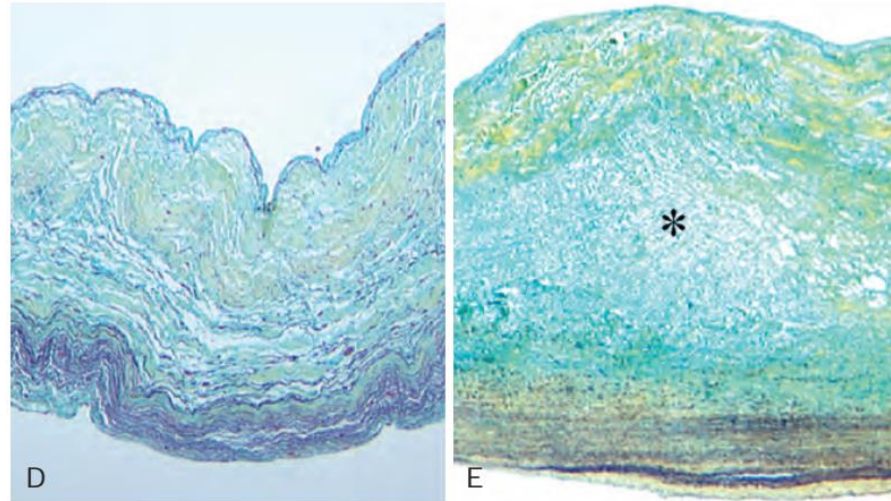


(A) Calcific aortic stenosis of a previously normal valve (viewed from aortic aspect). Nodular masses of calcium are heaped up within the sinuses of Valsalva (arrow). (B) Calcific aortic stenosis of a congenitally bicuspid valve. One cusp has a partial fusion at its center, called a raphe (arrow).

Mitral valve prolapse (Myxomatous degeneration)

- **Floppy** and prolapse, ballooning back into the left atrium during systole
- 2% to 3% of adults in the United States and more common in women
- The etiology is usually unknown (idiopathic)
- Clinical features: asymptomatic, chest pain, dyspnea, palpitation
- Increased risk of infective endocarditis

Normal heart valve (D) and myxomatous mitral valve (E). In myxomatous valves, collagen in the fibrosa layer is loose and disorganized, proteoglycan deposition (asterisk) in the central spongiosa layer is markedly expanded, and elastin in the atrialis layer is disorganized.



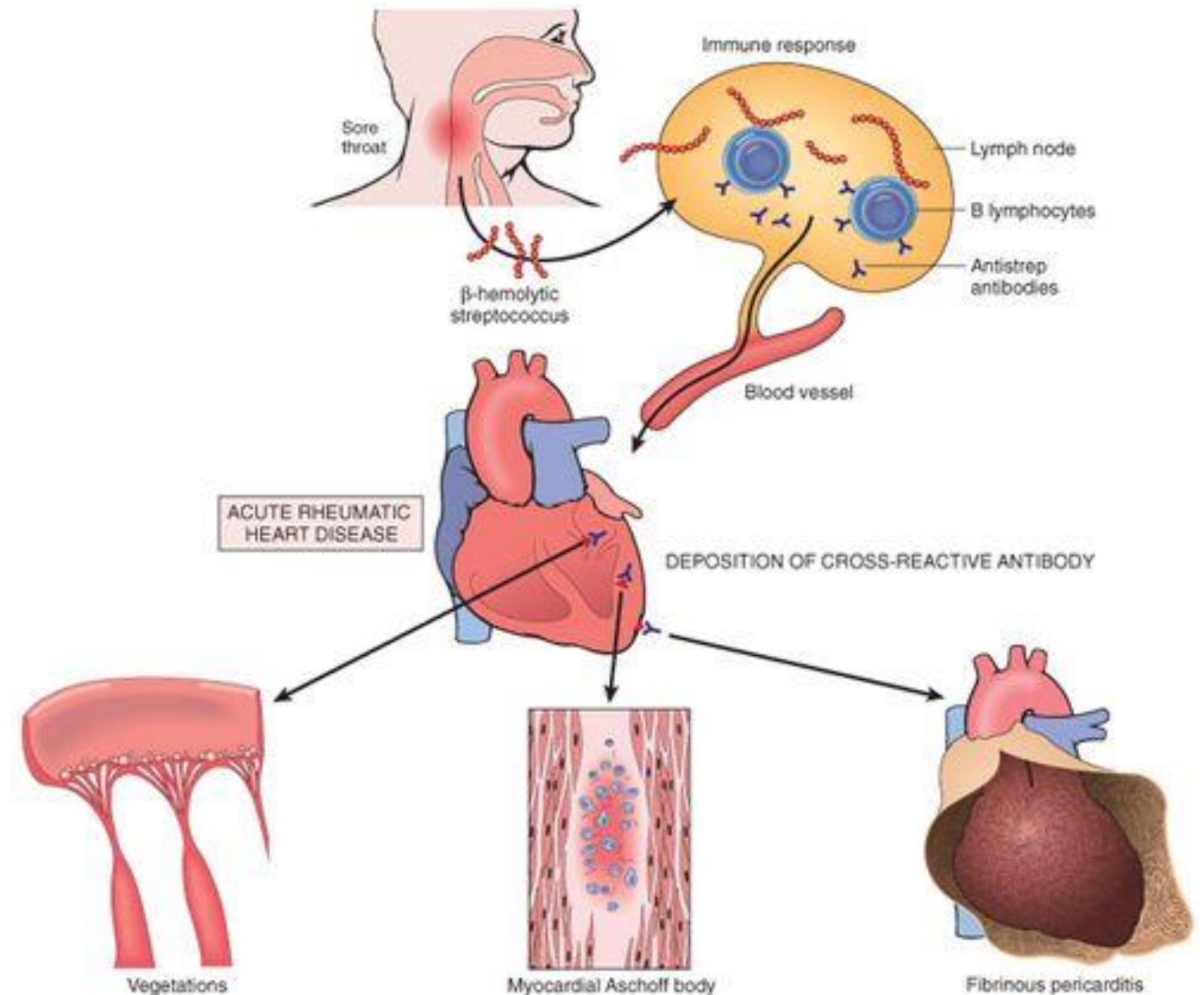
Long axis view (left ventricle is on the right) demonstrating hoarding with prolapse of the posterior mitral leaflet into the left atrium (arrow)

Rheumatic fever and Rheumatic heart disease

- **Rheumatic fever (RF)** is an acute, **immunologically mediated**, multisystem inflammatory disease classically occurring a few weeks after **group A streptococcal pharyngitis**.
- **Acute rheumatic carditis** is a common manifestation of active RF and may progress over time to **chronic rheumatic heart disease (RHD)**
- The most important consequence of RF are chronic valvular deformities especially **mitral stenosis**

Pathogenesis of Rheumatic fever

- Host immune responses to **group A streptococcal antigens** that **cross-react with host proteins**
(Type II Hypersensitivity)

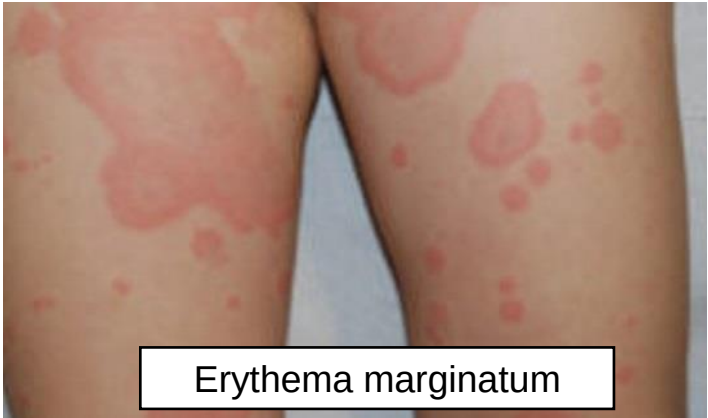


Clinical features of Rheumatic fever

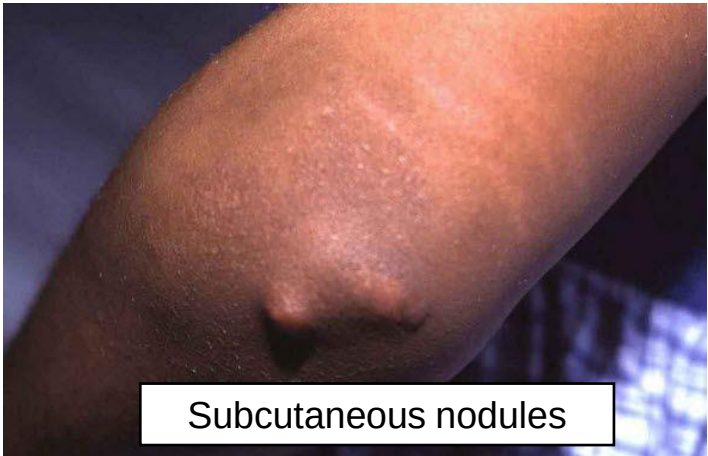
- Acute RF typically appears 10 days to 6 weeks after a group A streptococcal infection in about 3% of patients.
- **Revised Jones Criteria:** evidence of group A streptococcal infection with
 - 2 major manifestations, or
 - 1 major and 2 minor manifestations

Anti-streptolysin O (ASO)
or anti-DNase B

Major Diagnostic Criteria	Minor Diagnostic Criteria
Carditis	Fever
Polyarthrititis	Arthralgia
Chorea	Elevated acute phase reactant (ESR or CRP)
Erythema marginatum	
Subcutaneous nodules	Prolonged PR interval



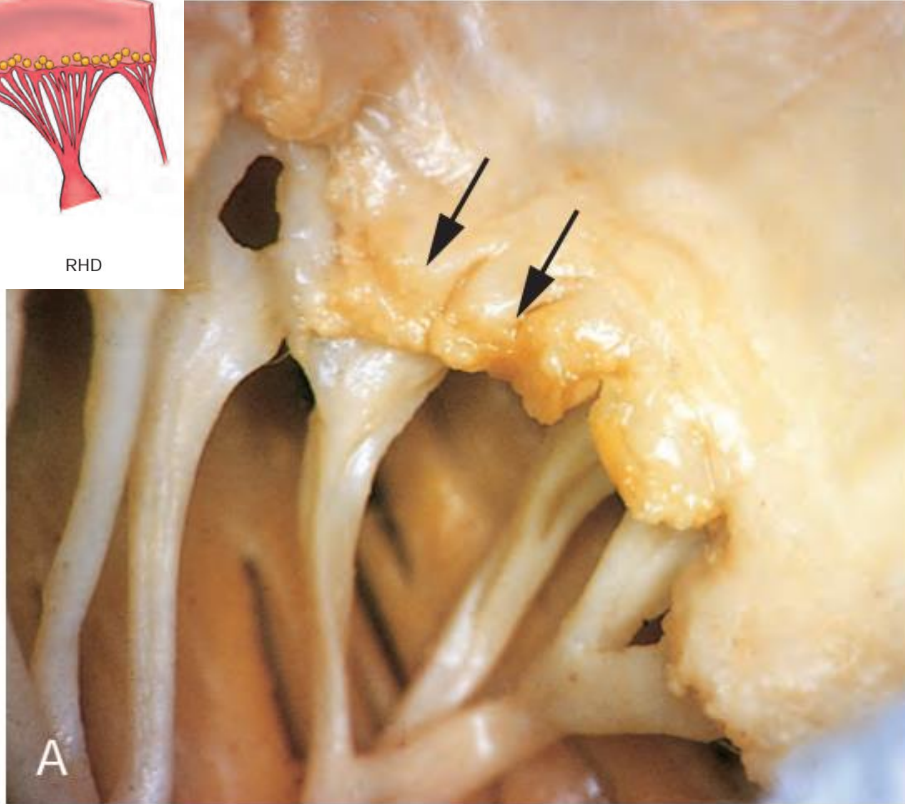
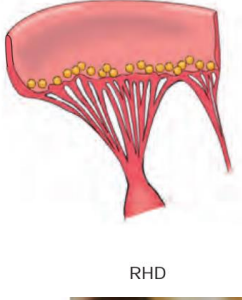
Erythema marginatum



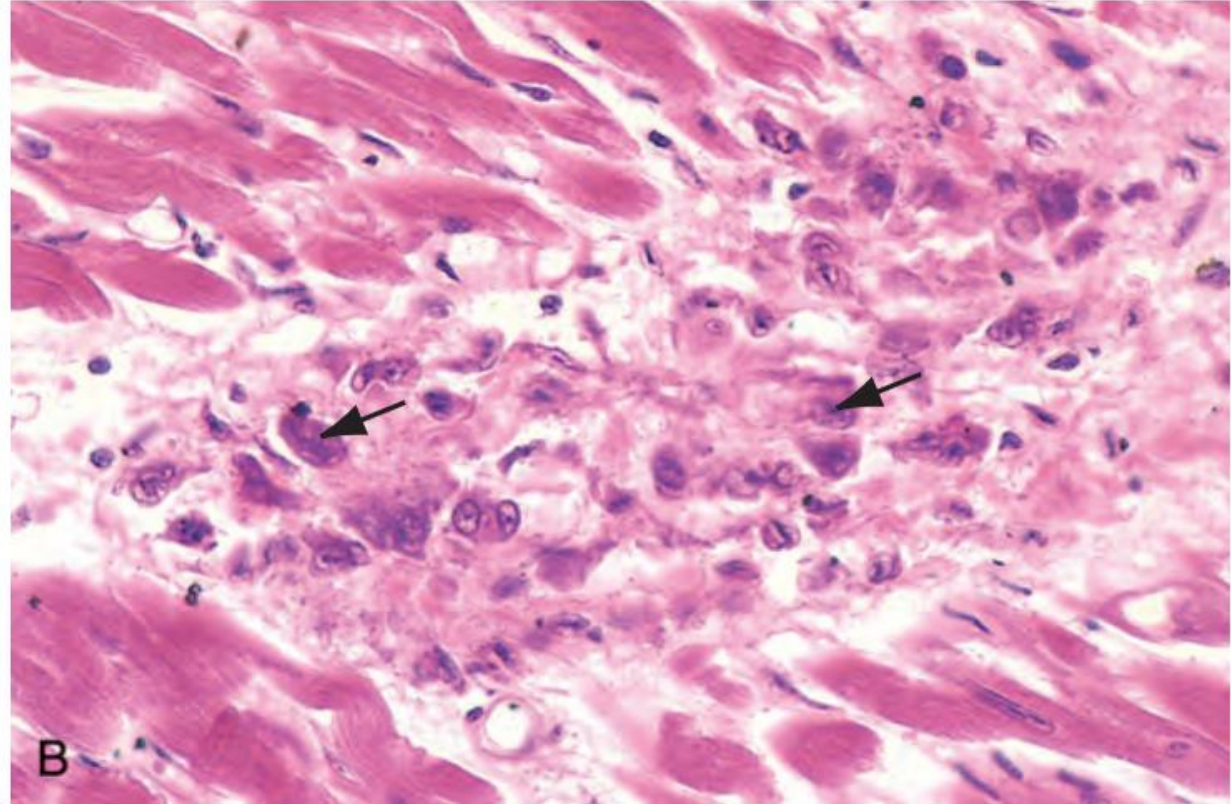
Subcutaneous nodules

ไข้ ข้อ คอ คาวี ผื่นแดง

Acute rheumatic carditis



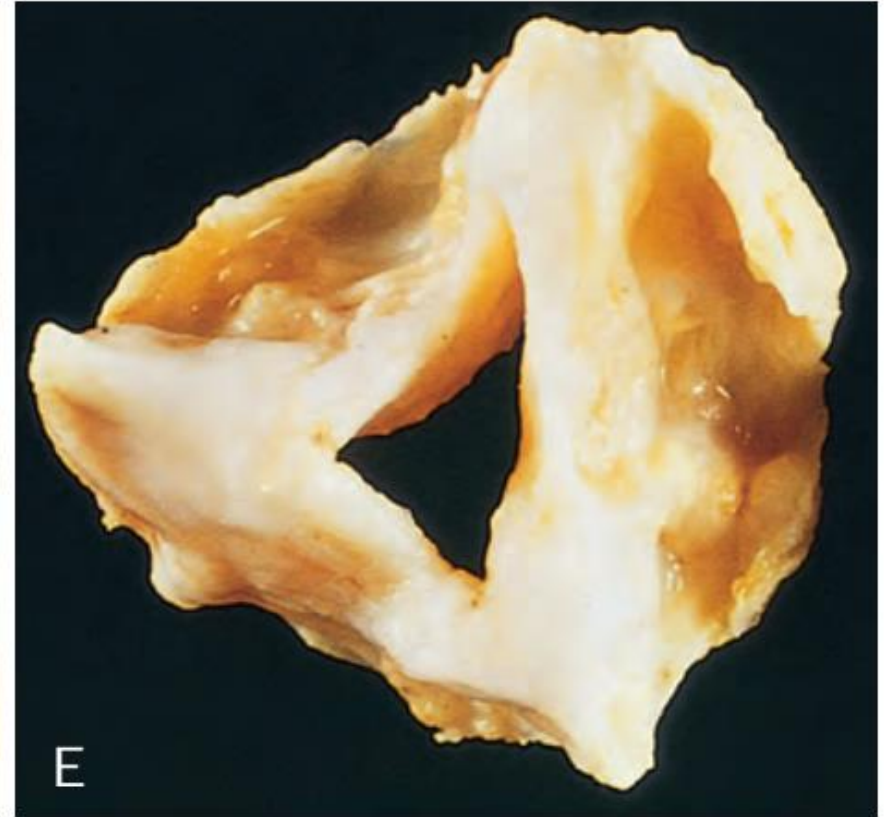
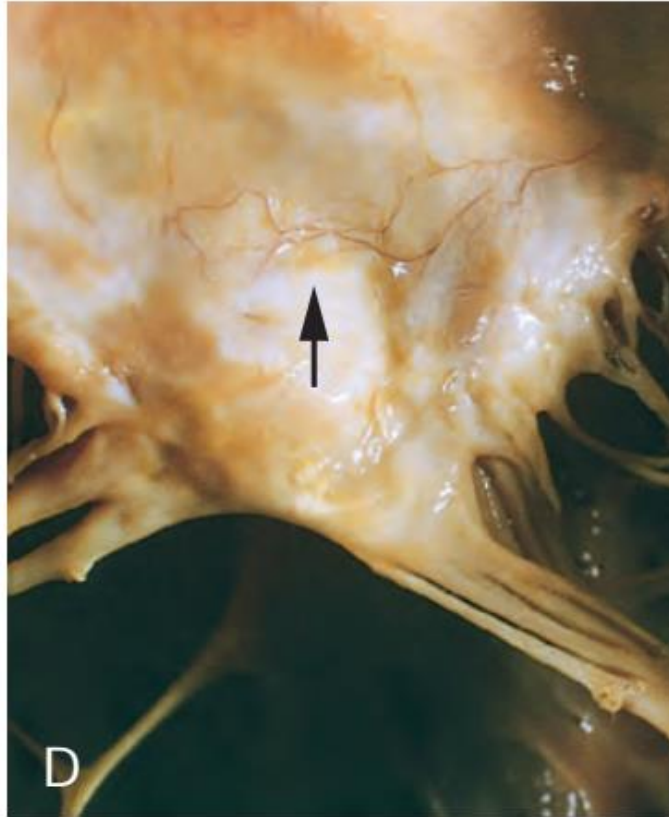
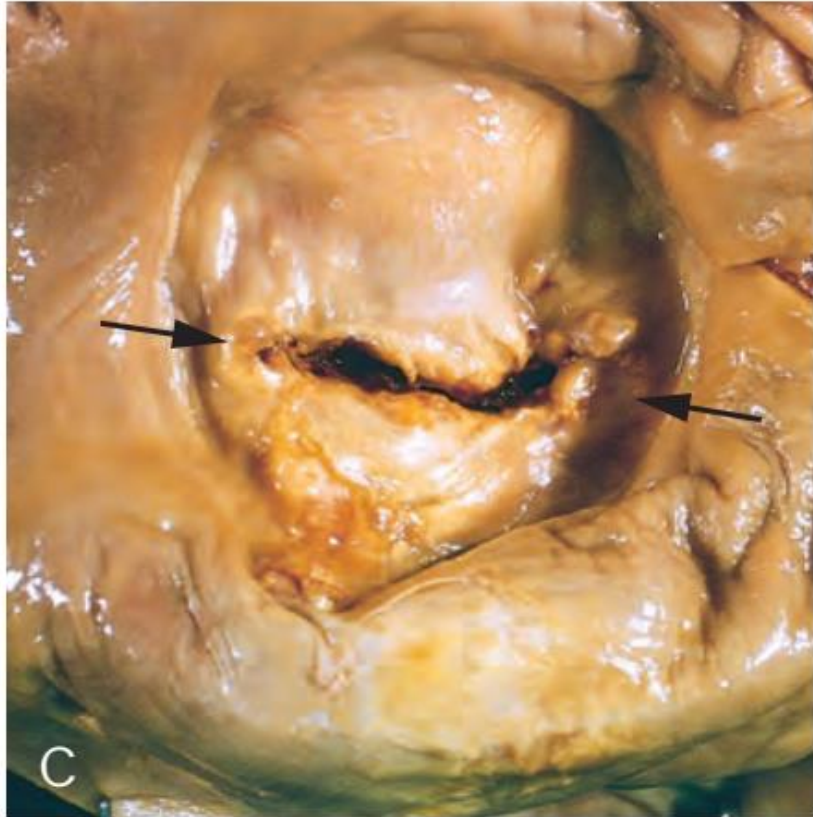
Vegetations
small, warty vegetations along
the lines of closure of the
valve leaflets



Aschoff bodies - are composed of foci of
T lymphocytes, occasional plasma cells, and
plump activated macrophages called
Anitschkow cells.



Chronic rheumatic heart disease



Mitral stenosis

Diffuse fibrous thickening and distortion of the valve leaflets and commissural fusion (arrows, C), and thickening of the chordae tendineae (D). (E) Surgically resected specimen of rheumatic aortic stenosis, demonstrating thickening and distortion of the cusps with commissural fusion.

Infective Endocarditis (IE)

- **Microbial infection** of the heart valves or the mural endocardium that leads to the formation of **vegetations** composed of **thrombotic debris and organisms**, often associated with destruction of the underlying cardiac tissues
- One of the most serious of all infections
- Cause of microorganism seeding into the blood stream
 - Dental or surgical procedure
 - Intravenous drug users
 - Occult sources from gut, oral cavity or trivial injuries
 - Obvious infection elsewhere
- Prevention by appropriate **antibiotic prophylaxis**

Infective Endocarditis (IE)

- Cardiac abnormalities increase the risk of developing IE – acquired valvular heart disease, prosthetic heart valves, and congenital defects
- Causative organisms
 - *Streptococcus viridans* (50-60%): native but previously damaged valves
 - *Staphylococcus aureus* (20-30%): healthy or deformed valves, intravenous drug abusers
 - *Enterococci* and the HACEK group (*Haemophilus*, *Actinobacillus*, *Cardiobacterium*, *Eikenella*, *Kingella*)
 - Gram negative bacilli and fungi
 - Prosthetic valves: *Staphylococcus aureus*, *Staphylococcus epidermidis*
 - “Culture negative” endocarditis (10%)
- High virulent organism (*S. aureus*) >> Acute IE >> 50% death in 2-4 weeks
- Low virulent organism (*S. viridans*) >> Subacute IE >> Insidious course

Clinical features of IE

- Acute IE
 - Fever, chills, weakness, fatigue
 - Murmur
- Subacute IE
 - Fever (Prolonged)
 - Non-specific symptoms: fatigue, weight loss, flu-like syndrome
 - Murmurs
 - Others: petechiae, subungual hemorrhage, Roth spots in the eyes

Heart diseases > Valvular heart disease

Splinter hemorrhages



Conjunctival hemorrhages

Janeway lesions

(painless spots on palms/soles of feet)



Osler's Nodes

(painful nodules in pulp of fingers/toes)

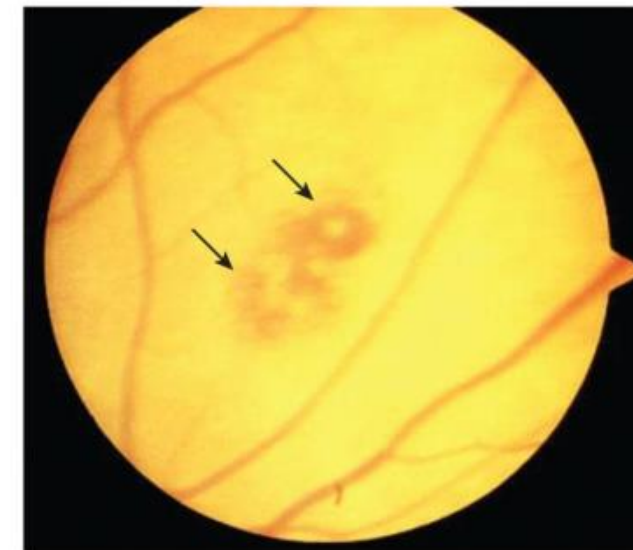


Fig. 62.3 Roth spots. Septic microemboli to the retinal arteries result in infarction. The ischemic cotton-wool centers are surrounded by hemorrhagic halos (arrows). These spots are not pathognomonic for endocarditis. (From Mandell GL, Bennett JE, Dolin R, editors. Principles and practice of infectious diseases. 6th ed. Philadelphia: Churchill Livingstone; 2005.)

Classic findings of endocarditis. And "classic" means *rare* (<5% sensitivity for all of these findings). They are more often seen in *subacute* endocarditis. Although rarely encountered, these features might be a red flag to investigate further.

Images from: Yang E and Frazee BW 2018 PMID 30296997; Mylonakis E & Calderwood SB 2001 PMID 11794152 ; Conjunctival hemorrhage image from Dr. Josh Fierer https://meded.ucsd.edu/clinicalimg/eye_petechiae.htm;

Diagnosis of IE

- **Modified Duke Criteria:** diagnosis of individuals with **suspected IE** by predisposing factors, physical findings, blood culture results, imaging findings, and laboratory information

IE Classification (at admission and during follow-up)

Definite:

- 2 major criteria.
- 1 major criterion and at least 3 minor criteria.
- 5 minor criteria.

Possible:

- 1 major criterion and 1 or 2 minor criteria.
- 3–4 minor criteria.

Rejected:

- Does not meet criteria for definite or possible at admission with or without a firm alternative diagnosis.

Diagnosis of IE

Major criteria

(i) Blood cultures positive for IE

- (a) Typical microorganisms consistent with IE from two separate blood cultures:
 - Oral streptococci, *Streptococcus gallolyticus* (formerly *S. bovis*), HACEK group, *S. aureus*, *E. faecalis*
- (b) Microorganisms consistent with IE from continuously positive blood cultures:
 - ≥ 2 positive blood cultures of blood samples drawn >12 h apart.
 - All of 3 or a majority of ≥ 4 separate cultures of blood (with first and last samples drawn ≥ 1 h apart).
- (c) Single positive blood culture for *C. burnetii* or phase I IgG antibody titre $>1:800$.

(ii) Imaging positive for IE:

Valvular, perivalvular/periprosthetic and foreign material anatomic and metabolic lesions characteristic of IE detected by any of the following imaging techniques:

- Echocardiography (TTE and TOE).
- Cardiac CT.
- [18F]-FDG-PET/CT(A).
- WBC SPECT/CT.

Minor criteria

(i) Predisposing conditions (i.e. predisposing heart condition at high or intermediate risk of IE or PWIDs)^a

(ii) Fever defined as temperature $>38^{\circ}\text{C}$

(iii) Embolic vascular dissemination (including those asymptomatic detected by imaging only):

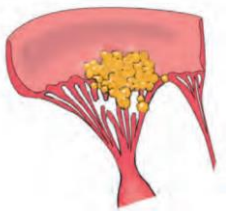
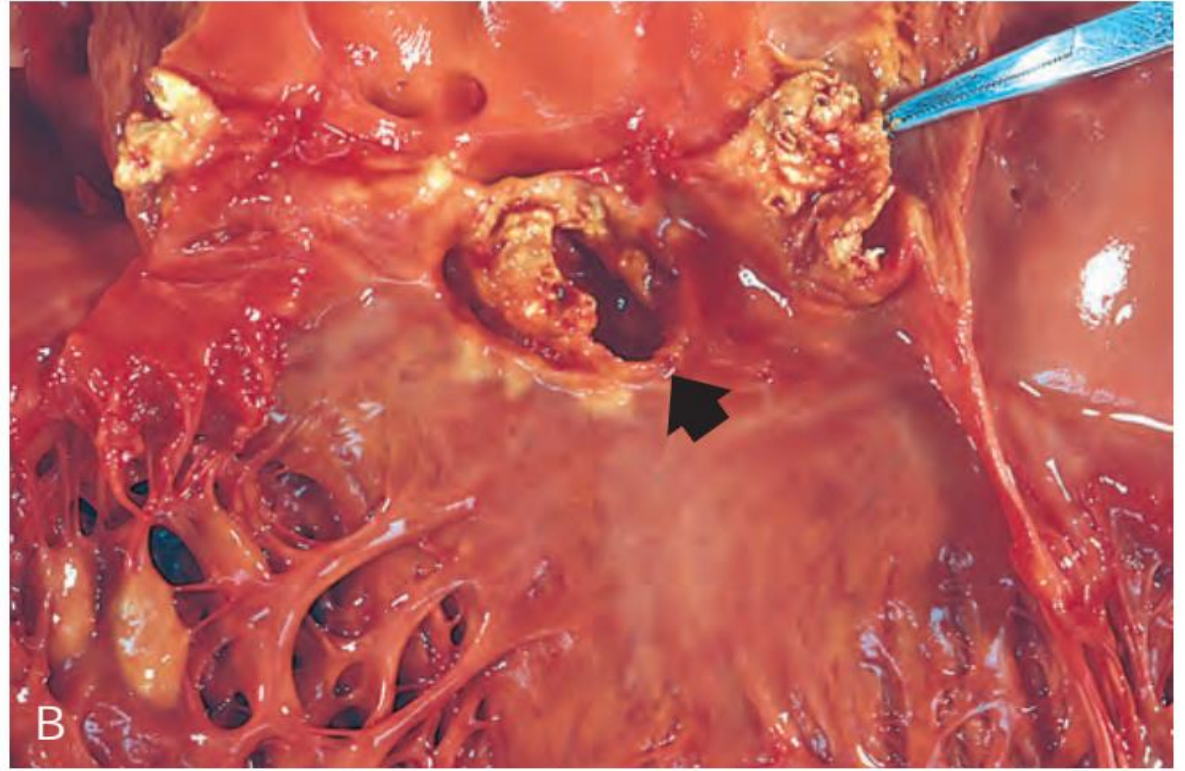
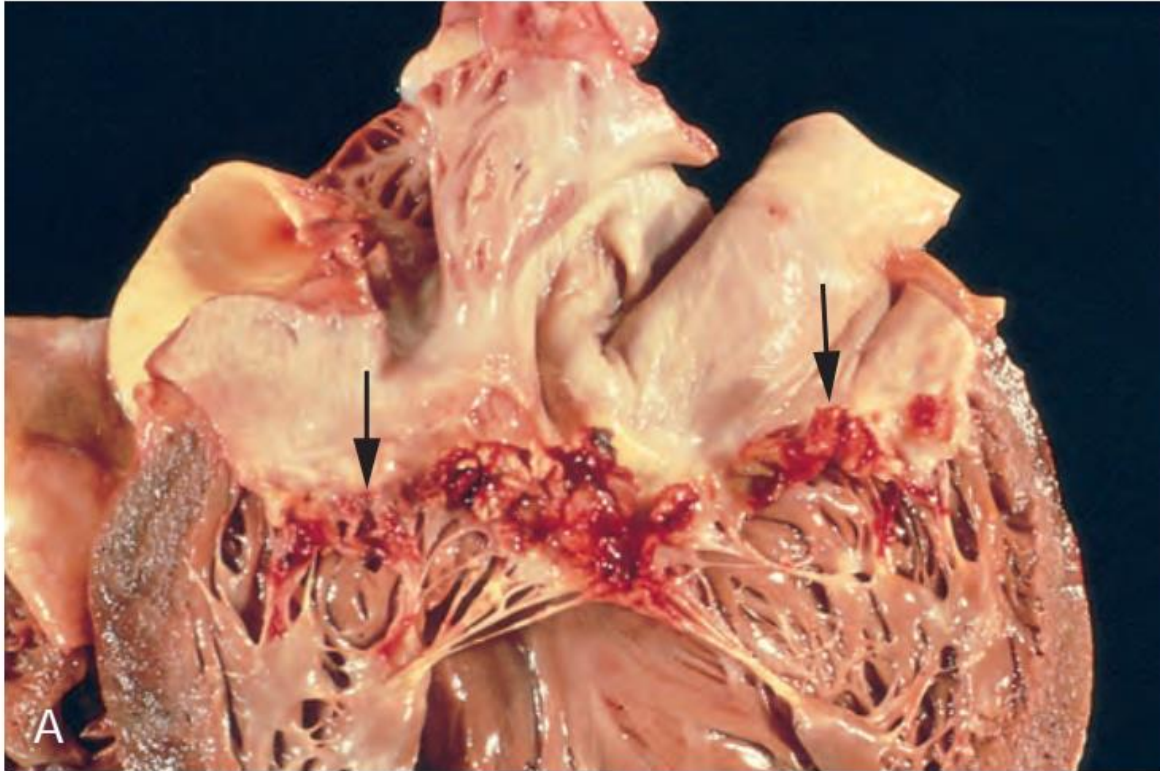
- Major systemic and pulmonary emboli/infarcts and abscesses.
- Haematogenous osteoarticular septic complications (i.e. spondylodiscitis).
- Mycotic aneurysms.
- Intracranial ischaemic/haemorrhagic lesions.
- Conjunctival haemorrhages.
- Janeway's lesions.

(IV) Immunological phenomena:

- Glomerulonephritis.
- Osler nodes and Roth spots.
- Rheumatoid factor.

(V) Microbiological evidence:

- Positive blood culture but does not meet a major criterion as noted above.
- Serological evidence of active infection with organism consistent with IE.

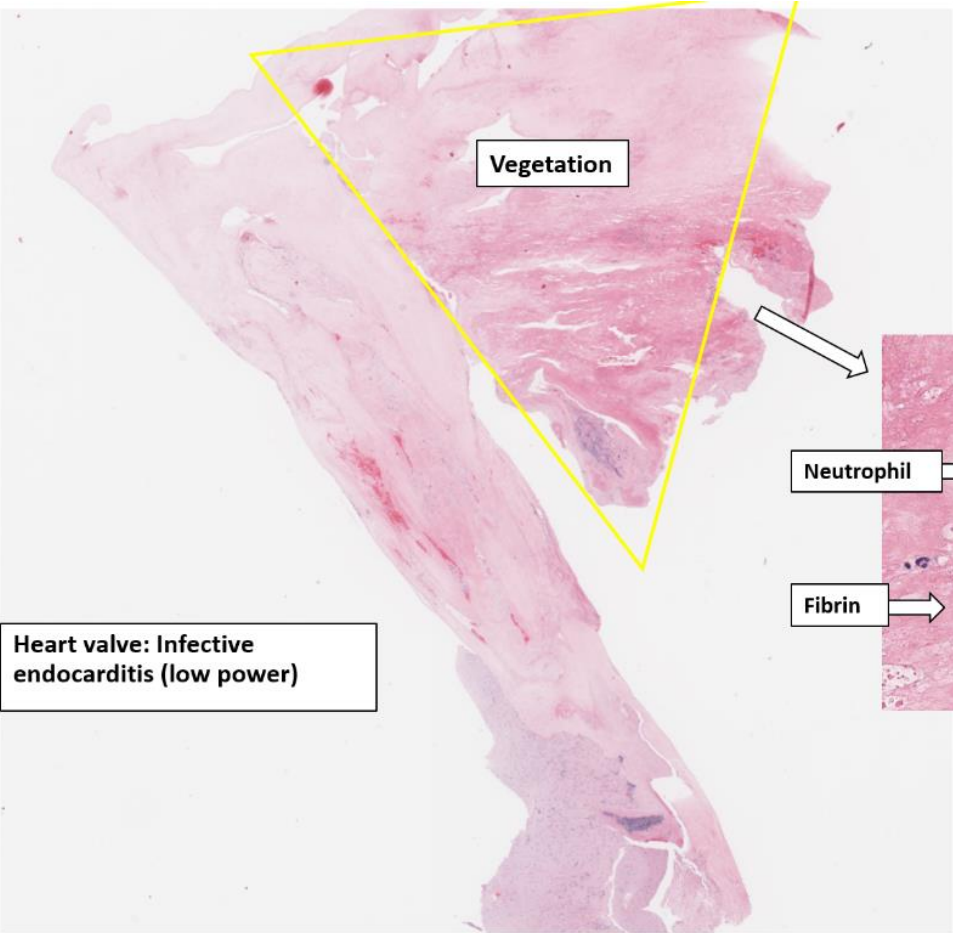


IE

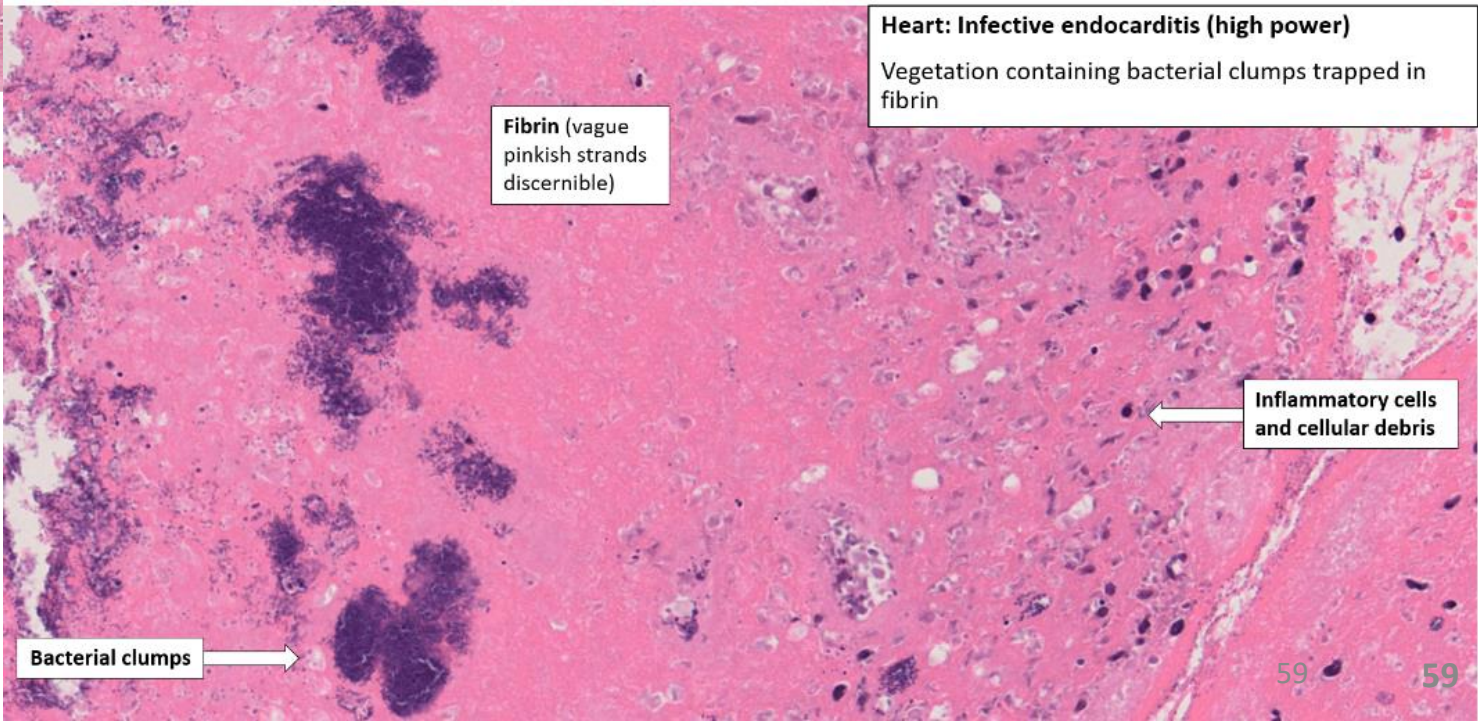
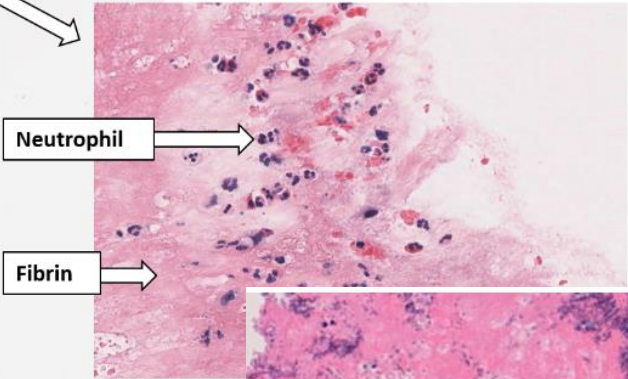
Vegetations
friable, bulky, potentially
destructive lesions

Ring abscess

Heart diseases > Valvular heart disease



Heart valve: Infective endocarditis (low power)



Complications of IE

- Cardiac complications:
 - Heart failure
 - Valvular insufficiency or stenosis
 - Myocardial ring abscess with perforation
 - Suppurative pericarditis
- Embolic complications:
 - Vegetation at left-sided: brain abscess, splenic abscess, kidneys abscess
 - Vegetation at right-sided : lung infection and abscess

Myocarditis

- A diverse group of pathologic entities in which **infectious microorganisms** and/or a **primary inflammatory process** cause myocardial injury
- **Viral infection** are the most common cause of myocarditis (**Coxsackieviruses A and B** and other **enteroviruses**)
- Clinical features vary from asymptomatic to abrupt onset of arrhythmia or CHF
 - fatigue, dyspnea, palpitations, precordial discomfort, and fever

Infections

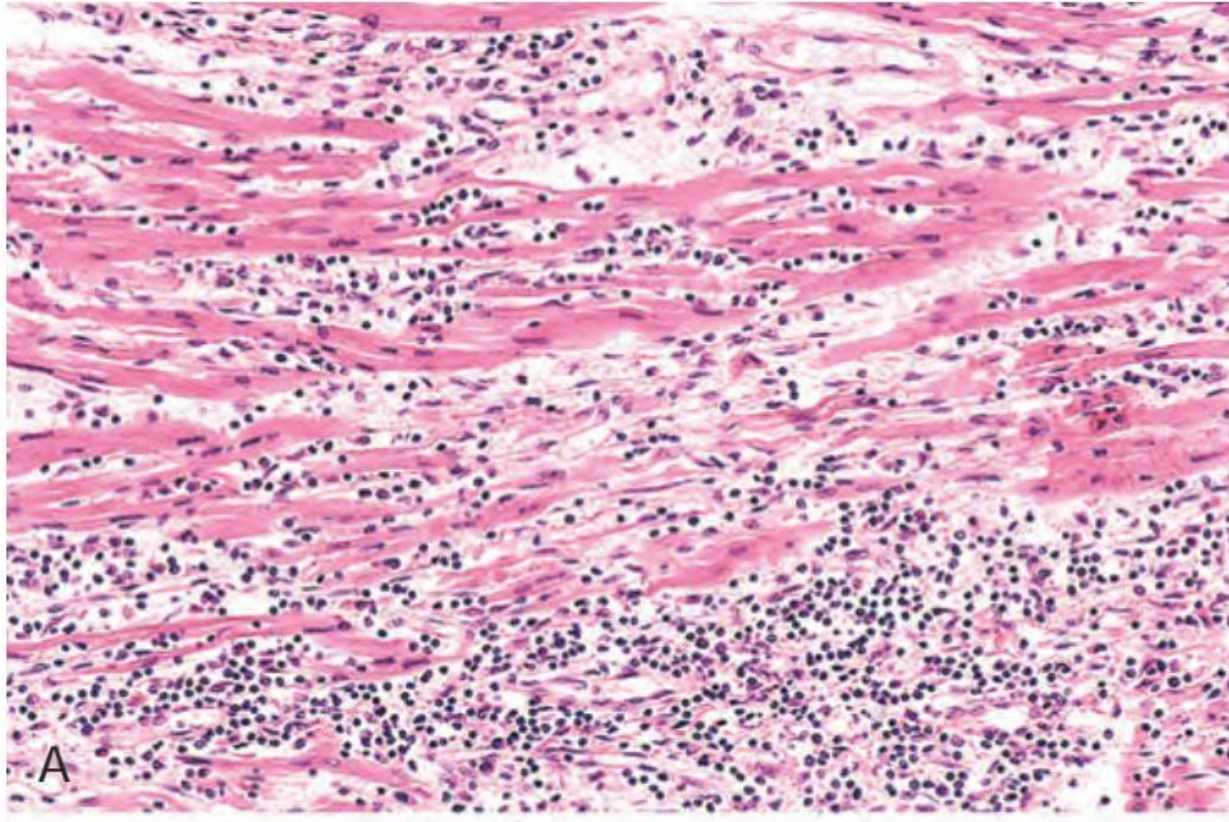
Viruses (e.g., coxsackievirus, echovirus, influenza, HIV, cytomegalovirus)
Chlamydiae (e.g., *Chlamydophila psittaci*)
Rickettsiae (e.g., *Rickettsia typhi*, typhus fever)
Bacteria (e.g., *Corynebacterium diphtheriae*, *Neisseria meningococcus*,
Borrelia (Lyme disease))
Fungi (e.g., *Candida*)
Protozoa (e.g., *Trypanosoma cruzi* [Chagas disease], toxoplasmosis)
Helminths (e.g., trichinosis)

Immune-Mediated Reactions

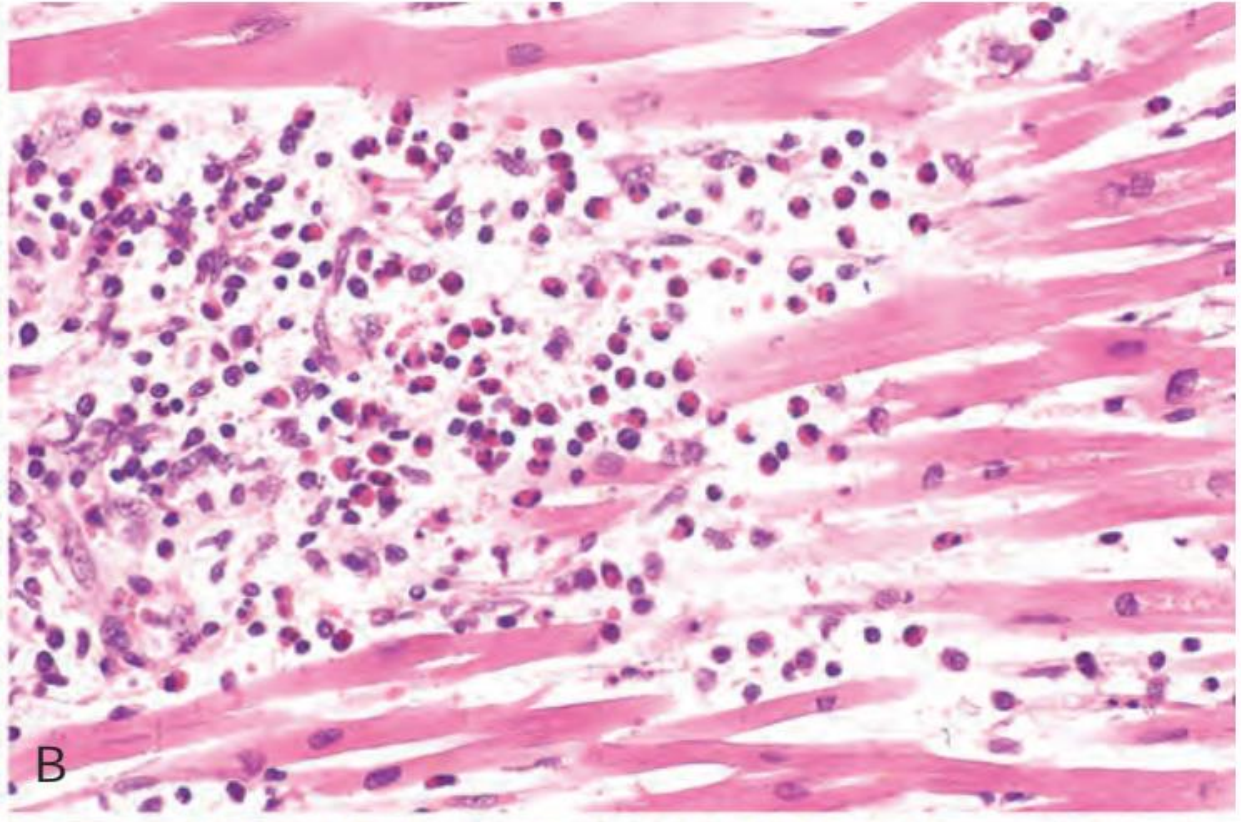
Postviral
Poststreptococcal (rheumatic fever)
Systemic lupus erythematosus
Drug hypersensitivity (e.g., methyldopa, sulfonamides)
Transplant rejection
Immune checkpoint inhibitor therapies

Unknown

Sarcoidosis
Giant cell myocarditis



Viral myocarditis
diffuse, mononuclear, predominantly
lymphocytic infiltrate



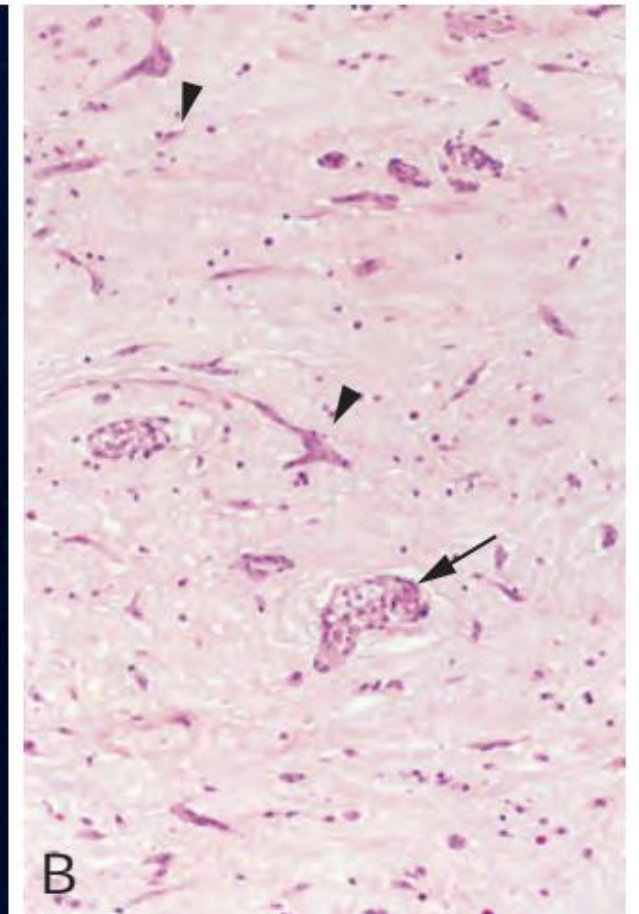
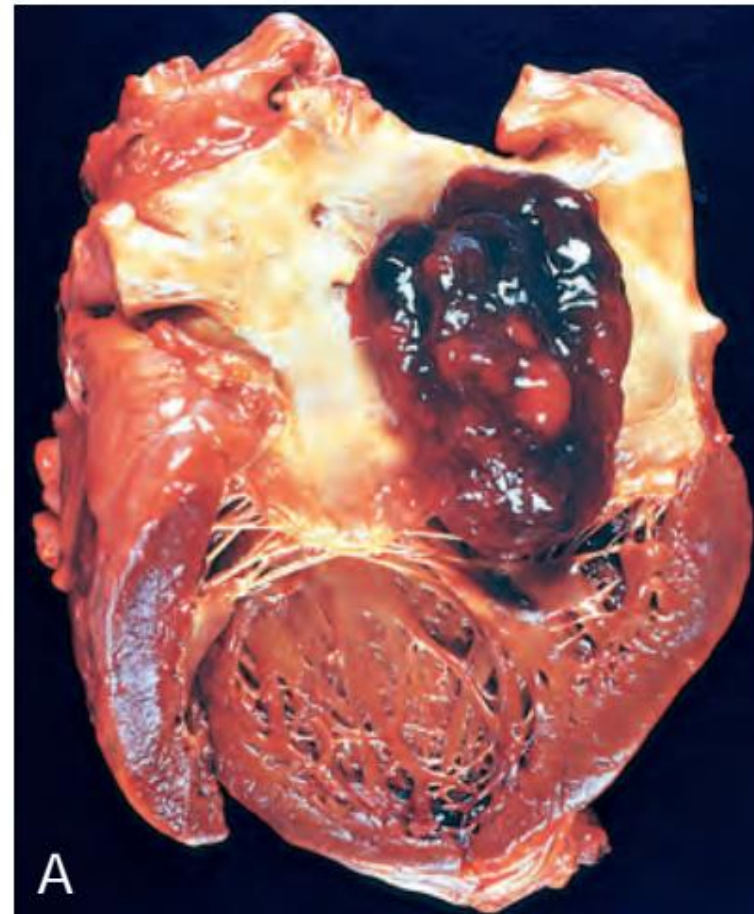
Hypersensitivity myocarditis
infiltrate composed largely of eosinophils and
mononuclear inflammatory cells

Tumors of the heart

- Benign tumor
 - Papillary fibroelastoma
 - Myxoma
 - Fibroma
 - Rhabdomyoma
 - Lipoma
 - Hemangioma
 - Hamartoma
 - Malignant tumor
 - Angiosarcoma
 - Leiomyosarcoma
 - Undifferentiated pleomorphic sarcoma
 - Hematolymphoid tumor
 - Diffuse large B cell lymphoma
-
- The diagram features two callout boxes. A yellow callout box points to 'Myxoma' in the list, containing the text 'the most common primary tumor of the **adult** heart'. A green callout box points to 'Rhabdomyoma' in the list, containing the text 'the most frequent primary tumor of the **pediatric** heart'.

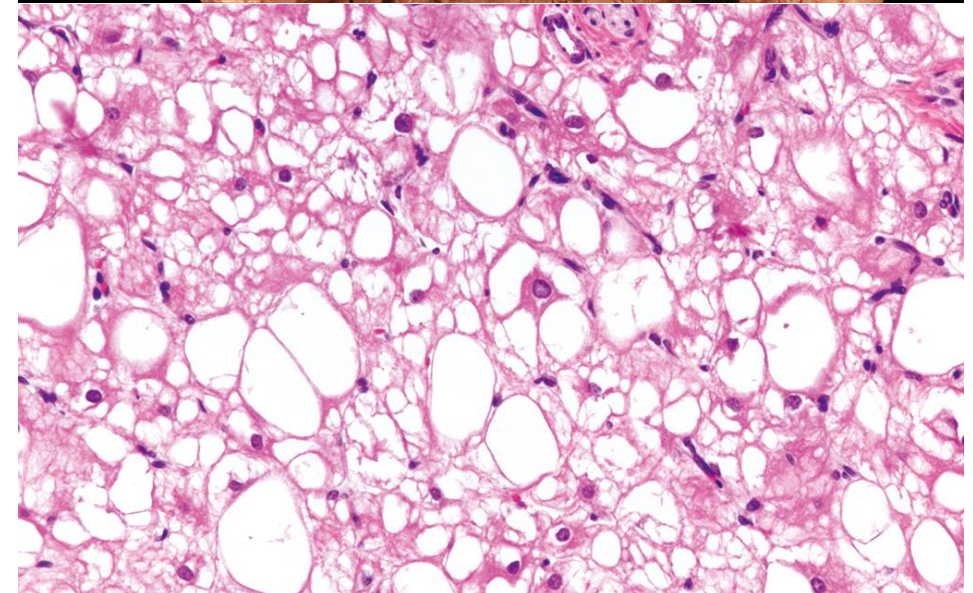
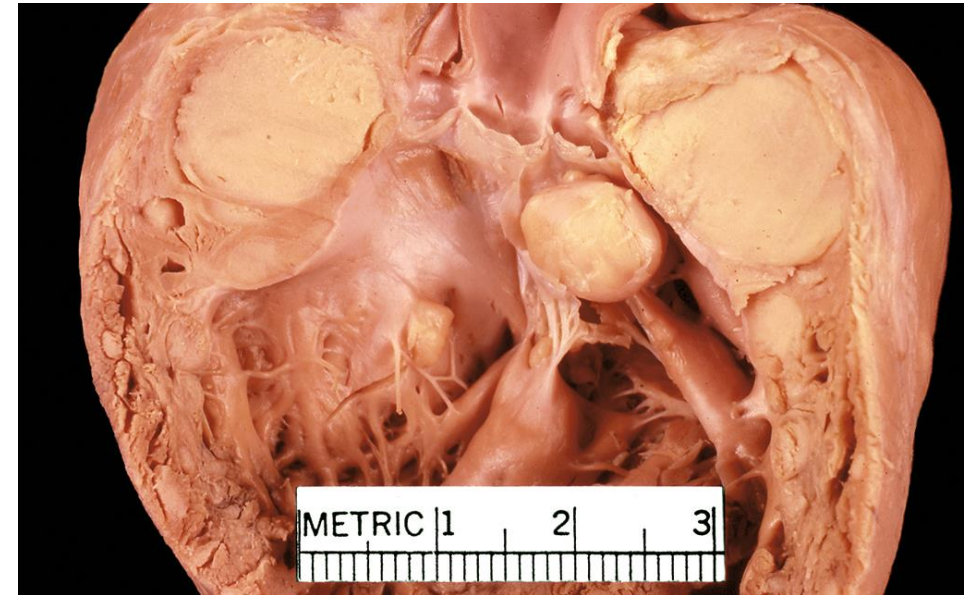
Myxoma

- About 90% of myxomas arise in the **atrium**, with a **left-to-right** ratio of approximately 4:1
- The major clinical manifestations are due to valvular “**ball-valve**” **obstruction**
- Sessile or pedunculated lesions with hemorrhage
- Abundant amorphous extracellular matrix contains scattered multinucleate myxoma cells in various groupings, including abnormal vessel-like formations



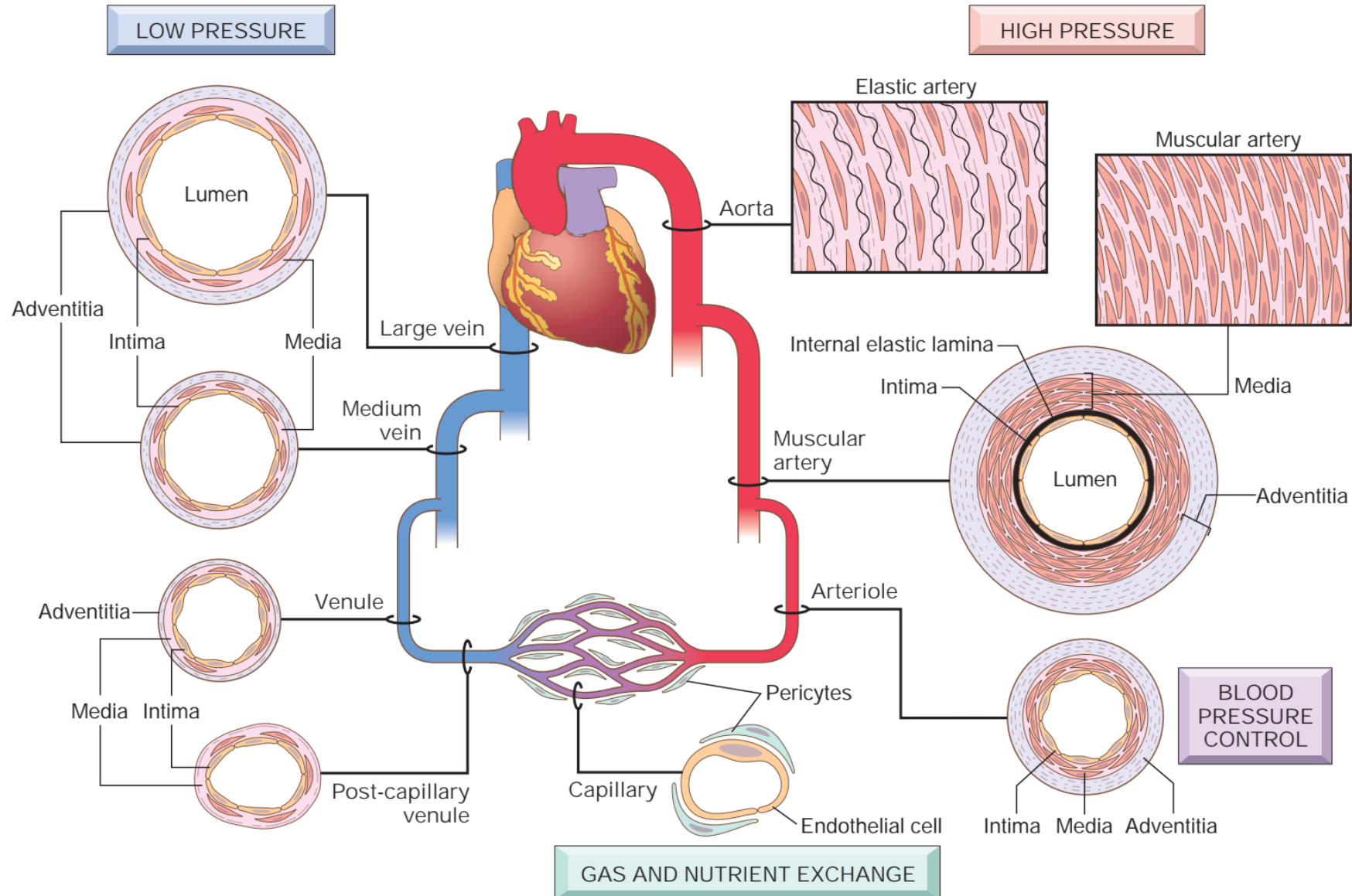
Rhabdomyoma

- Commonly discovered in **the first years of life**
- Location: **ventricle** > atrium
- The clinical features: asymptomatic to obstruction
- Well-defined, white-yellow mass
- Bizarre, markedly enlarged myocytes with large collections of glycogen, “**spider**” **cells**



VASCULAR DISEASES

Vascular structure



Vascular injury and response

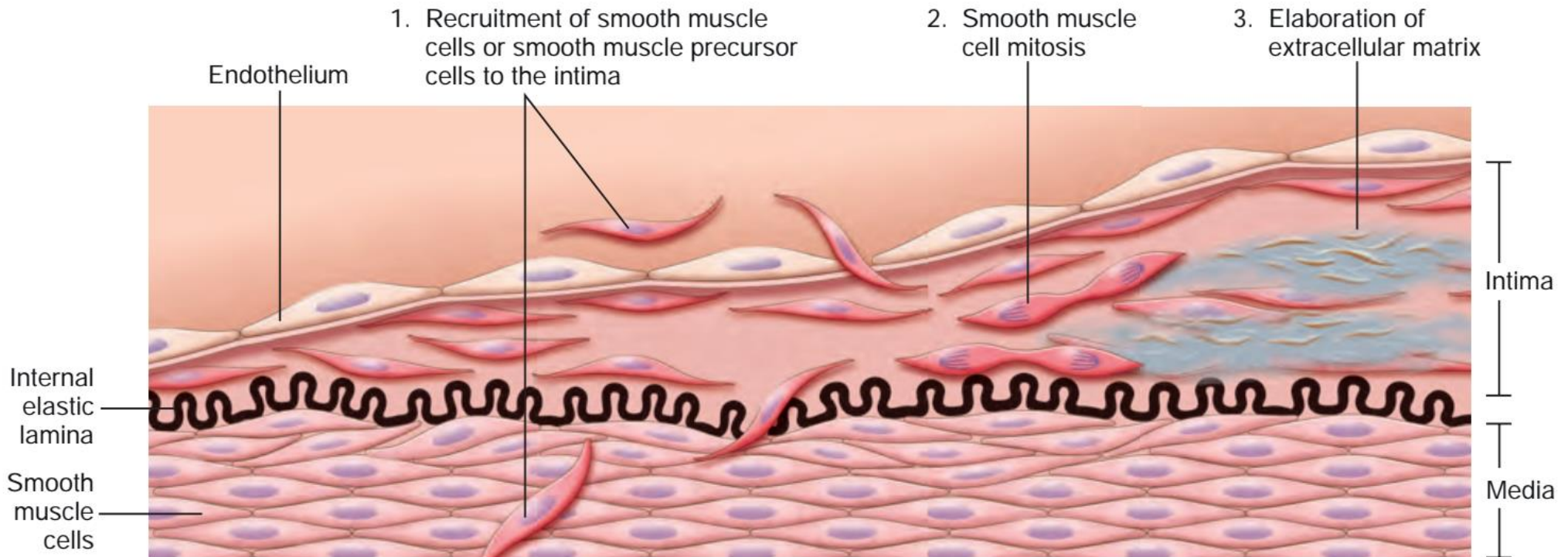


Figure 11.3 Stereotypical response to vascular injury. Schematic diagram of intimal thickening, emphasizing intimal smooth muscle cell migration and proliferation associated with extracellular matrix synthesis. Intimal smooth muscle cells can derive from the underlying media or can be recruited from circulating precursors; the intimal cells are shown in a darker shade to emphasize that they have a proliferative, synthetic, and noncontractile phenotype distinct from medial smooth muscle cells.

Excessive thickening of the intima can result in luminal stenosis and vascular obstruction

Blood pressure regulation

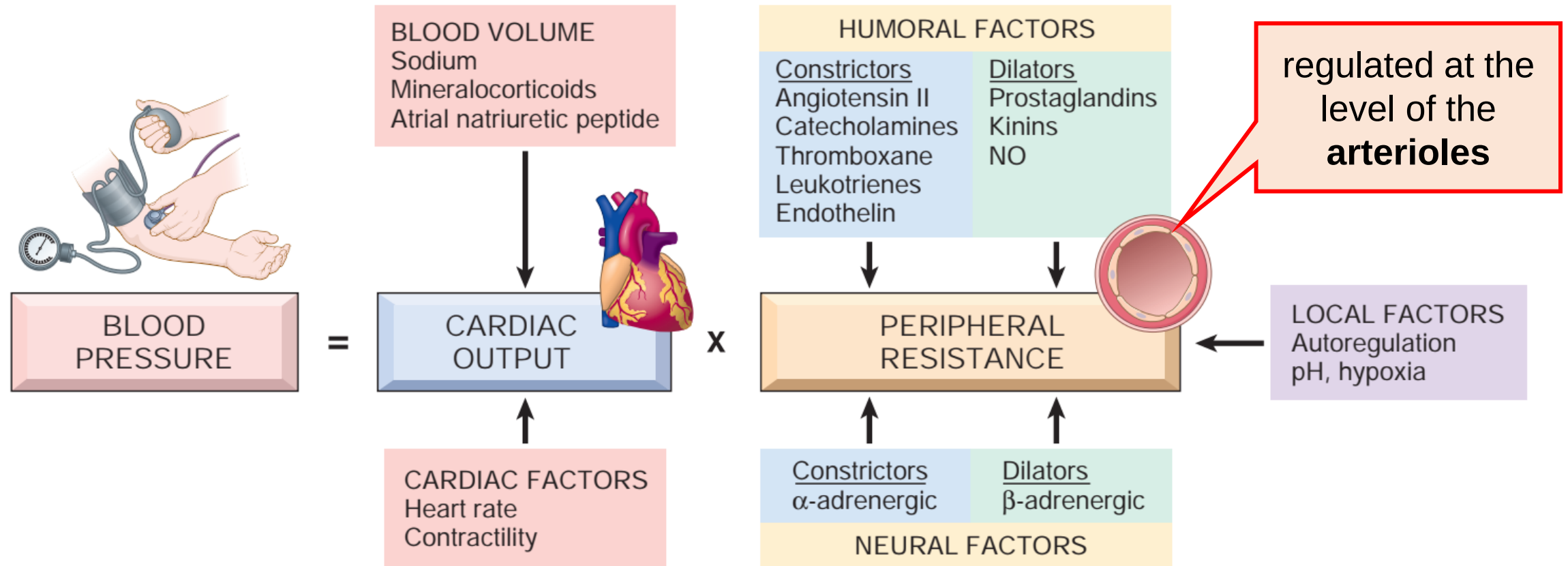


Figure 11.4 Blood pressure regulation. Diverse influences on cardiac output (e.g., blood volume and myocardial contractility) and peripheral resistance (neural, humoral, and local effectors) impact the output blood pressure.

Blood pressure regulation

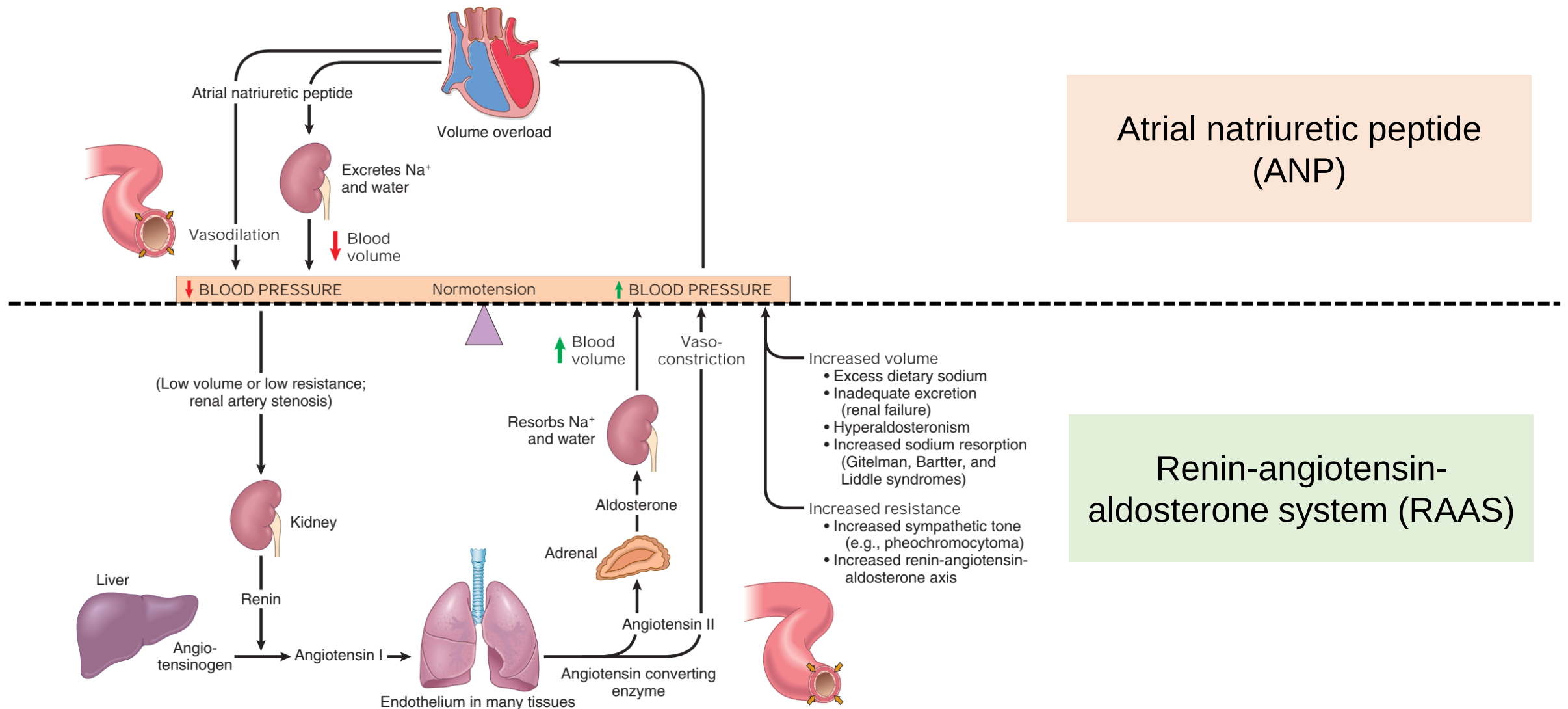


Figure 11.5 Interplay of renin-angiotensin-aldosterone and atrial natriuretic peptide in maintaining blood pressure homeostasis.

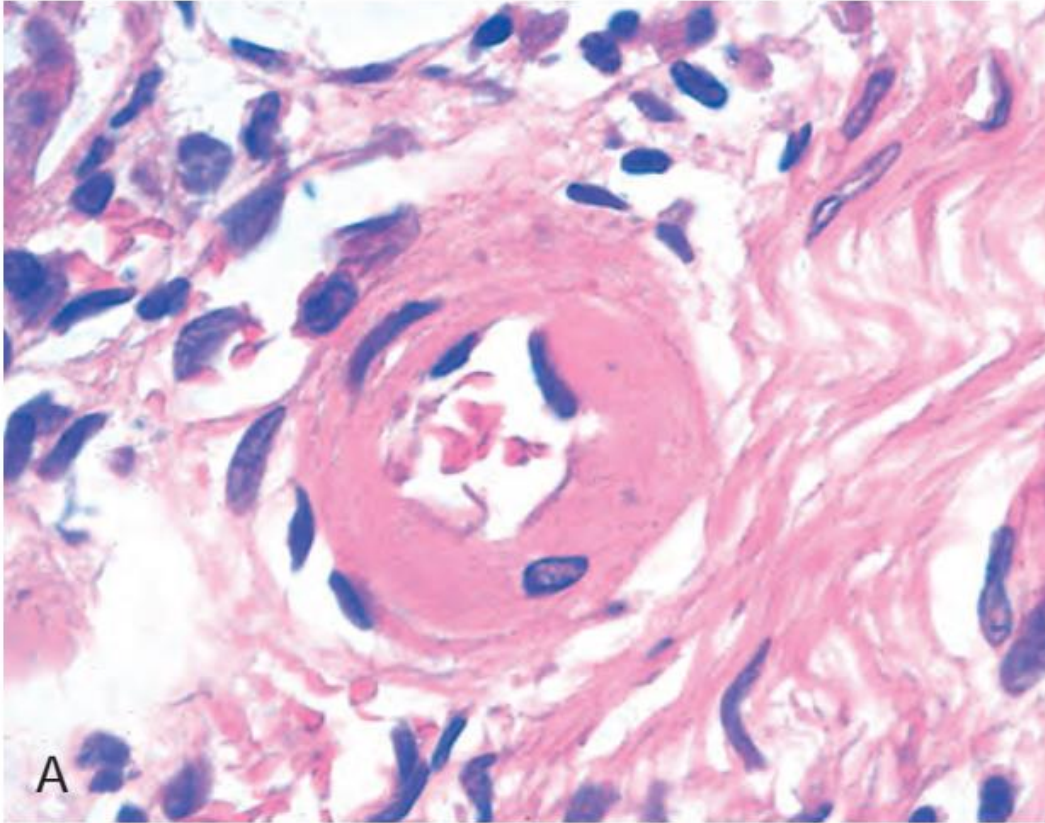
Hypertension

- Hypertension is a common disorder
- Blood pressure: SBP $\geq$ 140 and/or DBP $\geq$ 90 mm Hg***
- Major risk factor for atherosclerosis, heart failure, renal failure
- Essential hypertension represents 90% to 95% of cases and is a complex, multifactorial disorder
 - Genetic factors: altering sodium reabsorption in the kidney
 - Insufficient renal sodium excretion
 - Vasoconstrictive influences
 - Environmental factors: stress, obesity, smoking, physical inactivity, and heavy salt consumption

Table 11.2 Types and Causes of Hypertension (Systolic and Diastolic)

Essential Hypertension
Accounts for 90%–95% of all cases
Secondary Hypertension
Renal
Acute glomerulonephritis Chronic renal disease Polycystic disease Renal artery stenosis Renal vasculitis Renin-producing tumors
Endocrine
Adrenocortical hyperfunction (Cushing syndrome, primary aldosteronism, congenital adrenal hyperplasia, licorice ingestion) Exogenous hormones (glucocorticoids, estrogen [including pregnancy-induced and oral contraceptives], sympathomimetics and tyramine-containing foods, monoamine oxidase inhibitors) Pheochromocytoma Acromegaly Hyperthyroidism (thyrotoxicosis) Pregnancy-induced (preeclampsia)
Cardiovascular
Coarctation of the aorta Polyarteritis nodosa Increased intravascular volume Increased cardiac output Rigidity of the aorta
Neurologic
Psychogenic Increased intracranial pressure Sleep apnea Acute stress, including surgery

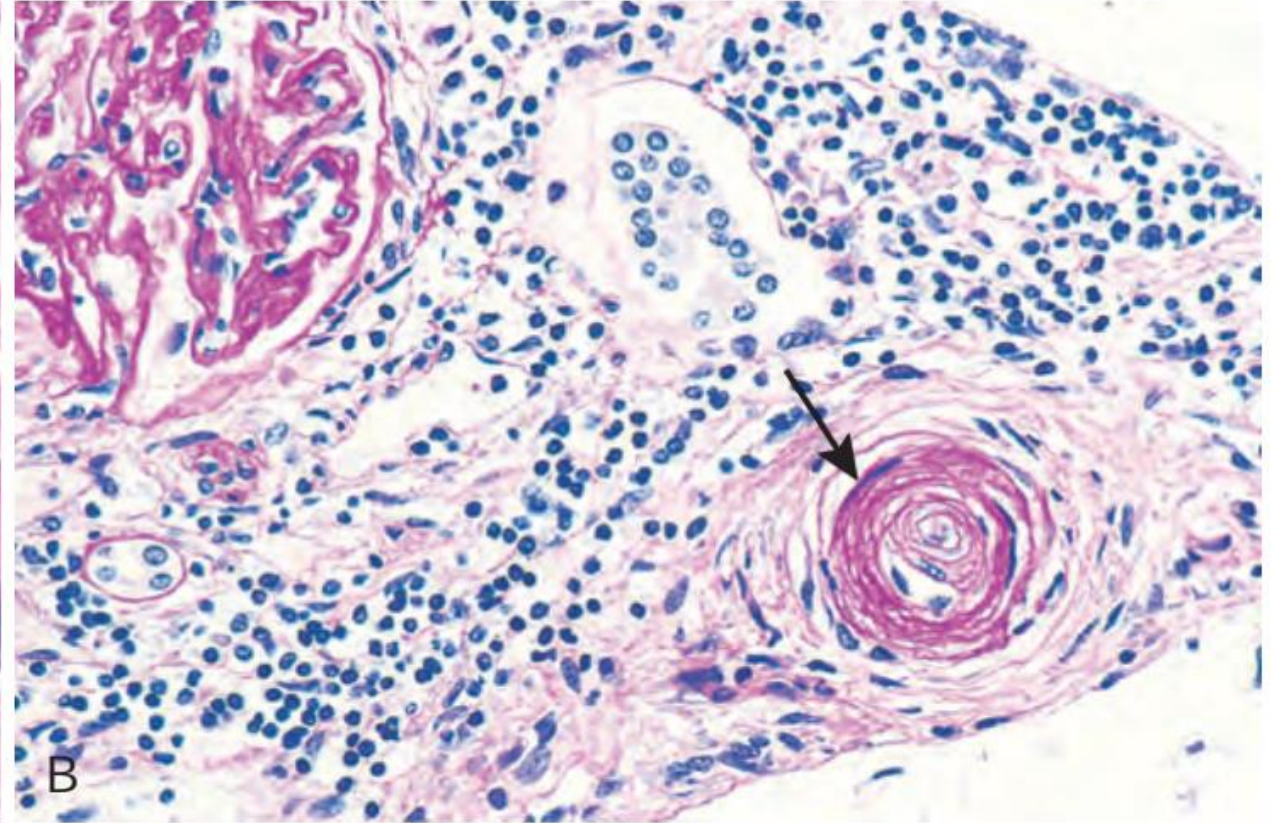
Vascular pathology in hypertension



A

Hyaline arteriosclerosis

The arteriolar wall is thickened with increased protein deposition (hyalinized)



B

Hyperplastic arteriosclerosis

concentric, laminated (“onion-skin”) thickening of the walls, consist of smooth muscle cells

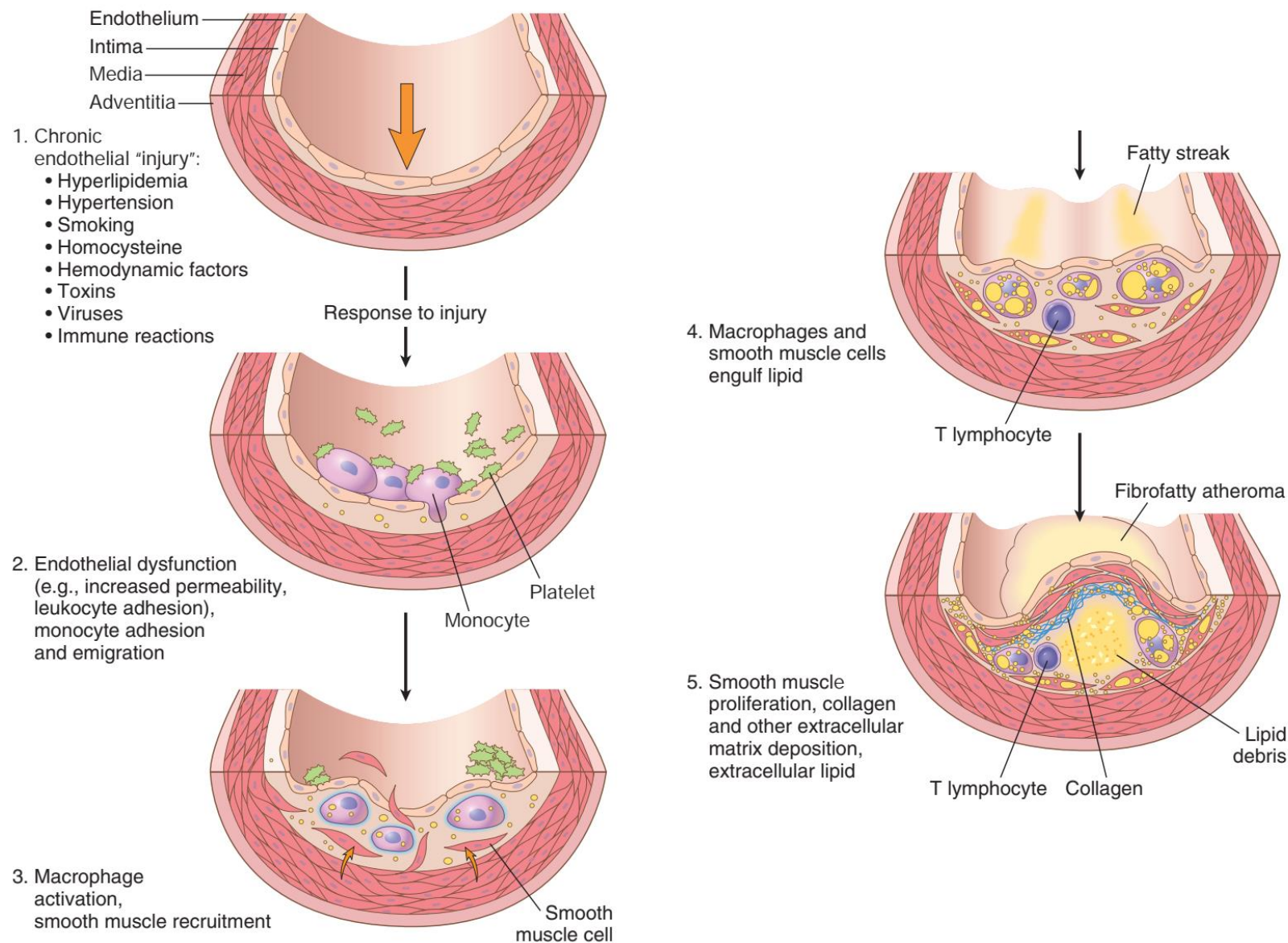
Atherosclerosis

- Hardening of the arteries
- Atherosclerosis is an intimal-based lesion composed of a fibrous cap and an atheromatous core
- Vessel wall injury and inflammation
- The multiple risk factors for atherosclerosis all cause **endothelial cells** dysfunction and influence **inflammatory cells** and **smooth muscle cells** recruitment and stimulation.
- Atherosclerotic plaques develop and generally grow slowly over decades

Table 11.3 Major Risk Factors for Atherosclerosis

Nonmodifiable (Constitutional)
Genetic abnormalities
Family history
Increasing age
Male gender
Modifiable
Hyperlipidemia
Hypertension
Cigarette smoking
Diabetes
Inflammation

Pathogenesis of Atherosclerosis



Pathogenesis of Atherosclerosis

1. Endothelial injury and dysfunction, causing (among other things) increased vascular permeability, leukocyte adhesion, and thrombosis.
2. Accumulation of lipoproteins (mainly LDL and its oxidized forms) in the vessel wall.
3. Monocyte adhesion to the endothelium, followed by migration into the intima and transformation into macrophages and foam cells.
4. Platelet adhesion.
5. Factor release from activated platelets, macrophages, and vascular wall cells, inducing SMC recruitment, either from the media or from circulating precursors.
6. SMC proliferation, ECM production, and recruitment of T cells.
7. Lipid accumulation both extracellularly and within cells (macrophages and SMCs).
8. Calcification of ECM and necrotic debris late in the pathogenesis.

SMC = smooth muscle cell, ECM = extracellular matrix

Atherosclerosis

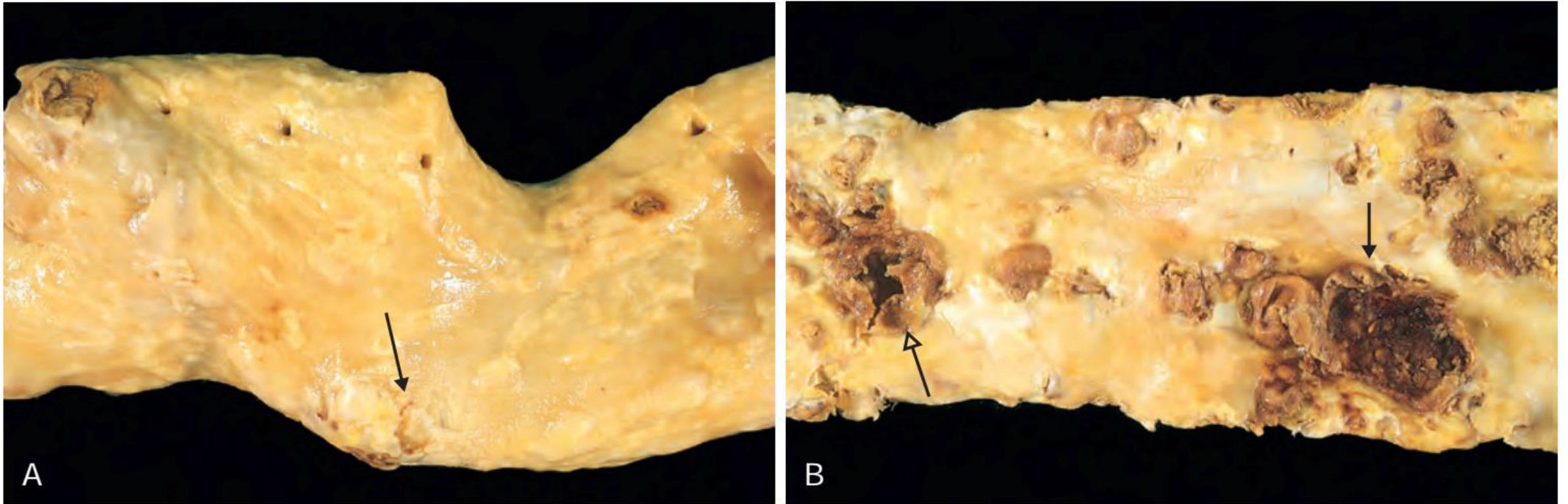


Figure 11.14 Gross views of atherosclerosis in the aorta. (A) Mild atherosclerosis composed of fibrous plaques, one of which is denoted by the arrow. (B) Severe disease with diffuse, complicated lesions including an ulcerated plaque (*open arrow*) and a lesion with overlying thrombus (*closed arrow*).

Atherosclerosis

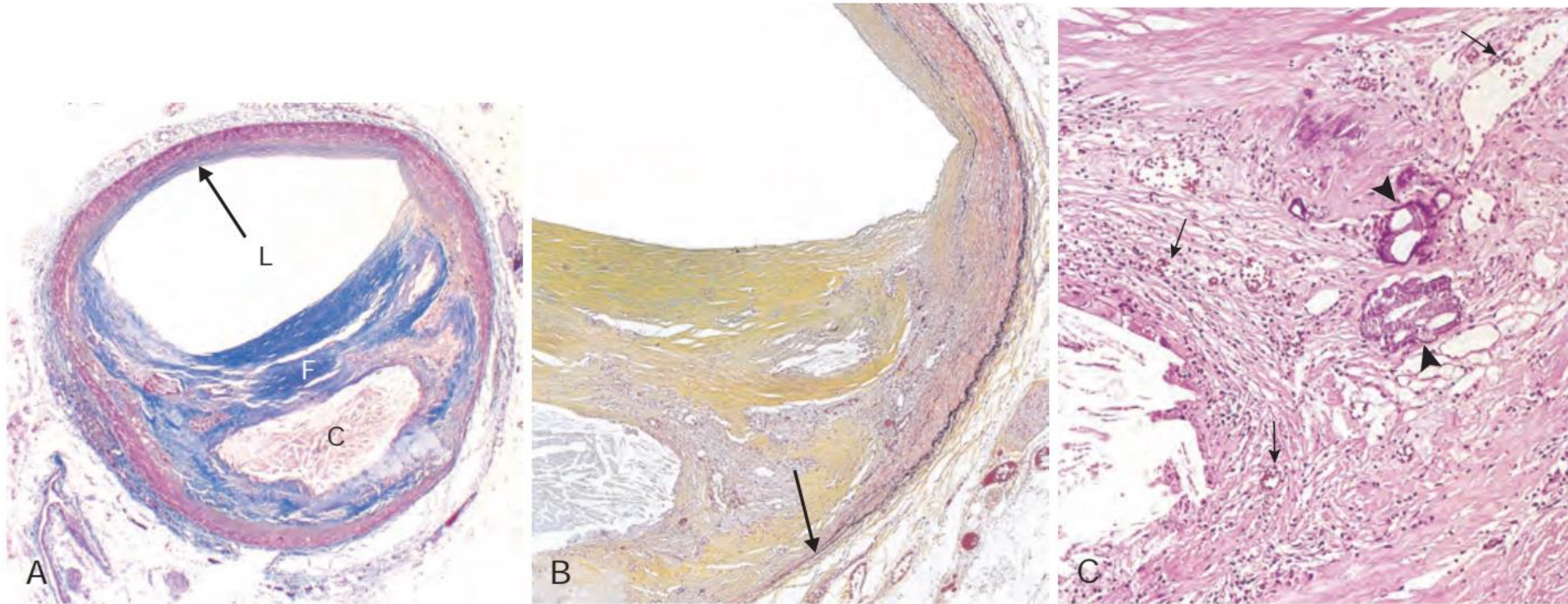
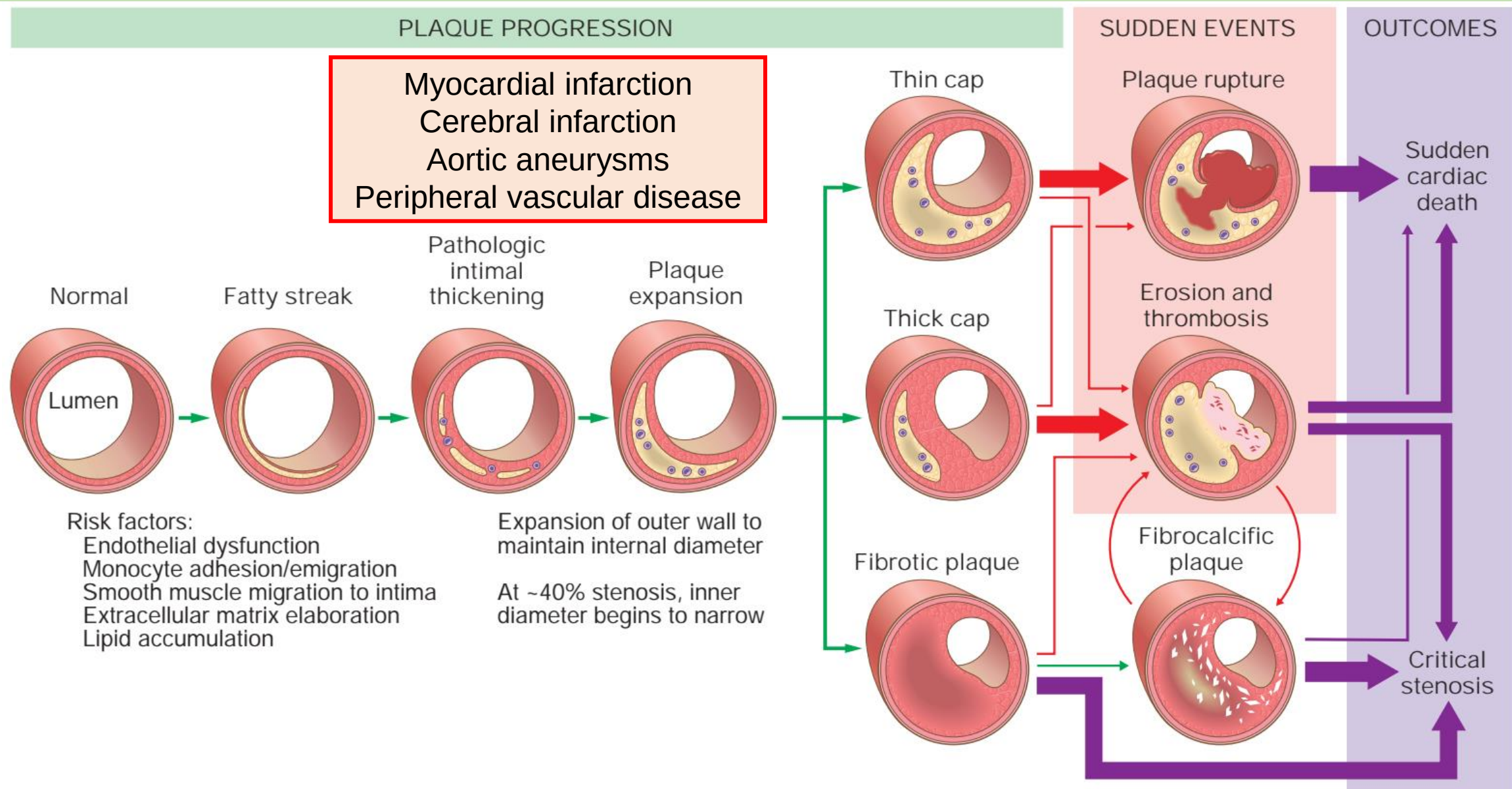


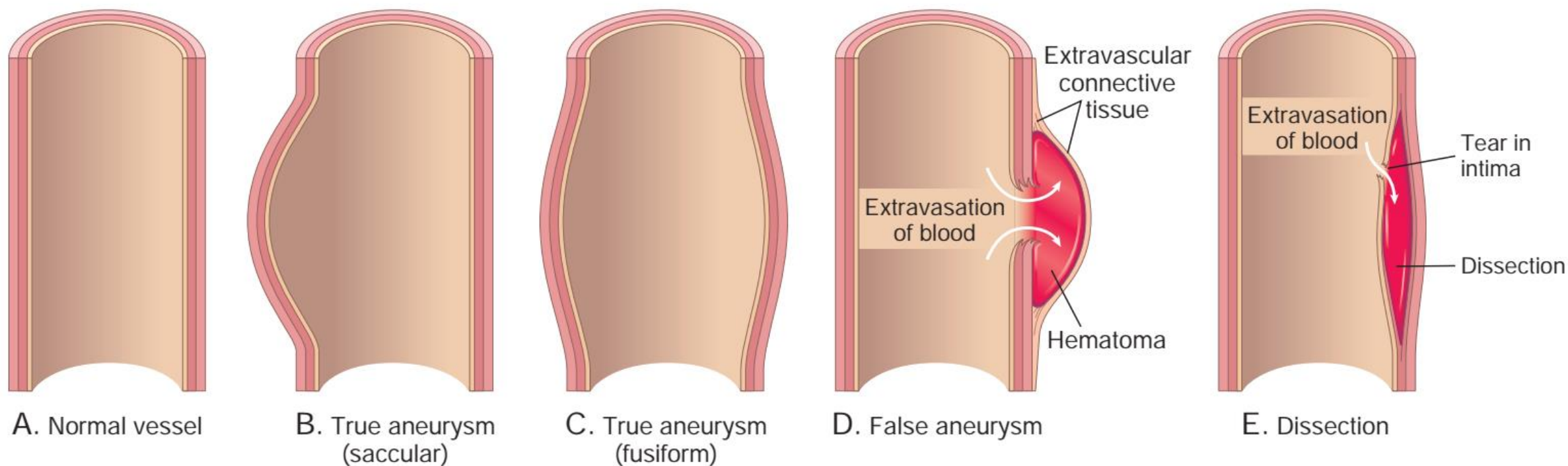
Figure 11.15 Histologic features of atheromatous plaque in the coronary artery. (A) Overall architecture demonstrating fibrous cap (F) and a central necrotic core (C) containing cholesterol and other lipids. The lumen (L) has been moderately compromised. Note that a segment of the wall is plaque-free (arrow); the lesion is therefore "eccentric." In this section, collagen has been stained blue (Masson trichrome stain). (B) Higher power photograph of a section of the plaque shown in (A), stained for elastin (black), demonstrating that the internal and external elastic membranes are attenuated and the media of the artery is thinned under the most advanced plaque (arrow). (C) Higher magnification photomicrograph at the junction of the fibrous cap and core showing scattered inflammatory cells, calcification (arrowhead), and neovascularization (small arrows).

Consequences of atherosclerotic disease



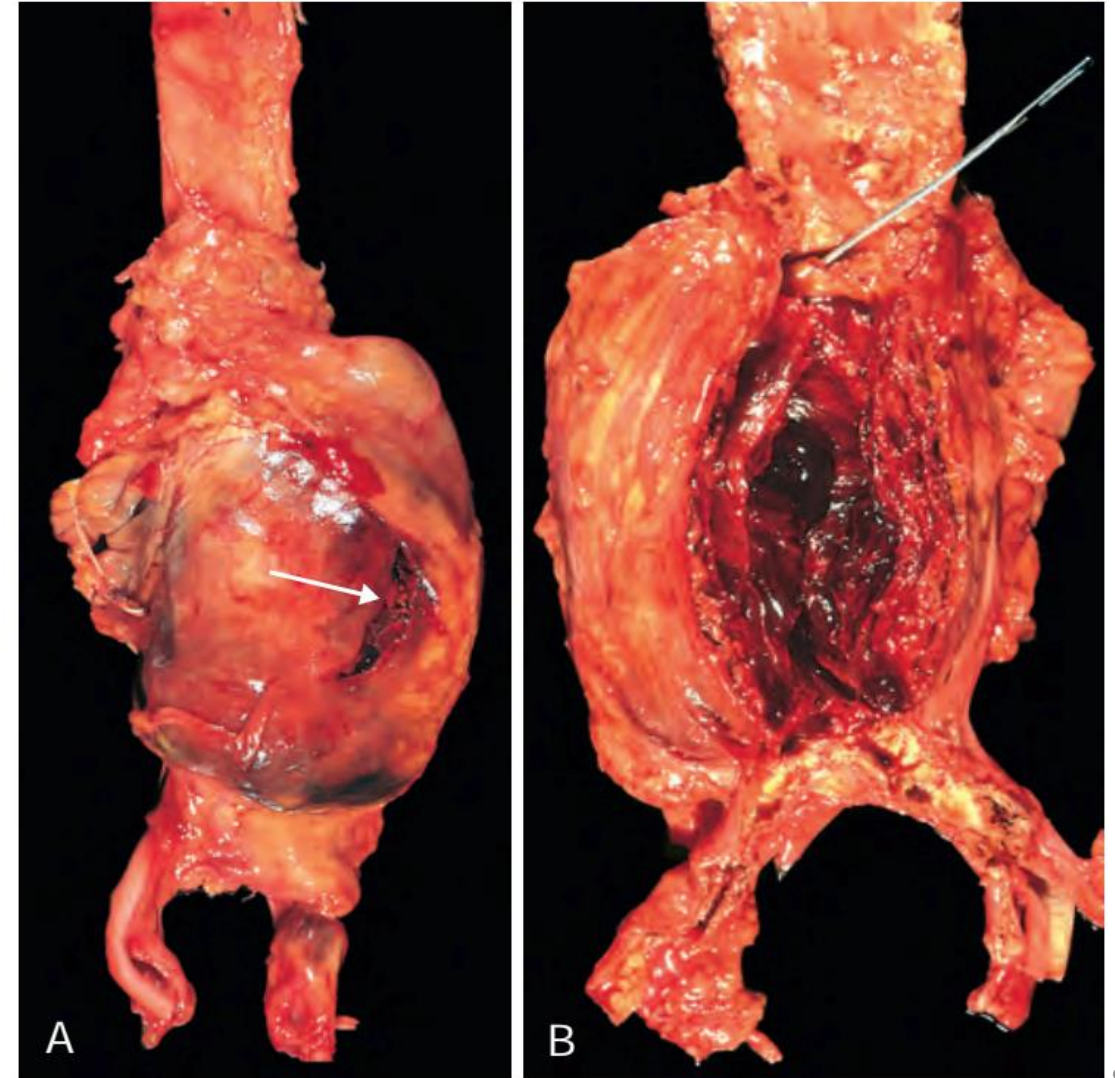
Aneurysm and Dissection

- **Vascular aneurysms** are congenital or acquired **dilations** of the blood vessels that involve the **entire thickness of the wall**
- **Dissections** occur when blood **enters** and **separates** the laminar planes of the **media** to form a blood-filled channel



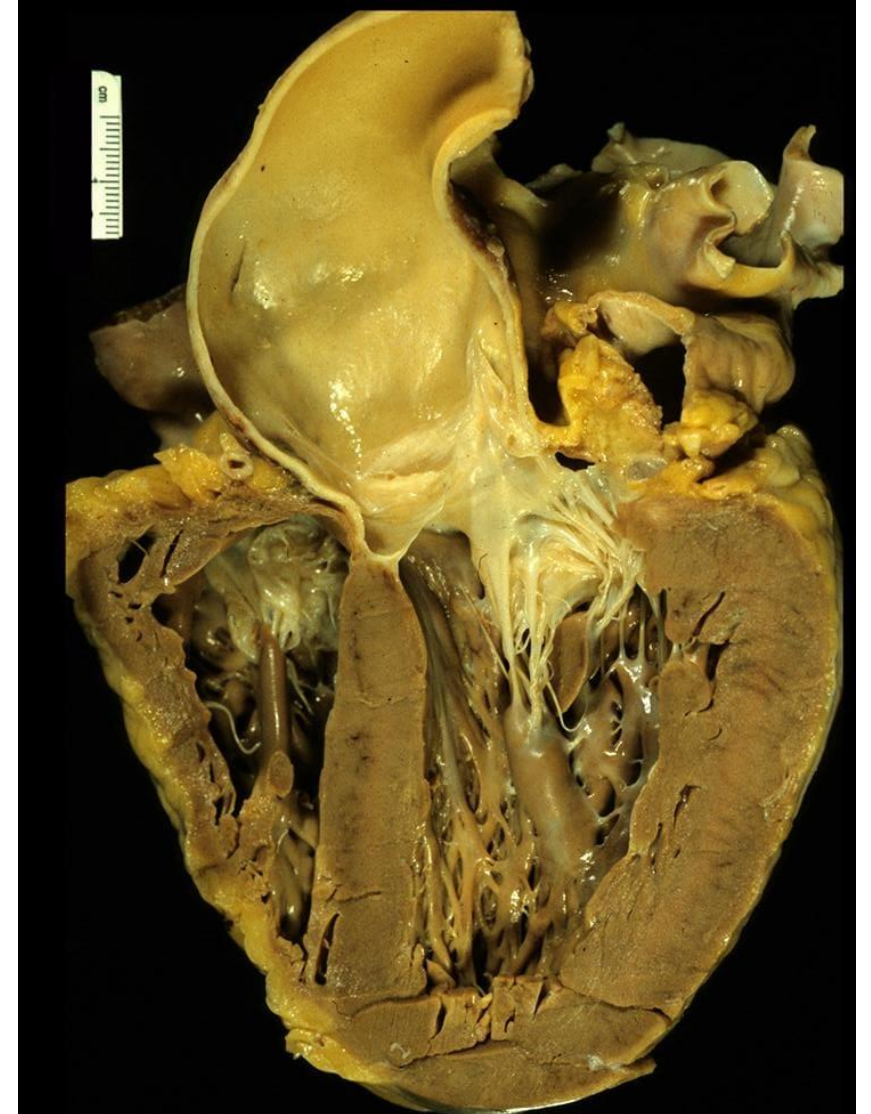
Abdominal aortic aneurysm

- Occurring as a consequence of **atherosclerosis** form most commonly in the abdominal aorta and common iliac arteries
- Most cases are asymptomatic and are discovered incidentally on physical examination as a pulsating abdominal mass
 - Rupture - abdominal pain, potentially fatal hemorrhage
 - Obstruction - leg, kidney, gastrointestinal tract, or spinal cord
 - Embolism
 - Compression an adjacent structure
- The risk of rupture is directly related to the size



Thoracic aortic aneurysm (TAA)

- Most commonly associated with **hypertension**
- Symptoms can include
 - chest pain - erosion into bone
 - myocardial ischemia - compression of a coronary artery
 - difficulty swallowing - compression of the esophagus
 - hoarseness - irritation of or pressure on the recurrent laryngeal nerves
 - respiratory complications - compression of the bronchi



Aortic Dissection

- Aortic dissection occurs principally in two groups of patients:
 1. men aged 40 to 60 years, with antecedent hypertension (more than 90% of cases)
 2. younger patients with syndromic diseases affecting the aorta (e.g. Marfan syndrome)
- **Hypertension** is the major risk factor
- The classic clinical symptoms are the **sudden onset** of **anterior chest pain and radiating to the back**
- The most common cause of death is rupture and bleeding

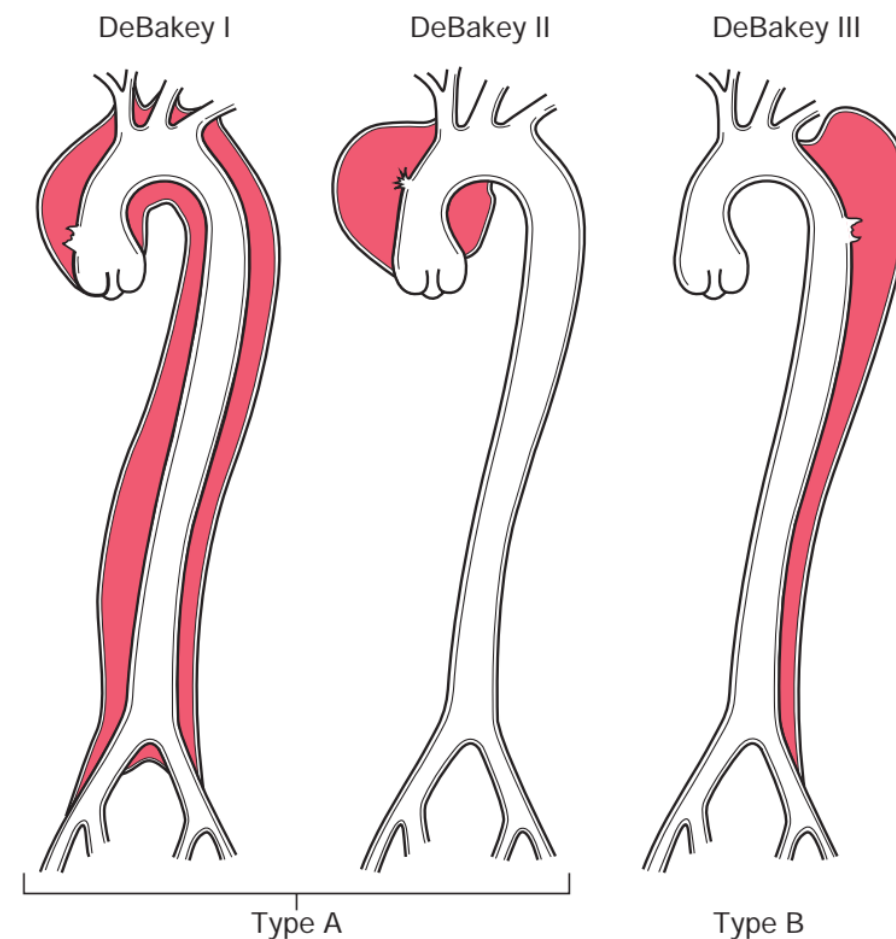


Figure 11.23 Classification of dissections. *Type A* (proximal) involves the ascending aorta, either as part of a more extensive dissection (DeBakey I) or in isolation (DeBakey II). *Type B* (distal or DeBakey III) dissections arise after the take-off of the great vessels. *Type A* dissections typically have the most serious complications and greatest mortality.

Aortic Dissection

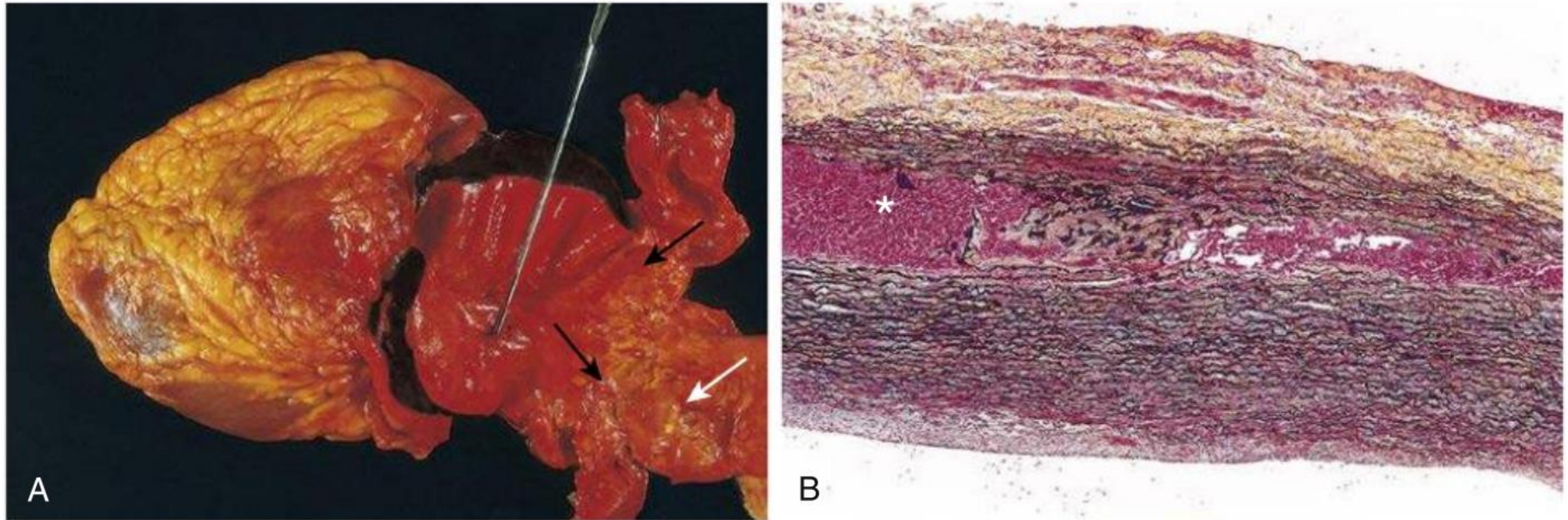


Figure 11.22 Aortic dissection. (A) Gross photograph of an opened aorta with proximal dissection originating from a small, oblique intimal tear (*probe*), allowing blood to enter the media and creating a retrograde intramural hematoma (*thin arrows*). Note that the intimal tear has occurred in a region largely free of atherosclerotic plaque and that propagation of the intramural hematoma distally is arrested where atherosclerosis begins (*thick arrow*). (B) Histologic view of dissection demonstrating an aortic intramural hematoma (*asterisk*). Aortic elastic layers are black and blood is red (Movat stain).

Vasculitis

- Vasculitis is defined as **inflammation of vessel walls**
- The two pathogenic mechanisms of vasculitis are **immune-mediated inflammation** and **direct invasion of vascular walls by infectious pathogens**
- It is frequently associated with **systemic manifestations** (including fever, malaise, myalgias, and arthralgias) and **organ dysfunction** that depends on the pattern of vascular involvement

Vasculitis

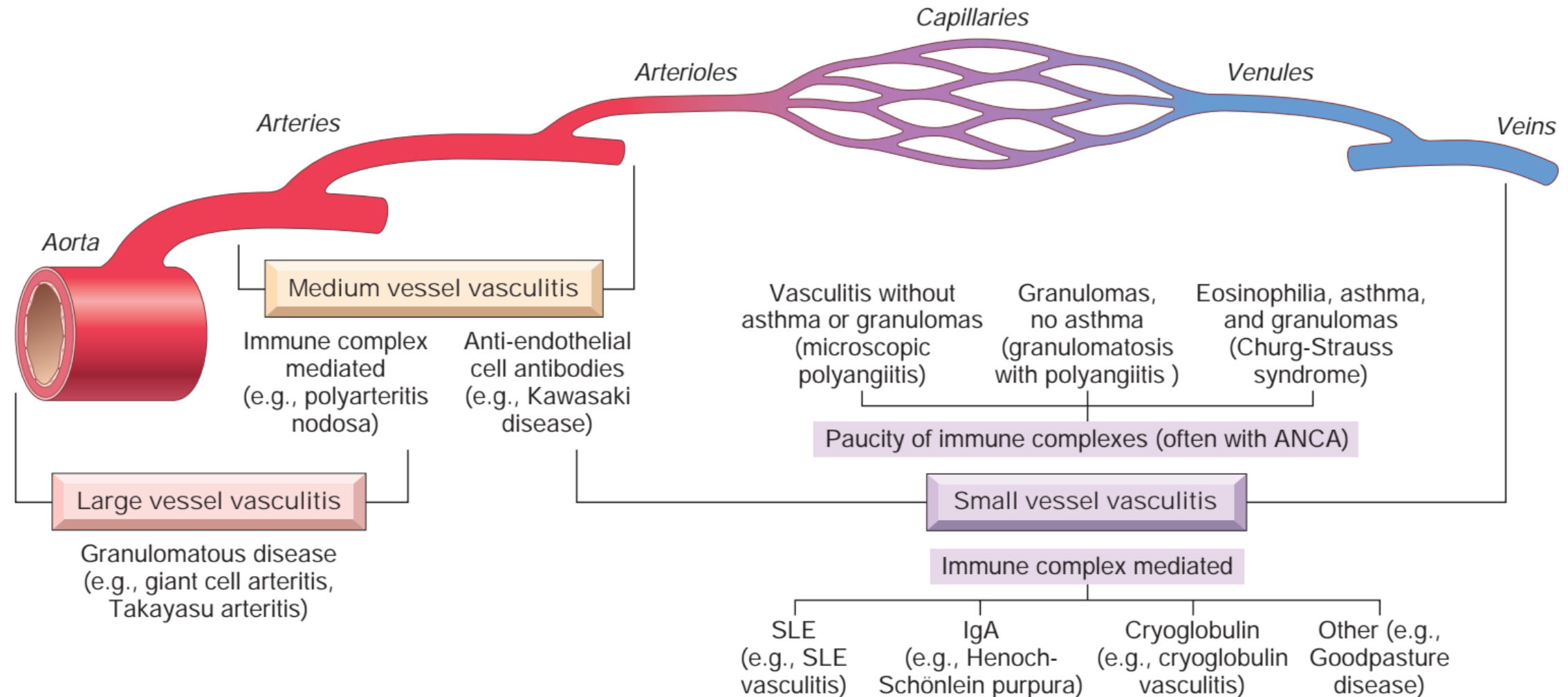


Figure 11.24 Vascular sites typically involved with the more common forms of vasculitis as well as their presumptive etiologies. Note that there is a substantial overlap in distributions. ANCA, Antineutrophil cytoplasmic antibody; SLE, systemic lupus erythematosus.

Table 11.4 Primary Forms of Vasculitis

	Giant Cell Arteritis	Granulomatosis With Polyangiitis	Churg- Strauss Syndrome	Polyarteritis Nodosa	Leukocytoclastic Vasculitis	Buerger Disease	Behçet Disease
Sites of Involvement							
Aorta	+	–	–	–	–	–	–
Medium-sized arteries	+	+	+	+	–	+	+
Small-sized arteries	–	+	+	+	+	+	+
Capillaries	–	–	–	–	+	–	+
Veins	–	–	–	–	+	+	+
Inflammatory Cells Present							
Lymphocytes	+	+	+	±	±	±	±
Macrophages	+	+	+	±	±	±	±
Neutrophils	Rare	+	+	±	±	±	Required
Eosinophils	Very rare	±	Required	±	±	±	±
Other Features							
Granulomas	+ ^a	Required ^a	±	–	–	–	–
Giant cells	Often; not required	±	–	–	–	–	–
Thrombosis	±	±	±	±	±	Required	±
Serum ANCA positivity	–	+	+	±	–	–	–
Clinical history	>40 years old, ± polymyalgia rheumatica	Any	Asthma, atopy	Any	Any	Young male smoker	Orogenital ulcers

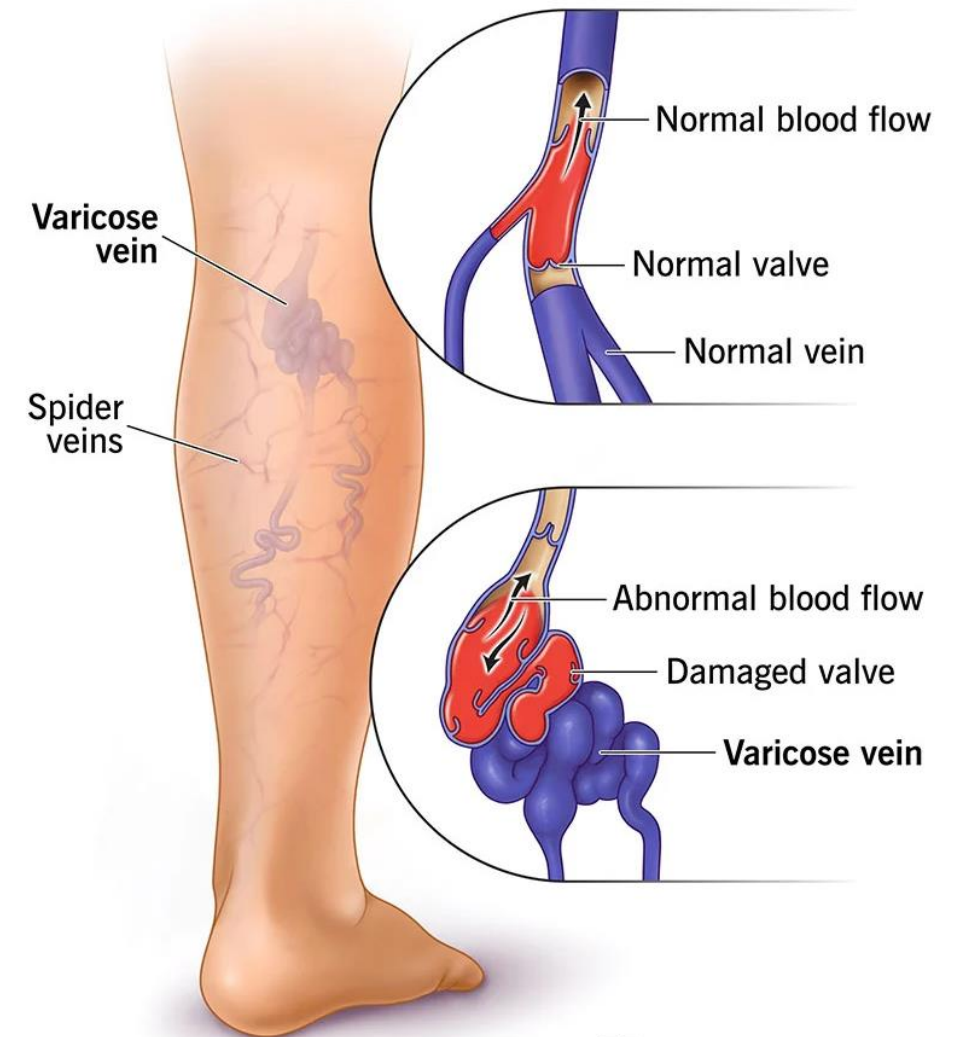
^aThe granulomas of giant cell arteritis are found within the vessel wall as part of the inflammation comprising the vasculitis but need not be present to render the diagnosis. The granulomas of granulomatosis with polyangiitis are larger, spanning between vessels, and associated with areas of tissue necrosis.

ANCA, Antineutrophil cytoplasmic antibody.

From Seidman MA, Mitchell RN: Surgical pathology of small- and medium-sized vessels, *Surg Pathol Clin* 5:435, 2012.

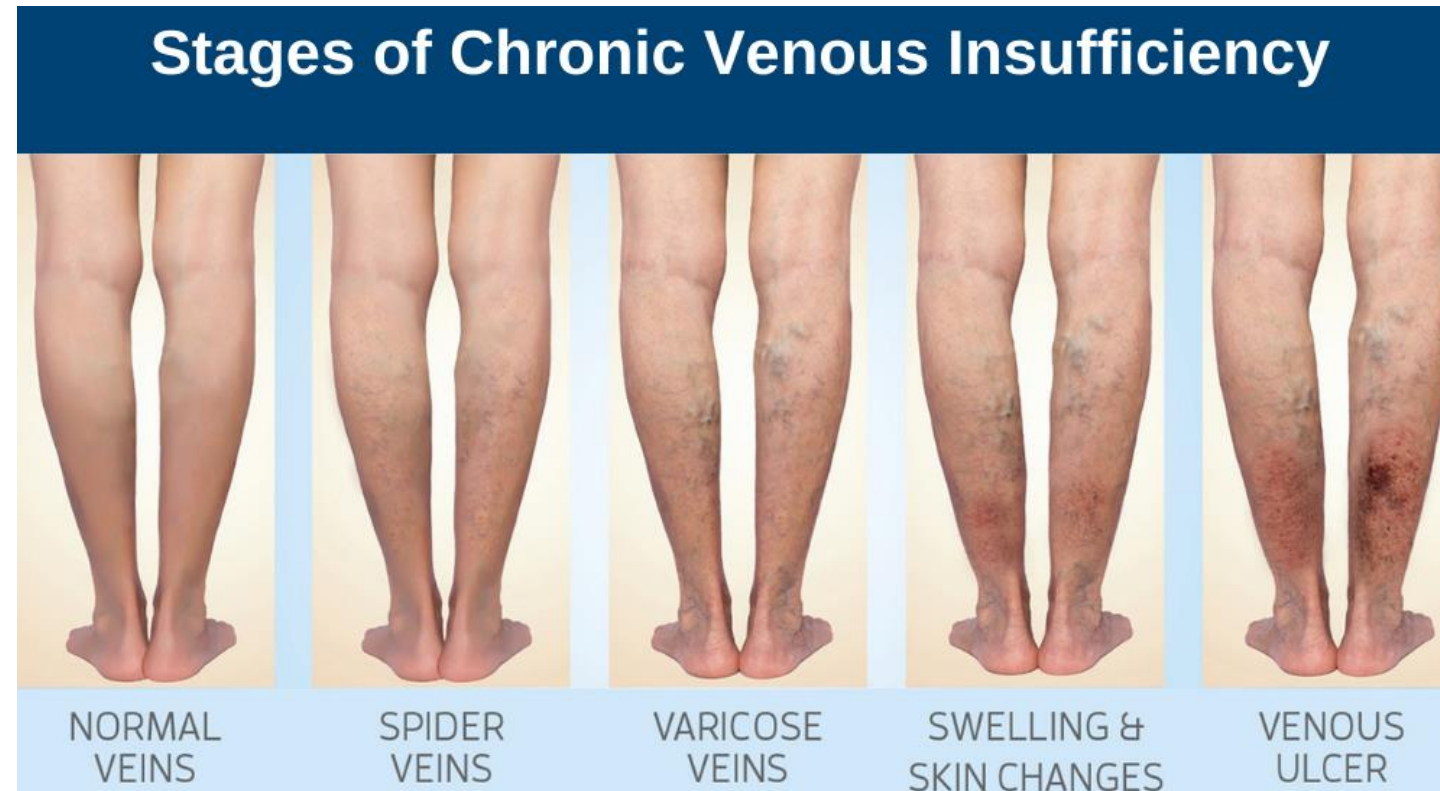
Varicose Veins

- Abnormally dilated, tortuous veins produced by prolonged, **increased intraluminal pressure** with vessel dilation and incompetence of the venous valves
- **Superficial veins of the upper and lower leg** are commonly involved because venous pressures in these sites can be markedly elevated (up to 10 times normal) by prolonged dependent posture



Varicose Veins

- Clinical Features
 - stasis, congestion, edema, pain, and thrombosis
 - secondary tissue ischemia and ulcerations
 - poor wound healing and superimposed infections



Vascular tumors

Table 11.5 Classification of Vascular Tumors and Tumor-Like Conditions

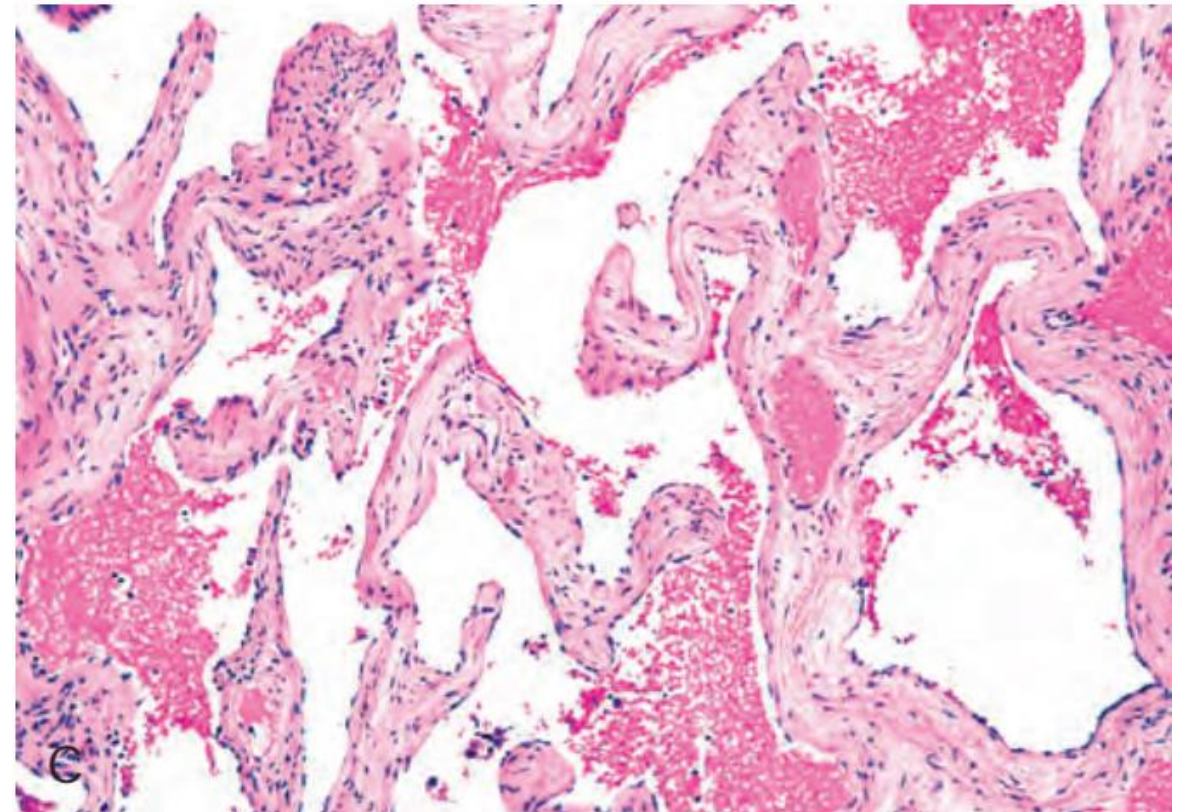
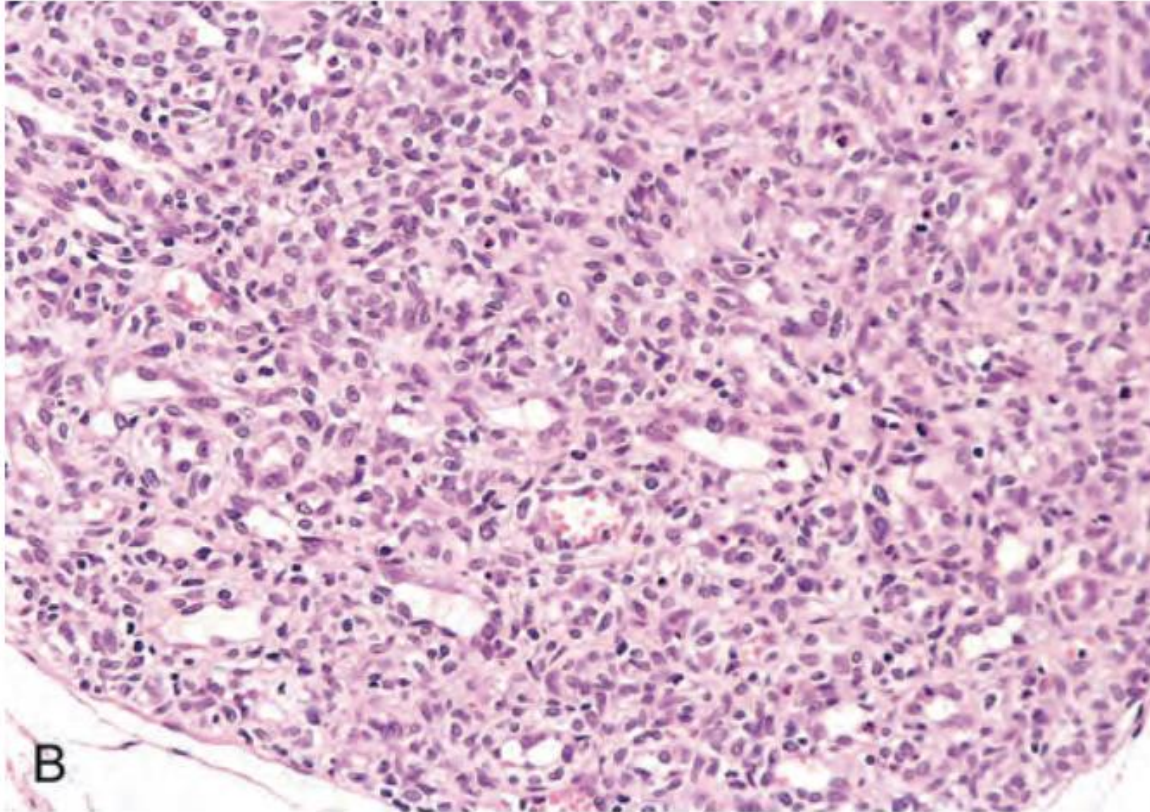
Benign Neoplasms, Developmental and Acquired Conditions
Hemangioma Capillary hemangioma Cavernous hemangioma Pyogenic granuloma
Lymphangioma Simple (capillary) lymphangioma Cavernous lymphangioma (cystic hygroma)
Glomus tumor
Vascular ectasias Nevus flammeus Spider telangiectasia (arterial spider) Hereditary hemorrhagic telangiectasis (Osler-Weber-Rendu disease)
Reactive vascular proliferations Bacillary angiomatosis
Intermediate-Grade Neoplasms
Kaposi sarcoma Hemangioendothelioma
Malignant Neoplasms
Angiosarcoma Hemangiopericytoma

Hemangioma

- Very common tumor composed of blood-filled vessels
- Typically confined to the head, neck, and thoracic skin, they can also arise internally
- Malignant transformation is rare



Hemangioma

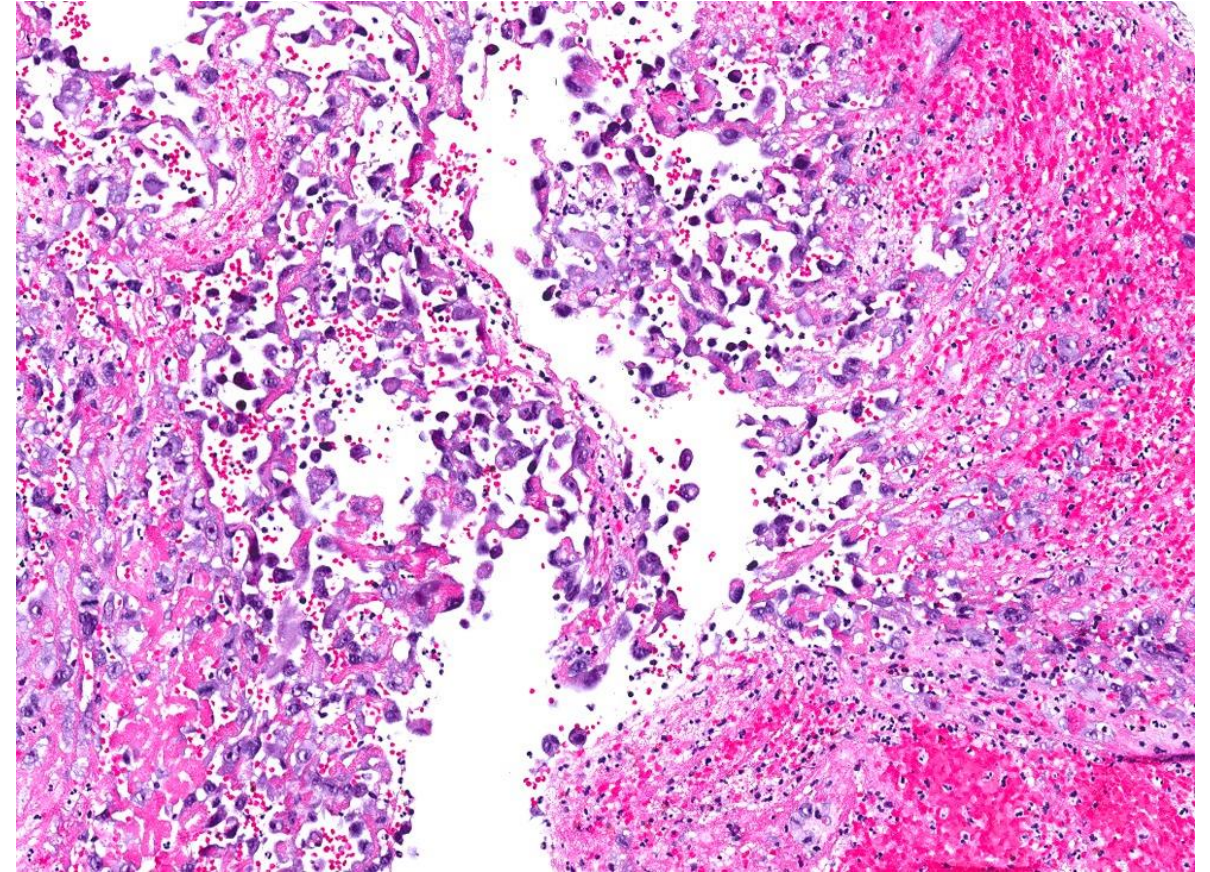
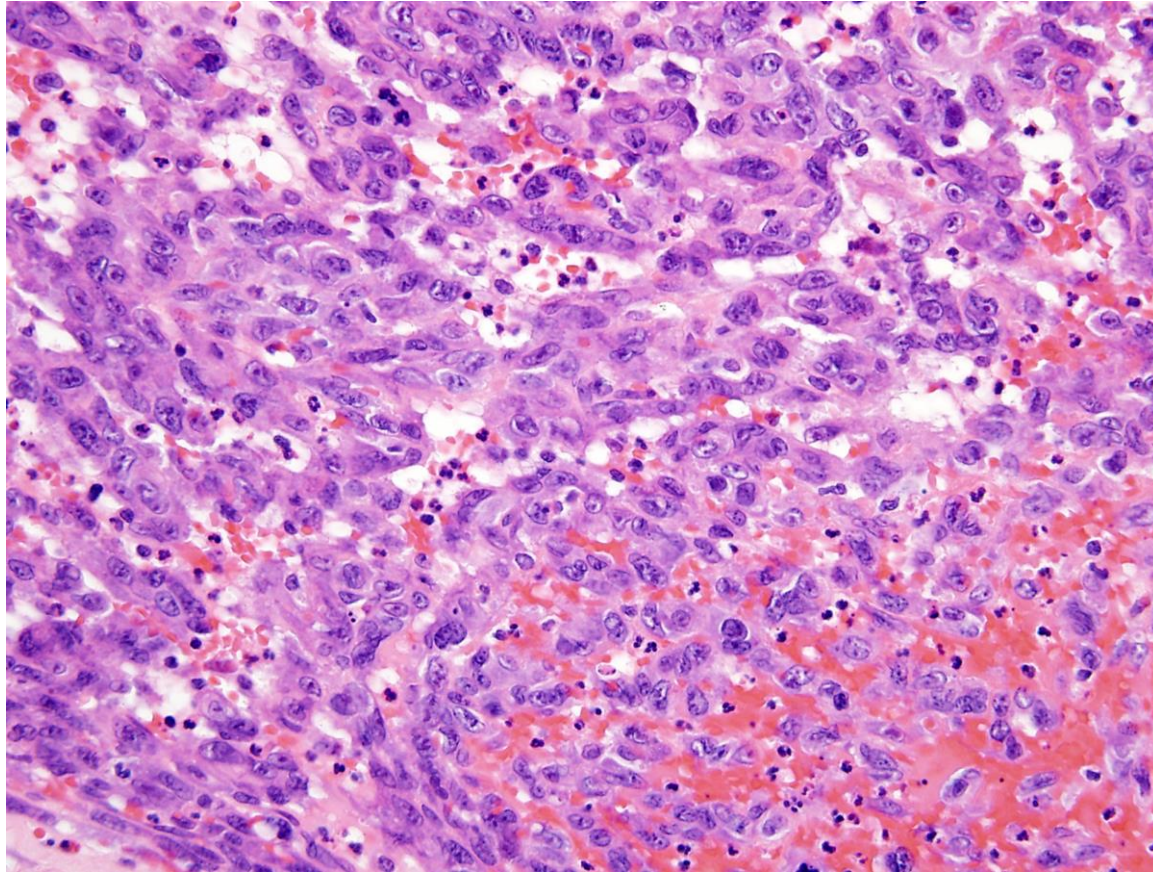


Angiosarcoma

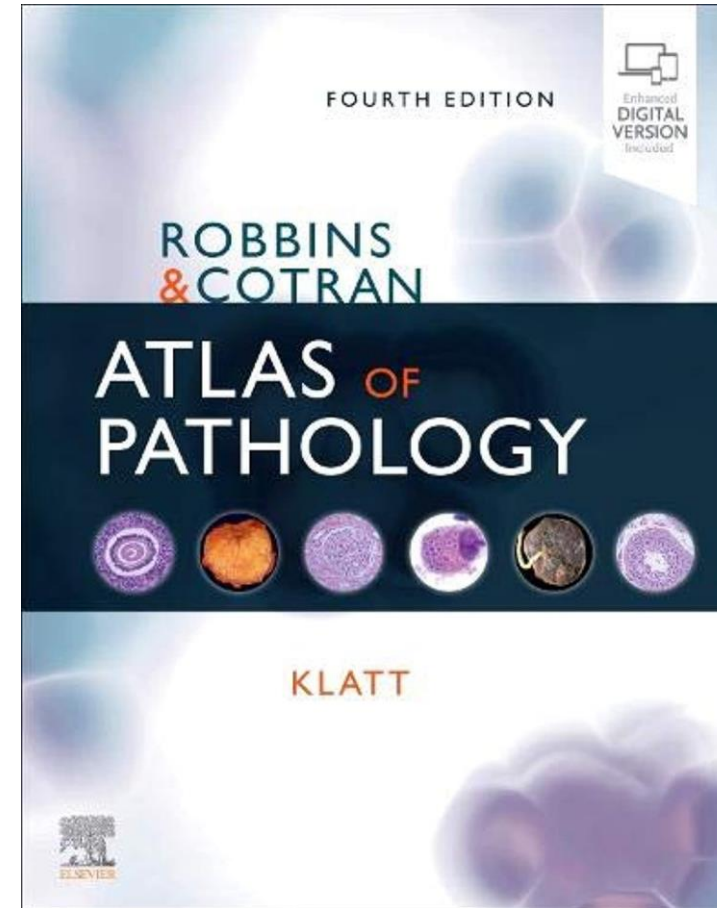
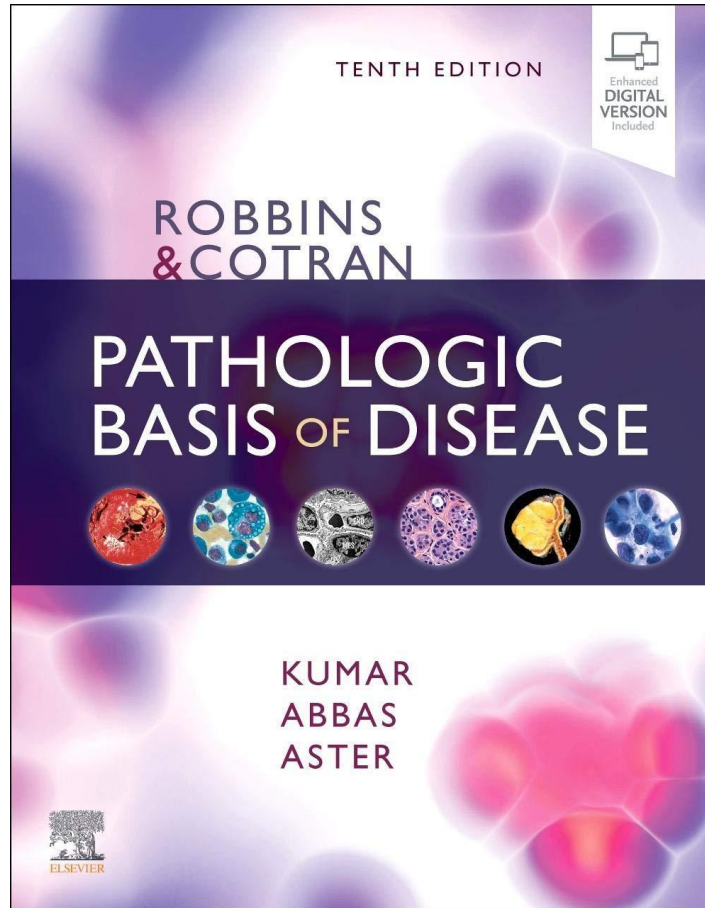
- Malignant endothelial neoplasms
- Older adults are more commonly affected
- Occur at any site, but most often involve skin, soft tissue, breast, and liver
- Angiosarcomas have also been induced by radiation



Angiosarcoma



References



<http://lib.med.psu.ac.th/eservice/ebook/index.php>